

MADE BY AFRINOVA SYSTEMS SOFTWARE ENGINEERING UFS RECRUITMENT SYSTEM (UNICONNECT)



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Disclaimer

It should be noted that different users interact with different features depending on their assigned roles and permissions (Student, Employer, Administrator). For the purposes of this project submission, the scope focuses primarily on the design and specification of the web-based application. Although the final system will also include a mobile application, this document concentrates on the web version.

Some diagrams and images in this document are AI-generated for illustrative purposes and should be verified for accuracy during implementation.

Taxonomy

Keyword Box	
Users of the Feature	Refers to specific roles who will interact with a functionality (Student, Employer, Administrator, UFS Database System).
Disclaimer	This indicates important conditions or limitations for certain features that must be noted before continuing.
Take Note:	This highlights important additional details to consider when reading feature descriptions

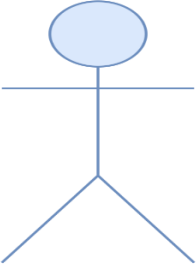
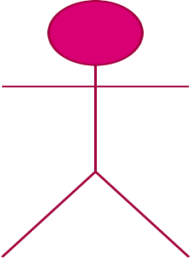
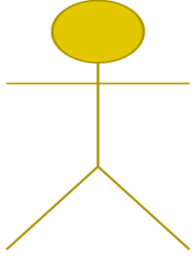
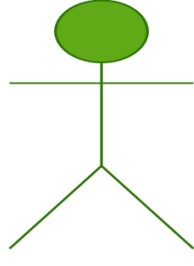
System Users			
Disclaimer: It should be noted that different users interact with different features according to their assigned roles and permissions (Student, Employer, Administrator). For the purposes of this project submission, the scope focuses primarily on the design and specification of the web-based application.			
Student	Employer	Administrator	UFS Database System
			

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Chapter 1: Project Description, Specifications, and Scope

Introduction:

In today's competitive job market, a critical gap exists between academic achievement and successful career placement. The University of the Free State (UFS) has identified the need for a modern, integrated platform to bridge this gap, enhancing student employability and streamlining employer engagement. This document outlines the specification for the development of a unified **2025 UFS Recruitment System**—a mobile and web-based application designed to connect students with potential employers seamlessly.

The proposed UFS Recruitment System aims to enhance student employability and employer engagement through a secure, integrated digital platform (**University of the Free State, 2025**). The system will serve as a dynamic bridge, allowing students to showcase their skills and academic credentials while providing employers with a powerful tool to discover and recruit qualified talent directly from the UFS community.

Problem Statement

The current process for connecting UFS students with employment opportunities is fragmented.

1. **For Students:** The journey from academic life to professional employment is disconnected. Students are forced to manage their professional identity across multiple, unverified platforms (like LinkedIn and generic job boards), leading to a diluted and inconsistent presentation to employers. Manually creating CVs, searching for relevant jobs, and tracking applications is time-consuming and often misses opportunities that align perfectly with their unique academic and skill profile. Furthermore, they lack a clear, data-driven understanding of the specific skills they need to develop to be competitive in their chosen field.
2. **For Employers:** Recruiting qualified talent from UFS is a manual and inefficient process. Verifying student credentials (degrees, GPA, skills) is difficult and non-standardized, increasing the risk and time-to-hire. They struggle to efficiently filter and find candidates who precisely match their requirements based on verified academic data and specific skill sets. The process of scheduling interviews and requesting academic records is cumbersome and conducted outside a secure, managed environment.
3. **For the University (UFS):** There is a critical lack of centralized, actionable data on graduate employment outcomes and market-driven skill trends. This hinders the university's ability to measure the effectiveness of its programs, report on graduate success, and proactively adapt curricula to meet evolving industry demands, potentially leading to a gap between academic offerings and market needs.

This fragmentation results in missed opportunities for students, inefficient recruitment for employers, and a limited ability for UFS to demonstrate and enhance the employability of its graduates. Similar university–industry platforms have demonstrated improved graduate placement outcomes through intelligent matching algorithms (Verma & Singh, 2023; Xu, Zhong & Wang, 2024).

Project Objectives

The primary objective is to develop a user-friendly, secure, and scalable "UFS Recruitment System" that:

1. **Empowers Students:** Provides tools for professional profile management, CV building, and direct job application tracking.
2. **Attracts and Support Employers:** Offers advanced search capabilities and a streamlined process for posting jobs and scheduling interviews.
3. **Supports Administration:** Delivers analytics on employment trends and manages system users and data integrity.
4. **Ensures Integration:** Securely connects with existing UFS databases for real-time academic data updates (with student consent).

Key Features & Scope

Disclaimer: This project specification focuses exclusively on the design and development of the web-based application for the UFS Recruitment System. While the final delivered product for the University of the Free State is envisioned to include a fully functional mobile application, the scope of this document and the subsequent development phase is limited to the web platform. All use cases, user interface designs, and functional specifications described herein pertain to the web version. The mobile application remains a consideration for future expansion and is not part of the current project deliverables.

In Scope:

- **Student Features:**
 - Registration and professional profile setup.
 - Automatic academic data integration from UFS databases.
 - Privacy-controlled sharing of grades and transcripts.
 - AI-assisted CV Builder.
 - Intelligent job matching and job board.

- Interview scheduling calendar.
- Portfolio/Project showcase (SkillStack).
- **Employer Features:**
 - Verified employer profile registration.
 - Job and internship posting.
 - Candidate search with filters (GPA, skills, graduation year).
 - Option to request academic records (with student approval).
 - Integrated interview scheduling.
- **Admin Features:**
 - User and role management.
 - Analytics dashboard for job placements and skill trends.
 - System oversight and support.
- **Selected Additional Features:**
 - Interview **Scheduling System**: Employers and students can confirm interview times within the system.
 - Skill **Gap Analysis Tool**: Suggests resources for students to align their skills with market demand.
 - **Portfolio Showcase (SkillStack)**: Allows students to upload and display projects and achievements.

Out of Scope:

- Processing of salaries or employment contracts.
- Full AI-based career path prediction.
- Direct messaging outside interview scheduling.
- Integration with external social platforms such as LinkedIn.

Target Audience

- Students: UFS Undergraduate and postgraduate students seeking internships, part-time jobs, and graduate positions.

- Employers: Companies and organisations looking to recruit UFS talent.
- Administrators: UFS staff overseeing platform performance, analytics, and user management.

Benefits / Value Proposition

- **For Students:** A single platform for professional growth, visibility to employers, and streamlined job applications.
- **For Employers:** Access to a verified pool of candidates with advanced search and scheduling tools.
- **For UFS:** Data-driven insights into employability trends and skill gaps to improve academic programmes.

Risks and Mitigation

- **Data Privacy Risks:** Risk of unauthorised data access. → Mitigation: Implement strict encryption and role-based access.
- **System Adoption Risks:** Students and employers may hesitate to adopt a new platform. → Mitigation: Offer onboarding tutorials and training.
- **Scalability Risks:** Increased users over time. → Mitigation: Design the system with scalable cloud infrastructure.

Non-Functional Requirements (NFRs).

- **Performance:** The system shall support up to 5,000 concurrent users with a page load time of under 3 seconds.
- **Security:** All sensitive data (passwords, academic records) shall be encrypted at rest and in transit. The system shall comply with POPIA guidelines.
- **Availability:** The system shall achieve 99.5% uptime, excluding scheduled maintenance.
- **Usability:** The system shall adhere to WCAG 2.1 AA guidelines for accessibility.

According to ISO standards, quality attributes such as functionality, reliability, usability, and security must be embedded early in the design (ISO, 2011).



Additionally, compliance with the Protection of Personal Information Act ensures that student data are processed lawfully and transparently (Republic of South Africa, 2013).

Presentation Information:

The Culture Compass

As AfriNova Systems, we are excited to introduce The Culture Compass a tool designed to bring the heart of a company to life and help students connect with organisations beyond job titles and tasks.

Concept: The Culture Compass is a dedicated, interactive space within the platform where companies can showcase more than just a vacancy. It functions as a digital open house, giving students an authentic feel for a company's personality, core values, daily work environment, and employee experiences.

Functionality

For Employers: A dedicated dashboard allows them to create and update their Culture Compass profile. They can upload photos/videos of their workspace, describe their mission and benefits, and include testimonials or video clips from current employees.

For Students: Accessible from any job listing or company profile, students can explore this visual and emotional snapshot of an organisation to assess cultural fit before applying.

Purpose: This feature adds depth to the recruitment process, encouraging better alignment between students and employers and supporting longer-lasting, meaningful employment connections. This innovation will be a key highlight of our final presentation and demo.

Conclusion

The UFS Recruitment System is a strategic solution designed to enhance employability, strengthen employer engagement, and empower UFS in aligning academic outcomes with market needs. With its scalable, secure, and user-friendly design, the system will serve as a bridge between academia and employment, ensuring lasting benefits for students, employers, and the university.

Chapter 2: Specification Document

Introduction

The Specification Document outlines the functional requirements, user roles, and key features of the UFS Recruitment Platform, such as login, profile management, job search, and candidate matching. It details how users, including Students, Employers, and Administrators, interact with the system for authentication, recruitment activities, and administrative oversight.

Use Case Specifications

2.1 Summary Use Case Diagram



Figure 1: Summary Use case

Actors: Student, Employer, Administrator, UFS database System (PPS external system).

Core Functionalities: User Authentication, Profile Management, Job Posting & Search, Candidate Search & Filtering, Application Management, Interview Scheduling, Administrative Analytics & Control.

Disclaimer: All users, except those browsing public job postings, must be logged in to access functionality. Different users interact with functionalities in different ways based on their permissions.

System Design (UML, Entities, and Architecture)

The object-oriented approach aligns with established software engineering principles that encourage modularity, encapsulation, and reuse (**Sommerville, 2015; Pressman & Maxim, 2020**).

UML modelling techniques provide a standardised visual language for specifying, constructing, and documenting software systems (**IEEE, 1998**).

2.2 Detailed Use Cases

Use Case UC1: User Login & Authentication

Actor: Student, Employer, Administrator

Description:

This is the universal authentication functionality that allows all users (Students, Employers, Administrators) to securely access the system using their registered credentials. It includes security measures such as account lockout after multiple failed attempts.

UML

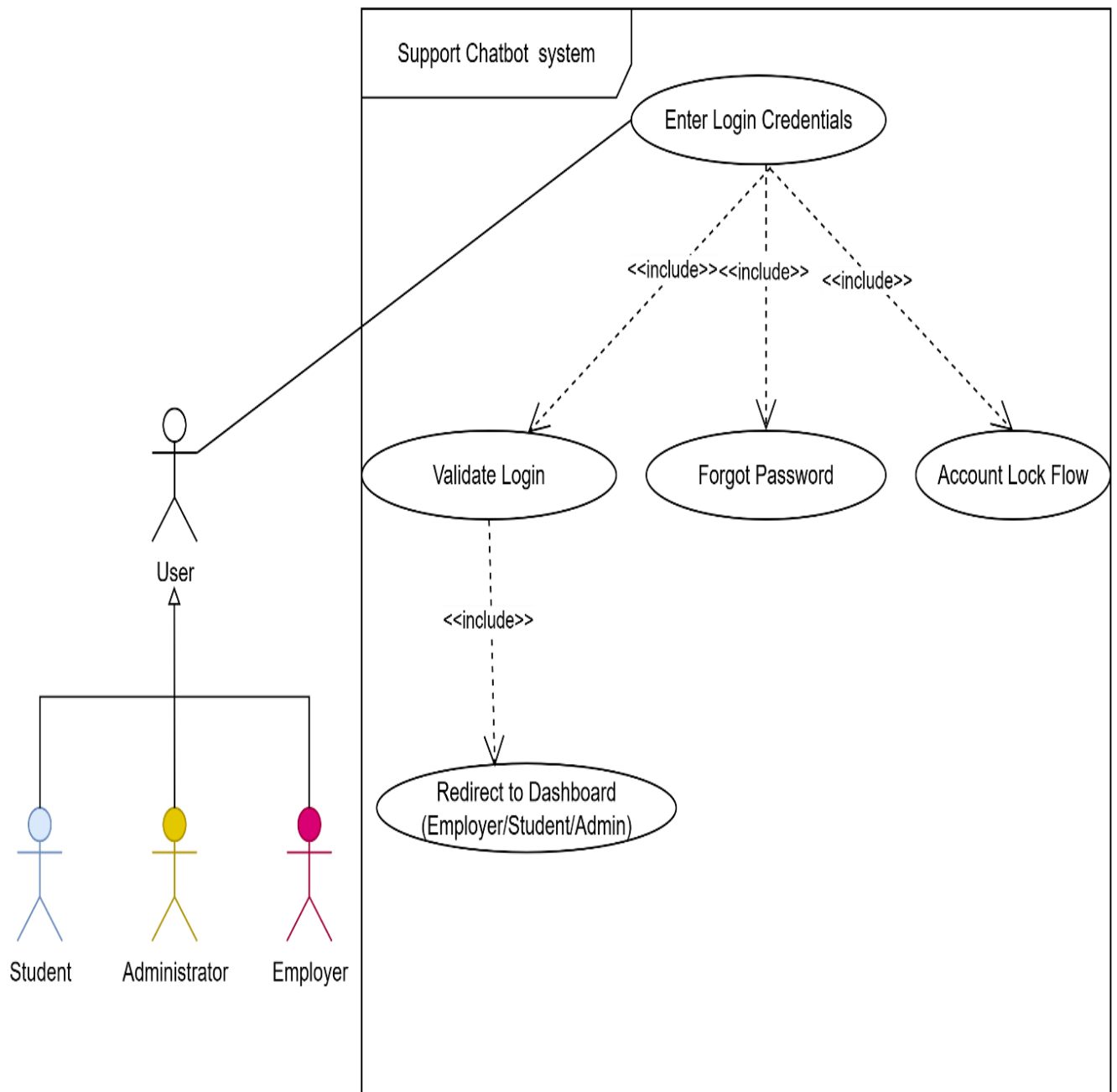


Figure 2: Use Case UC1: User Login & Authentication

Step-by-Step:

1. User navigates to the UFS Recruitment System login page.
2. User enters their registered email address into the "Email" text box.
3. User enters their password into the "Password" text box.
4. User clicks the "Sign In" button.
5. System validates the credentials against the user database.
 - 5.1. If credentials are correct: The system checks the user's role, logs the login, and redirects them to their appropriate dashboard (Student, Employer, Admin).
 - 5.2. If credentials are incorrect: The system displays an error message below the password field: "Invalid email or password." A login attempt counter increments, showing "2 login attempt(s) remaining."
6. After a third failed attempt, the system displays a new message: "Account locked due to multiple failed attempts. Please reset your password or contact support." The "Sign In" button is disabled.
7. User clicks the "Forgot your password?" link.
8. System redirects to a password recovery page where the user can enter their email to receive a reset link.

Best-Case Scenario:

1. The admin navigates to the University Job Portal login page.
2. They enter their registered email: admin@university.edu and the correct password: admin123.
3. They click the "Sign In" button.
4. The system instantly verifies the credentials, recognizes the user's role as "Admin," and redirects them to the Admin Analytics Dashboard.
5. The login is successful and logged by the system.

Worst-Case Scenario:

1. The student goes to the login page and enters their email (student@university.edu) but an old, incorrect password.
2. They click "Sign In." The system displays: "Invalid email or password." The counter updates: "2 login attempt(s) remaining."
3. They try again with another wrong password. The counter updates: "1 login attempt(s) remaining."
4. On the third failed attempt, the system displays a new error message: "Account locked due to multiple failed attempts. Please reset your password via email or contact support."
5. The student is now completely locked out and must use the "Forgot your password?" link to regain access.

Alternative Scenario:



1. The employer enters their correct credentials (employer@company.com / employer123) and clicks **"Sign In."**
2. Instead of granting access, the system displays a specific message: "This account is currently locked. Please contact system administrator for assistance."
3. The employer cannot proceed with the password reset because the account was locked by an admin, not by failed attempts.
4. The employer must follow the instruction and email the support team to unlock their account.

Login Authentication UI

University Job Portal

Sign in to access your account

⚠ Invalid email or password.

Email

2023420041.sethu@uniconnect.ac.za

Password

.....

👁

Sign In

Forgot your password?

2 login attempt(s) remaining

Demo Accounts:
Admin: admin@university.edu / admin123
Student: student@university.edu / student123
Employer: employer@company.com / employer123

Figure 3:Login Authentication UI

Use Case: UC02 - Student Registration & Profile Setup

Actor: Student (primary), UFS Database (supporting).

Brief Description:

This functionality allows a new UFS student to register for the system by verifying their university credentials and subsequently create a comprehensive professional profile. The system automatically imports academic data from secure UFS databases with the student's consent, streamlining the profile setup process.

UML

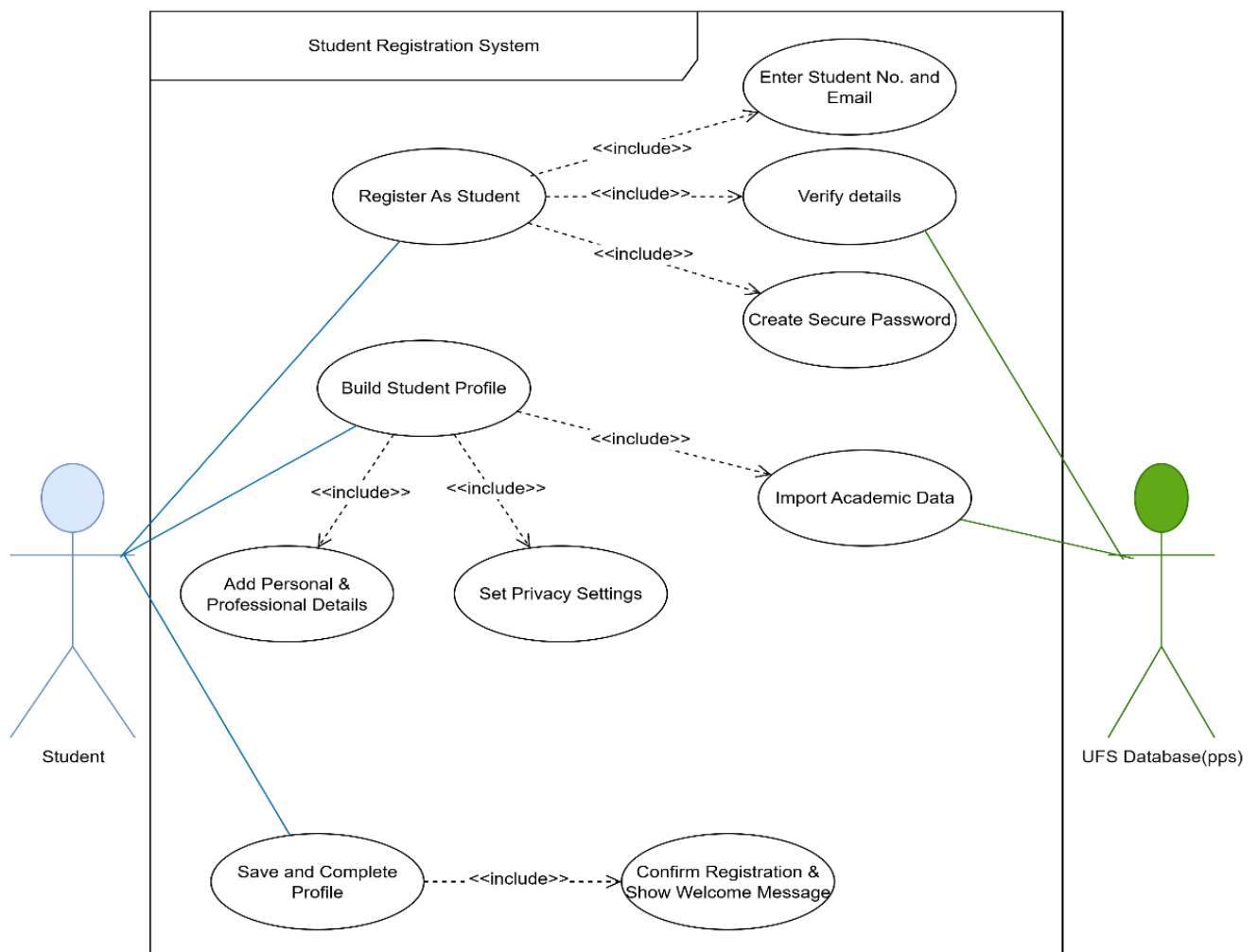


Figure 4: UC02 - Student Registration & Profile Setup

Step-by-Step:

1. Student navigates to the system's registration page.
2. System displays registration type selection screen with two options: "Register as Student" and "Register as Employer"
3. Student clicks "Register as a Student" button.
4. System redirects to Student Registration Form page
5. Student enters their 9-digit UFS student number in the "Student Number" field
6. Student enters their official university email address (format: 2023420041@ufs.ac.za) in the "UFS Email" field
7. Student clicks "Verify Details" button.
8. System verifies the credentials against a secure UFS database API.
9. Upon verification, the system prompts the student to create a secure password.
10. Student enters new password in "Password" field
11. System displays real-time password strength indicator (Weak/Medium/Strong) and shows password requirements checklist:
 - Minimum 12 characters
 - At least 1 uppercase letter
 - At least 1 lowercase letter
 - At least 1 number
 - At least 1 special character (! @#\$%^&*)
12. Student re-enters password in "Confirm Password" field
13. System validates passwords match in real-time
14. System prompts the student to build their profile.

Part 2: Profile Setup & Academic Integration

- 2.1 Student clicks the "Proceed to Profile Setup" button.
- 2.2 System redirects to the "Complete Your Profile" page. The system automatically pre-fills the "Degree Program" and "Major" fields using data from the UFS database. A toggle switch labelled "Auto-import Academic Record" is set to ON by default.
- 2.3 Student reviews the auto-imported academic data. The student can toggle the "Auto-import Academic Record" switch to OFF if they wish to manually enter their academic information.
- 2.4 Student manually fills in the "Professional Bio" text area.
- 2.5 Student adds skills by typing into the "Add Skills" text box and pressing Enter, creating visual skill tags.
- 2.6 Student clicks the "Save & Complete Profile" button at the bottom of the form.
- 2.7 System validates all mandatory fields.
 - 2.7.1 If validation is successful, the system saves the profile, displays a confirmation pop-up "Registration Complete! Welcome to the UFS Recruitment System." and redirects the student to their personal dashboard.
 - 2.7.2 If validation fails (e.g., bio is empty), the system displays an error message at the top of the form "Please complete all required fields." and highlights the empty/mandatory fields in red. The user must correct the errors before proceeding.

Best-Case Scenario:

1. Thabo, a new UFS student, navigates to the system's registration page.
2. He selects "Register as a Student" and enters his student number and official UFS email.
3. The system instantly verifies his credentials against the UFS database.
4. Thabo creates a strong password and grants permission to import his academic data.
5. The system auto-populates his degree and courses. Thabo adds his bio and skills, sets his privacy settings, and clicks "Save."
6. The system confirms successful registration, and Thabo can immediately access his dashboard.

Worst-Case Scenario:

1. Thabo enters his student number but makes a typo in his UFS email address (e.g., @fs.ac.za).

2. The system fails to verify his credentials and displays an error message.
3. After several failed attempts, Thabo realizes his mistake and corrects the email.
4. The system verifies him, but during academic data import, there's a temporary connection issue with the UFS database.
5. The system displays a "Data Unavailable" message. Thabo must complete his profile manually and contact support to resolve the data sync issue later.

Alternative Scenario:

1. Thabo successfully registers and imports his academic data.
2. While reviewing his auto-populated profile, he notices that a recently completed course is missing.
3. He uses the "Edit Profile" feature to manually add the missing course information before finalizing his profile.
4. He submits a support ticket via the system to report the data discrepancy for future syncs.

Registration UI

[← Back to Login](#)

**UniConnect**
Create your account

 Student

 Employer

Student Registration

Register with your university student details

First Name

Last Name

Student Number

University Email Address

Password

Confirm Password

Register as Student

Student Registration
Step 1 of 7

Registration Type
UFS Verification
Password Setup
Academic Data
Profile Building
Privacy Settings
Complete



Welcome to UniConnect

Join the University of the Free State's career development platform

Register as a Student

Connect with potential employers, showcase your academic achievements, and explore career opportunities.

Previous
Next

Figure 5:Registration UI

Student Registration
Step 4 of 7

Registration Type
UFS Verification
Password Setup
Academic Data
Profile Building
Privacy Settings
Complete

Academic Data Integration

Allow us to pull your academic information to enhance your profile

 We'll securely access your academic records to auto-populate your profile. You can review and edit all information before it becomes visible.

☒ I grant permission to pull my academic data (courses, marks)

Previous
Next

Figure 6: Academic data integration UI

Use Case UC03: Employer Registration & Profile Setup

Actor: Employer, Admin (secondary actor)

Description:

This functionality allows a representative of a company to register for the system as a verified employer. The process involves submitting company details and documentation for administrative review and verification, ensuring that only legitimate organizations can access the university's talent pool.

UML

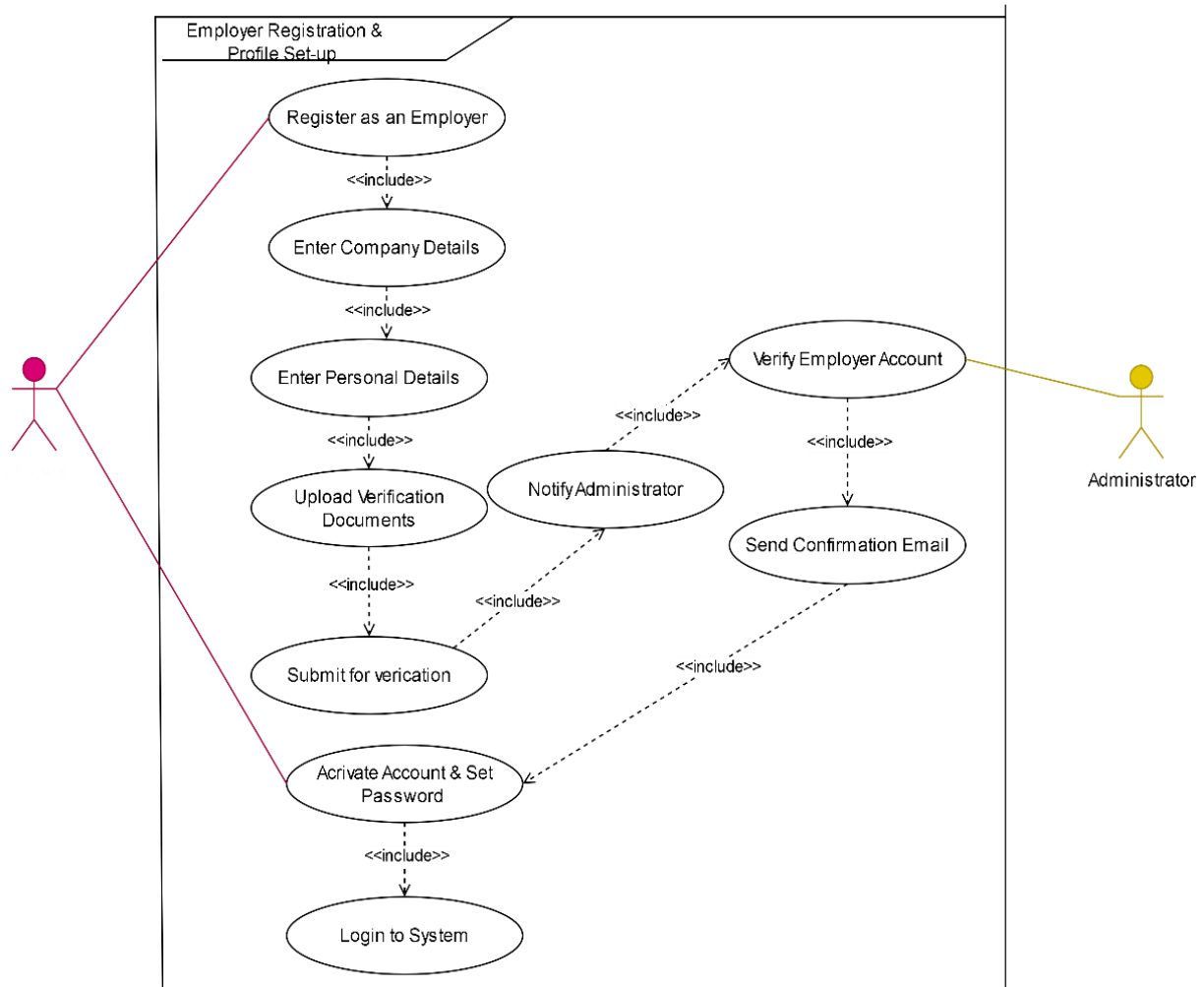


Figure 7: Use Case UC03: Employer Registration & Profile Setup

Step-by-Step:

1. Employer navigates to the registration page.
2. Employer selects "Register as an Employer".
3. Employer fills in company details: "Legal Company Name", "Address", "Website", and "Company Description".
4. Employer fills in primary contact details: "Your Full Name", "Job Title", "Work Email", "Phone Number".
5. Employer clicks the "Upload Verification Documents" button and selects a company registration document (PDF) from their computer.
6. Employer clicks the "Submit for Verification" button at the bottom of the form.
7. System displays a message: "Thank you for your registration. Your application is under review. You will receive an email once your account has been verified by our administration team." The form is disabled.

Best-Case Scenario:

1. Sarah, from "InnovateTech," navigates to the registration page and selects "Register as an Employer."
2. She accurately fills in all company details and uploads the required registration documents.
3. She clicks "Submit for Verification."
4. The next day, an Administrator reviews and approves the application. Sarah receives a confirmation email, sets her password, and gains immediate access to the platform.

Worst-Case Scenario:

1. Sarah submits the registration form but forgets to upload the required company documentation.
2. The system does not flag this missing requirement clearly, and her submission is stuck in a "Pending" state.
3. After a week with no response, she contacts support and is informed her application is incomplete.
4. She must resubmit the entire form with the correct documents, causing a significant delay.

Alternative Scenario:

1. Sarah starts the registration process but is interrupted before she can upload the documents.
2. The system has a "Save Draft" feature, which she uses.
3. She returns later, retrieves the draft, completes the information, and submits it successfully.

Employer Registration & Profile Setup UI

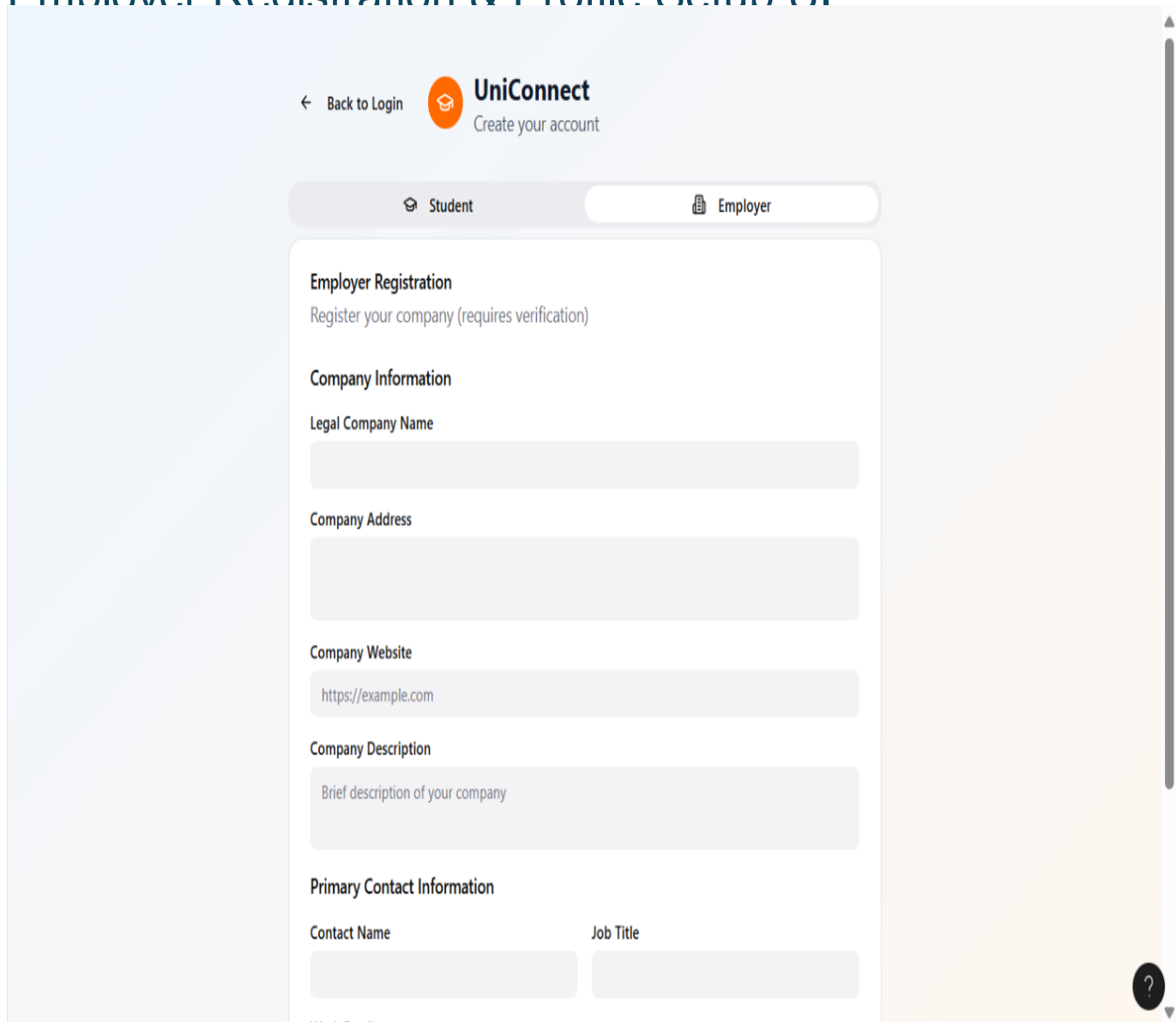
The image shows a web interface for employer registration. At the top, there's a navigation bar with a "Back to Login" link and the "UniConnect" logo with the tagline "Create your account". Below this is a tabbed interface with "Student" and "Employer" tabs; the "Employer" tab is active. The main content area is titled "Employer Registration" with a subtitle "Register your company (requires verification)". It contains several form sections: "Company Information" with fields for "Legal Company Name", "Company Address", and "Company Website" (pre-filled with "https://example.com"); "Company Description" with a text area (pre-filled with "Brief description of your company"); and "Primary Contact Information" with fields for "Contact Name", "Job Title", and "Work Email". A vertical scrollbar is on the right, and a help icon (?) is in the bottom right corner.

Figure 8: Employer Registration & Profile Setup UI

[← Back to Selector](#)

Company Information

Enter your official company details as they appear on legal documents

Legal Company Name *
e.g., ABC Technologies (Pty) Ltd

Trading Name (if different)
e.g., ABC Tech

Company Registration Number *
e.g., 2023/123456/07

Year Founded
e.g., 2020

Industry *
Select Industry

Company Size *
Select Company Size

Business Address

Street Address *
e.g., 123 Business Street, Business Park

City *
e.g., Cape Town

Province *
e.g., Western Cape

Postal Code *
e.g., 8001

Company Website
https://www.yourcompany.com

Company Description *
Describe your company, its mission, and what makes it unique...

This description will be visible to students browsing job opportunities.

[← Back](#)
[Next: Contact Information →](#)

Figure 9: Employer Information UI

Employer Profile Set Up:

[← Back to Selector](#)

Contact Information

Name: Sesethu Mngxozana

Job Title: Project Manager

Work Email: AfriNovaSystems10@gmail.com

Phone: +27632107908

Alternate Email: sesethumngxozana@gmail.com

LinkedIn: https://www.linkedin.com/in/AfriNovaSystems

Uploaded Documents

Company Registration Certificate

(CompTIA Network+ Guide to Networks 9th edition -- Jill West -- 9, 2021 -- Cengage Learning -- 9780357508138 -- f7a52aef9143dd905ea1ca5e2de45ca -- Anna's Archive.pdf)

31.19 MB

Tax Compliance Certificate

(CompTIA Network+ Guide to Networks 9th edition -- Jill West -- 9, 2021 -- Cengage Learning -- 9780357508138 -- f7a52aef9143dd905ea1ca5e2de45ca -- Anna's Archive.pdf)

31.19 MB

Proof of Business Address

(Final Team Project submission_Best Project 2024_compressed-1-127-1-123.pdf)

1.21 MB

By submitting this application, you confirm that all information provided is accurate and complete. Our team will review your application and contact you within 2-3 business days.

[← Back](#)
[Submit for Verification](#)

Figure 10: Employer Profile Set Up

Use Case UC04: Manage User Accounts & Roles

Actor: Administrator

Description:

This functionality grants administrators' full control over system user management. It includes the ability to verify new employer accounts, reset user passwords, lock/unlock accounts due to security concerns, and manage administrative privileges, ensuring system integrity and security.

UML

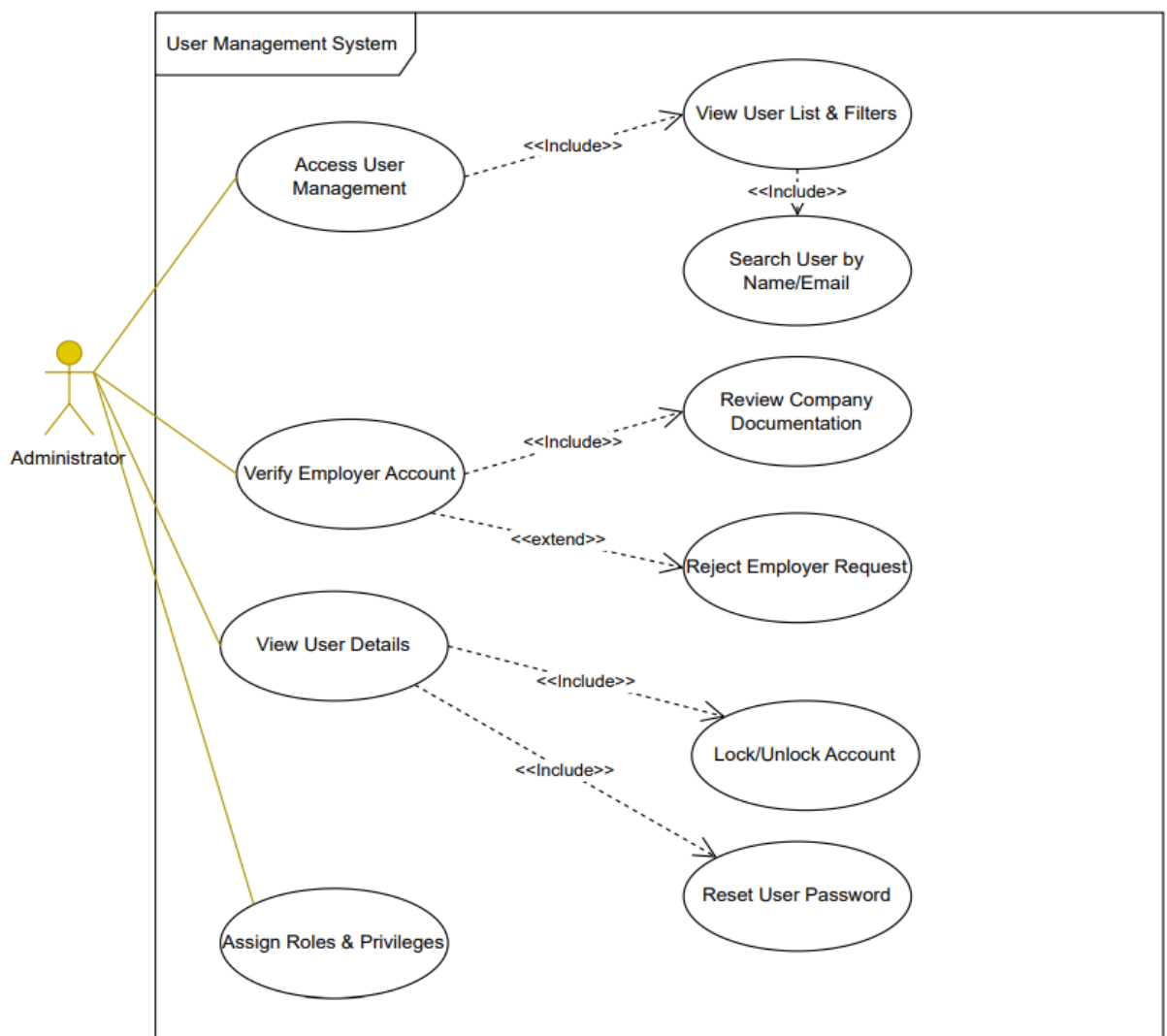


Figure 11: Use Case UC04: Manage User Accounts & Roles

Step-by-Step:

1. Administrator logs in and is directed to the Admin Dashboard.
2. Administrator clicks the "User Management" link in the main admin navigation sidebar.
3. System loads the User Management page, displaying a table of all users. There are filter dropdowns for "Role" (Student, Employer, Admin) and "Status" (Active, Pending, Locked).
4. Administrator notices a new employer, "TechCorp Inc.," has a status of "Pending". The admin clicks on the "TechCorp Inc." row.
5. System opens a detailed view sidebar showing the company's registration details and uploaded documents.
6. Administrator reviews the documents, finds them valid, and clicks the "Verify Account" button at the bottom of the sidebar.
7. System displays a confirmation pop-up: "Verify this employer account? This will grant them full access to the platform." The admin clicks "Confirm".
8. System updates the employer's status to "Active" in the table, sends a confirmation email to the employer, and displays a toast notification: "TechCorp Inc. has been successfully verified."

Best-Case Scenario:

1. The admin navigates to the User Management section from the main menu.
2. The system instantly loads the user list. The admin notices "TechCorp Inc." has a Status of "Pending."
3. The admin clicks on the "TechCorp Inc." row to view details and review the uploaded company documentation.
4. The documents are complete and valid. The admin clicks the "Verify Account" button.
5. The system updates the employer's status to "Active" immediately and sends a confirmation email to the employer. The task is completed successfully.

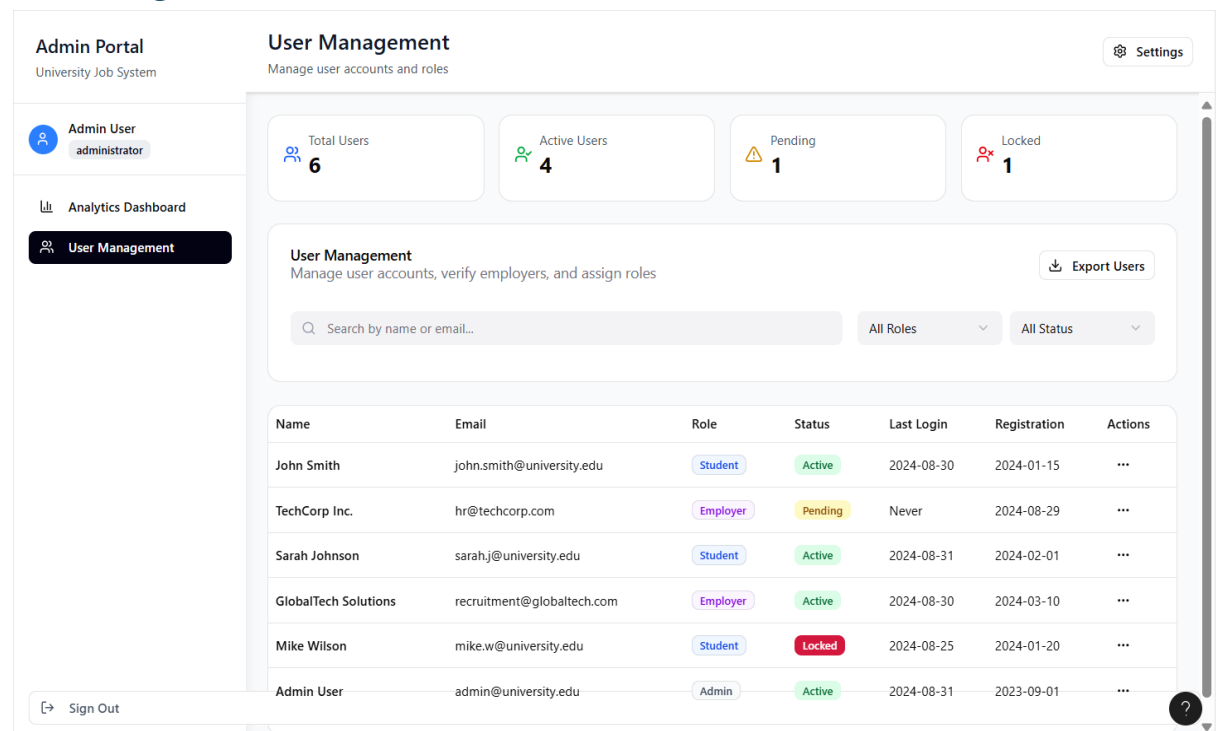
Worst-Case Scenario:

1. The admin navigates to User Management, but the page loads very slowly and the user list is empty.
2. An error message appears: "Cannot connect to database."
3. The admin tries the Search bar to find a specific user, but typing causes the page to freeze.
4. After a refresh, the page loads but some user data is corrupted (e.g., missing names, incorrect roles).
5. Clicking any button (like "Reset Password") does nothing. The admin cannot perform any management tasks and must report a critical system failure.

Alternative Scenario:

1. The admin goes to User Management and uses the Search bar to quickly find "Mike Wilson" by name.
2. The system filters the list instantly. The admin sees Mike's account is "Active" but notices an unusual login time in the "Last Login" column.
3. The admin clicks on Mike's row to view detailed login history, which confirms suspicious activity from an unknown location.
4. Instead of resetting the password, the admin selects the "Lock Account" action from the Actions menu to secure it immediately.
5. The system locks the account, changing its status to "Locked," and logs the action. The admin then emails Mike to initiate a secure account recovery. [User](#)

Management UI



The screenshot displays the 'User Management' interface within an 'Admin Portal' for a 'University Job System'. The interface includes a sidebar with navigation options: 'Admin User administrator', 'Analytics Dashboard', and 'User Management' (selected). The main content area shows a summary of user statistics: Total Users (6), Active Users (4), Pending (1), and Locked (1). Below this is a search bar and filters for 'All Roles' and 'All Status'. A table lists users with columns for Name, Email, Role, Status, Last Login, Registration, and Actions. The table includes entries for John Smith, TechCorp Inc., Sarah Johnson, GlobalTech Solutions, Mike Wilson (locked), and Admin User. An 'Export Users' button is located in the top right of the table area. A 'Sign Out' button is in the bottom left corner.

Name	Email	Role	Status	Last Login	Registration	Actions
John Smith	john.smith@university.edu	Student	Active	2024-08-30	2024-01-15	...
TechCorp Inc.	hr@techcorp.com	Employer	Pending	Never	2024-08-29	...
Sarah Johnson	sarah.j@university.edu	Student	Active	2024-08-31	2024-02-01	...
GlobalTech Solutions	recruitment@globaltech.com	Employer	Active	2024-08-30	2024-03-10	...
Mike Wilson	mike.w@university.edu	Student	Locked	2024-08-25	2024-01-20	...
Admin User	admin@university.edu	Admin	Active	2024-08-31	2023-09-01	...

Figure 12: Management UI

Use Case UC05: Manage Portfolio (SkillStack)

Actor: Student

Description:

This functionality enables a student to create and maintain a digital portfolio of their projects and work. Students can upload project details, descriptions, and files to showcase their practical skills and achievements to potential employers, providing evidence of their capabilities beyond academic records.

UML

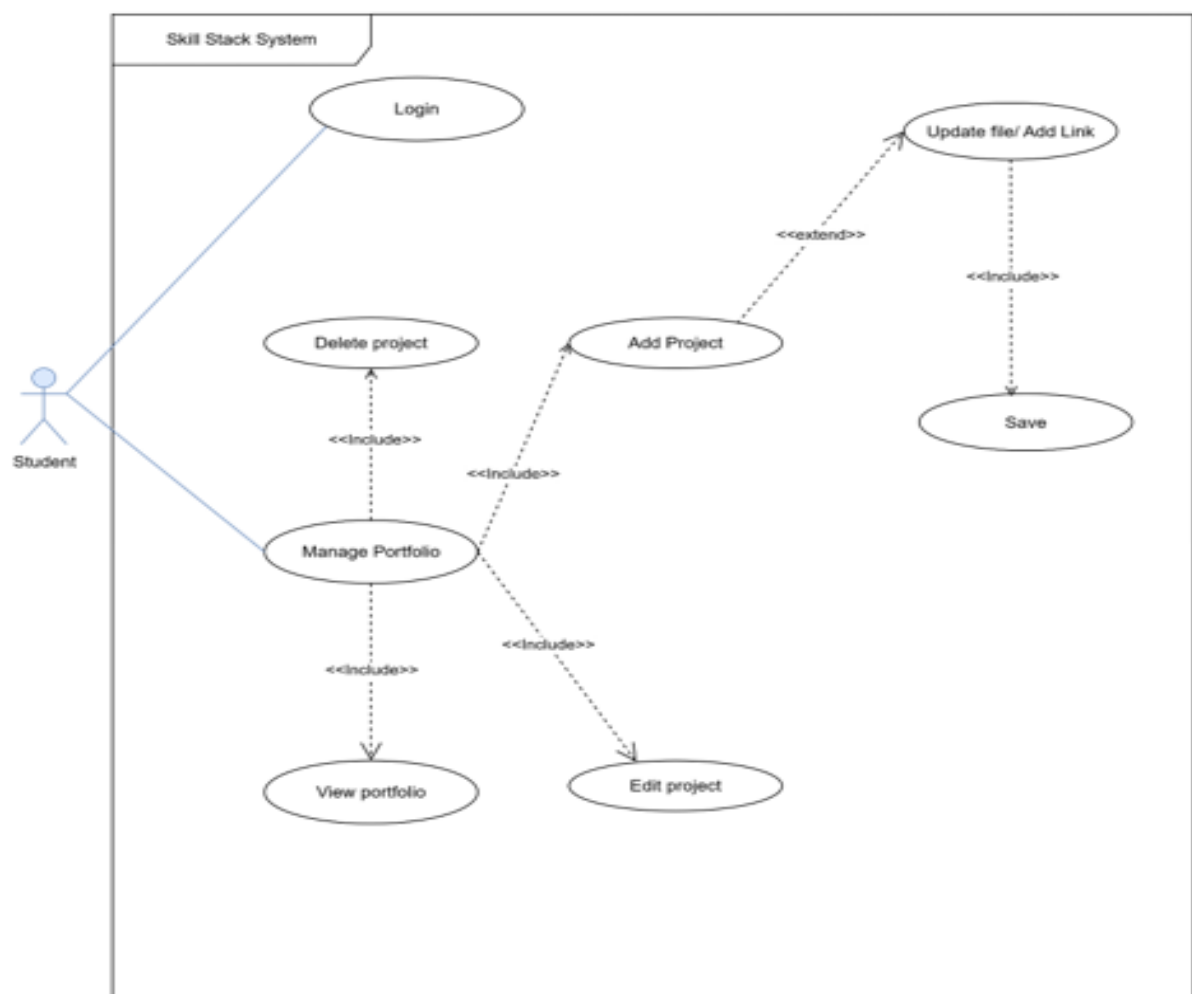


Figure 13: Use Case UC05: Manage Portfolio (SkillStack)

Step-by-Step:

- 1 Student logs in and navigates to their "Profile" page by clicking their name in the top-right corner and selecting "My Profile" from the dropdown.
- 2 Student scrolls down to the "SkillStack Portfolio" section and clicks the "+ Add Project" button.
- 3 System opens a "Add New Project" modal form with fields: "Project Name", "Description", "Date", "Skills Demonstrated", and a "File Upload" area.
- 4 Student fills in the "Project Name" and "Description" text boxes.

- 5 Student clicks the "Browse Files" button and selects a PDF report from their computer.
- 6 System uploads the file and displays the filename below the upload button with a red "X" to remove it.
- 7 Student clicks the "Save Project" button at the bottom of the modal.
- 8 System validates the input.
 - 1.8.1 If successful the system closes the modal, saves the project, and displays the new project as a card in the SkillStack section. A toast notification appears: "Project 'Mobile App UI' added successfully!"
 - 1.8.2 If validation fails (e.g., no project name) the system highlights the empty "Project Name" field in red and displays an inline error message: "Please provide a name for your project."

Best Scenario:

This flow uses the portfolio form fields and upload functionality successfully.

1. Student logs in and navigates to the "Profile" page via the main navigation menu.
2. Student clicks the "Add Project" button in the "SkillStack Portfolio" section.
3. System displays a full-screen modal or page containing the Project Submission Form.
4. Student fills out all text fields (Project Name, Description, Skills) and uses the "Browse Files" button to successfully upload a compliant file (e.g., a small JPEG image).
5. Student clicks the "Save Project" button at the bottom of the form.
6. System displays a toast notification "Project Successfully Added!" and the new project card appears instantly in the student's portfolio list on the profile page.

Worst Scenario:

This flow results in a failure due to file validation and causes the loss of all data entered the form fields.

1. Student Navigates to the Project Submission Form.
2. Student Fills out all required text fields in the form.
3. Student Clicks "Browse Files" and selects a file that is too large (e.g., 250MB video) or an invalid type (e.g., .exe).
4. System (Failure) After clicking "Save Project," the system throws a persistent error message at the top of the form "Error File Size Limit Exceeded (50MB Max)".

5. Student Observes that the page has reloaded, or the form has reset, and all previously entered text data (Project Name, Description, Skills) has vanished, forcing a complete restart.

Alternative Scenario:

This flow involves modifying an existing project entry using the edit controls.

- 1 Student Navigates to the "Profile" page and scrolls to the "SkillStack Portfolio" section.
- 2 Student Clicks the "Edit" icon (e.g., a pencil icon) on an existing project card.
- 3 System Displays the Project Submission Form, pre-filled with the existing project's data.
- 4 Student Modifies the "Description" text box and clears the existing "File/Link" field.
- 5 Student Enters a new URL for a public repository (e.g., GitHub) into the designated link field.
- 6 Student Clicks "Save Changes".
- 7 System Displays a confirmation message "Project Updated!" and the updated project card is displayed in the portfolio list.



Manage Portfolio UI



UniConnect
Student Portal

Dashboard

Find Jobs

Applications

Interviews ²

Profile

⁵

AJ

← Back to Portfolio

Add New Project

Showcase your skills and work to employers

Basic Information

Project Name *

e.g., E-commerce Platform

Description *

Describe your project...

Project Date *

2025/08/31 

Category

Web Development 

Status

Completed 

☐ Feature this project

Skills & Technologies

Skills Demonstrated

Type a skill... 

Technologies Used

e.g., React, Node.js... 

 Save Project

Cancel

Figure 14: Manage Portfolio UI

Use Case UC06: Manage Skill Tags & Badges

Actors: Student, System (secondary)

Description

This functionality allows students to add and manage their skill tags (e.g., “Python”, “Public Speaking”) and view badges earned automatically through academic milestones, certifications, or extracurricular achievements. Employers can see these tags and badges as part of the student’s profile.

UML

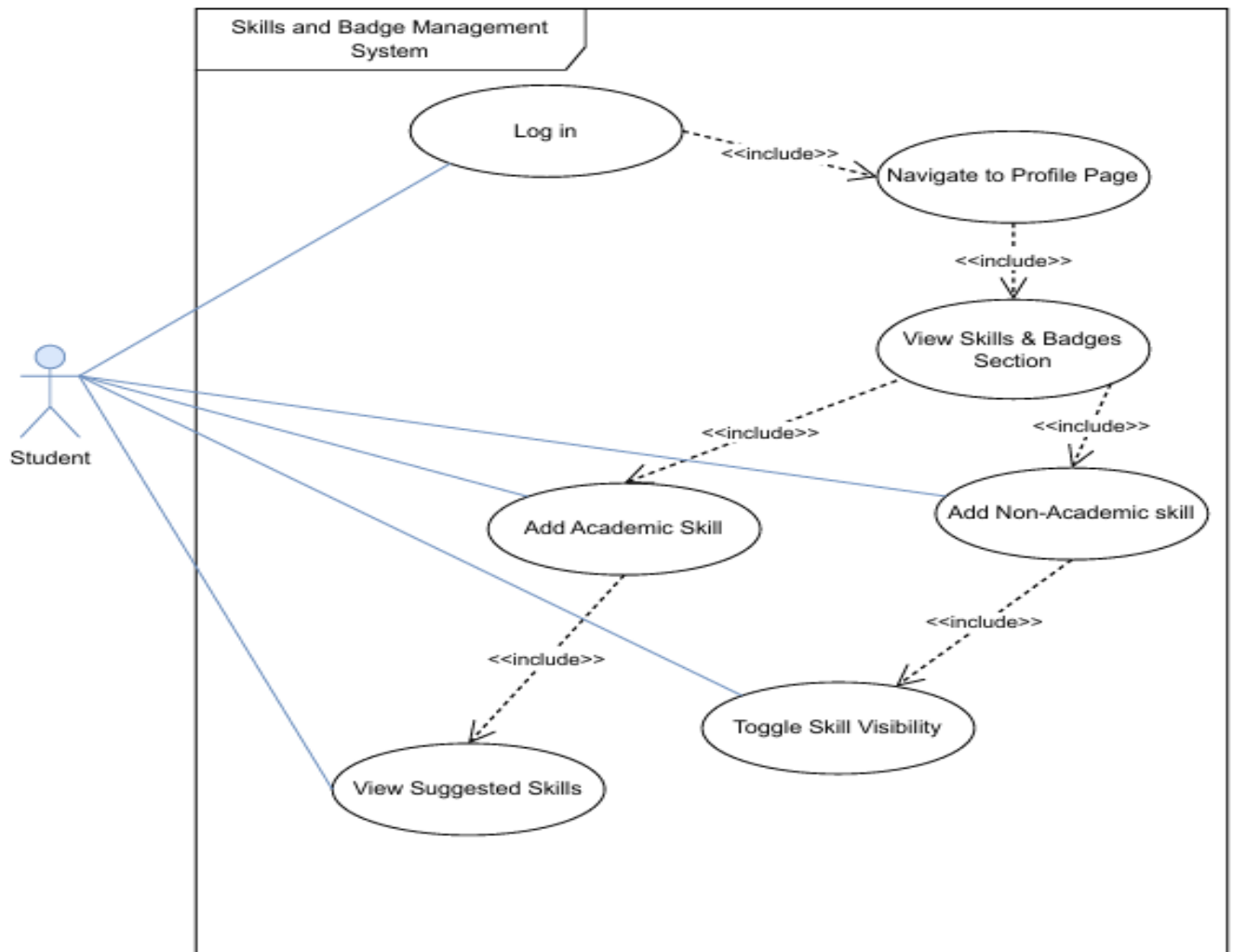


Figure 15: Use Case UC06: Manage Skill Tags & Badges

Step-by-Step:

1. Student logs in and navigates to their "Profile" page.
2. Student scrolls to the "Skills & Badges" section and clicks the "Add Skill" button.
3. System opens a modal with a text box for "Add a Skill" and a dropdown list of "Suggested Skills" pulled from their academic courses (e.g., "Java Programming," "Financial Accounting").
4. Student clicks on "Java Programming" from the suggested list.
5. System adds "Java Programming" as a visual tag in the student's skill list. Each tag has a visibility toggle switch (default Public).
6. System (Automated), upon detecting the student has completed their internship module, the system automatically awards and displays the "Internship Ready" badge in the badge subsection. This badge cannot be manually removed.
7. Student wants to add a non-academic skill. They type "Public Speaking" into the "Add a Skill" text box and press Enter.
8. System adds "Public Speaking" as a new tag. The student then toggles its visibility switch to Private so employers cannot see it.

Best-Case Scenario (Automatic and Manual Skill Management)

1. Thabo finishes his "Data Structures" course, and the system auto-tags "Java Programming."
2. He manually adds "Leadership" and "Public Speaking" to his skills.
3. His profile now shows a rich mix of system-verified and self-reported skills, making him more attractive to employers.

Worst-Case Scenario (System Error and Incorrect Tagging)

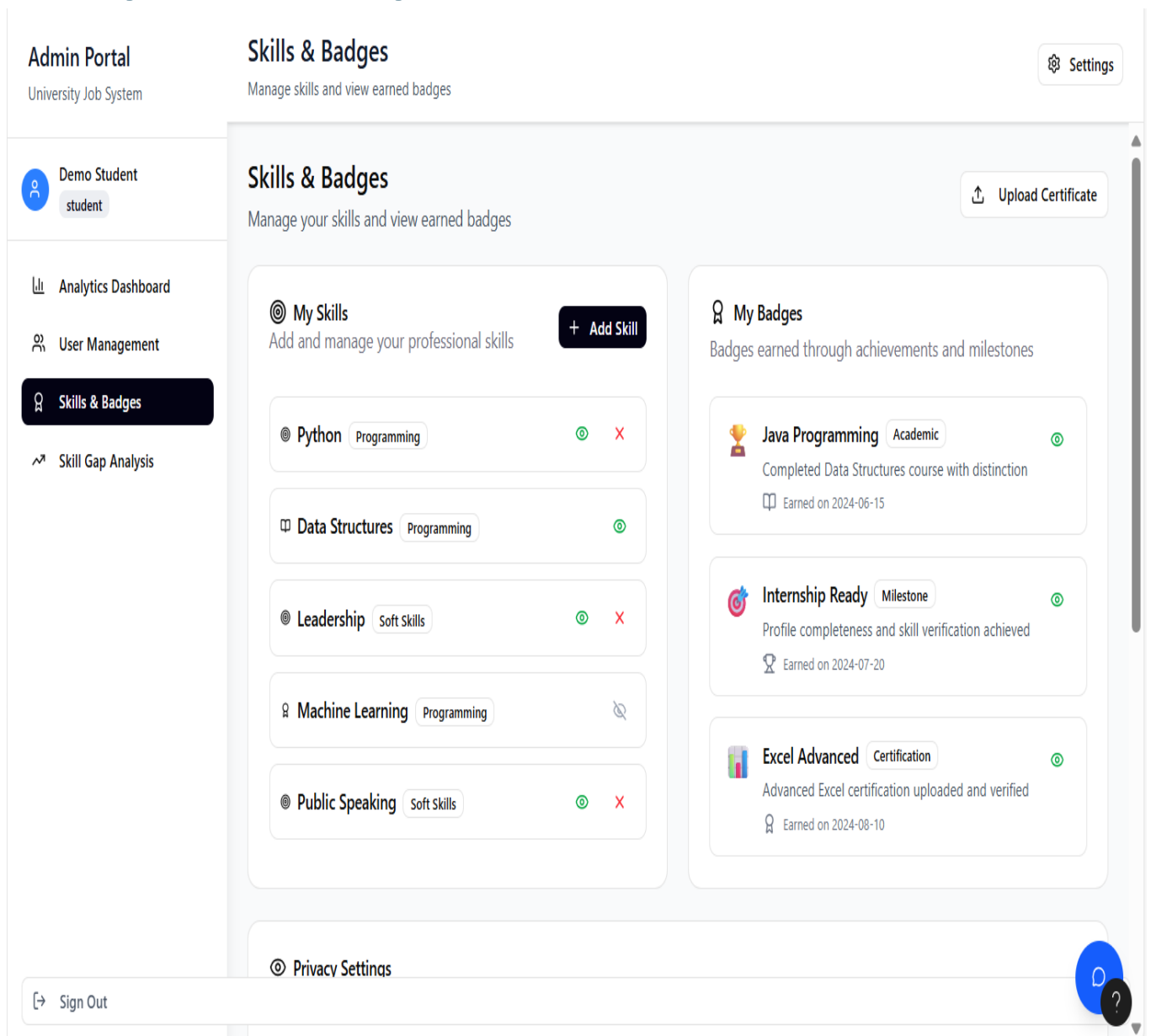
1. The system incorrectly tags Thabo with "Machine Learning" due to a faulty data integration.
2. Thabo manually deletes the tag, but the system's nightly sync re-adds it.
3. He must contact support to resolve the underlying data issue, as he cannot permanently remove the incorrect tag.

Alternative Scenario (Skill via Certification Upload)

1. Instead of typing a skill, Thabo clicks "Upload Certificate" in the Skills section.
2. He uploads a PDF of his "Microsoft Excel Advanced" certificate.

- The system scans the document, verifies its authenticity, and automatically adds an "Excel Advanced" skill tag to his profile, marked with a small verification checkmark.

Manage Skills & Tags UI



The screenshot displays the 'Manage Skills & Tags' interface within an 'Admin Portal' for a 'University Job System'. The user is logged in as 'Demo Student' (student). The interface is divided into a left sidebar with navigation options: 'Analytics Dashboard', 'User Management', 'Skills & Badges' (selected), and 'Skill Gap Analysis'. At the bottom of the sidebar is a 'Sign Out' button. The main content area is titled 'Skills & Badges' with the subtitle 'Manage skills and view earned badges'. A 'Settings' button is located in the top right corner. The main area is split into two columns. The left column, 'My Skills', allows users to 'Add and manage your professional skills' and includes an 'Add Skill' button. It lists five skills: 'Python' (Programming), 'Data Structures' (Programming), 'Leadership' (Soft Skills), 'Machine Learning' (Programming), and 'Public Speaking' (Soft Skills). Each skill entry has a status indicator (green circle with a checkmark or red X) and a delete icon. The right column, 'My Badges', shows 'Badges earned through achievements and milestones' and includes an 'Upload Certificate' button. It lists three badges: 'Java Programming' (Academic), 'Internship Ready' (Milestone), and 'Excel Advanced' (Certification). Each badge entry includes a description, a date earned, and a status indicator. A 'Privacy Settings' button is located at the bottom of the main content area. A user profile icon with a question mark is visible in the bottom right corner.

Figure 16: Manage Skills & Tags UI

Use Case UC07 Use CV Builder Tool

Actor: Student

Description

This functionality guides a student through the automated generation of a professional CV. The tool uses the student's profile and academic data to pre-populate a CV, which the student can then customize with various templates and edits before downloading it for applications or saving it to their profile.

UML

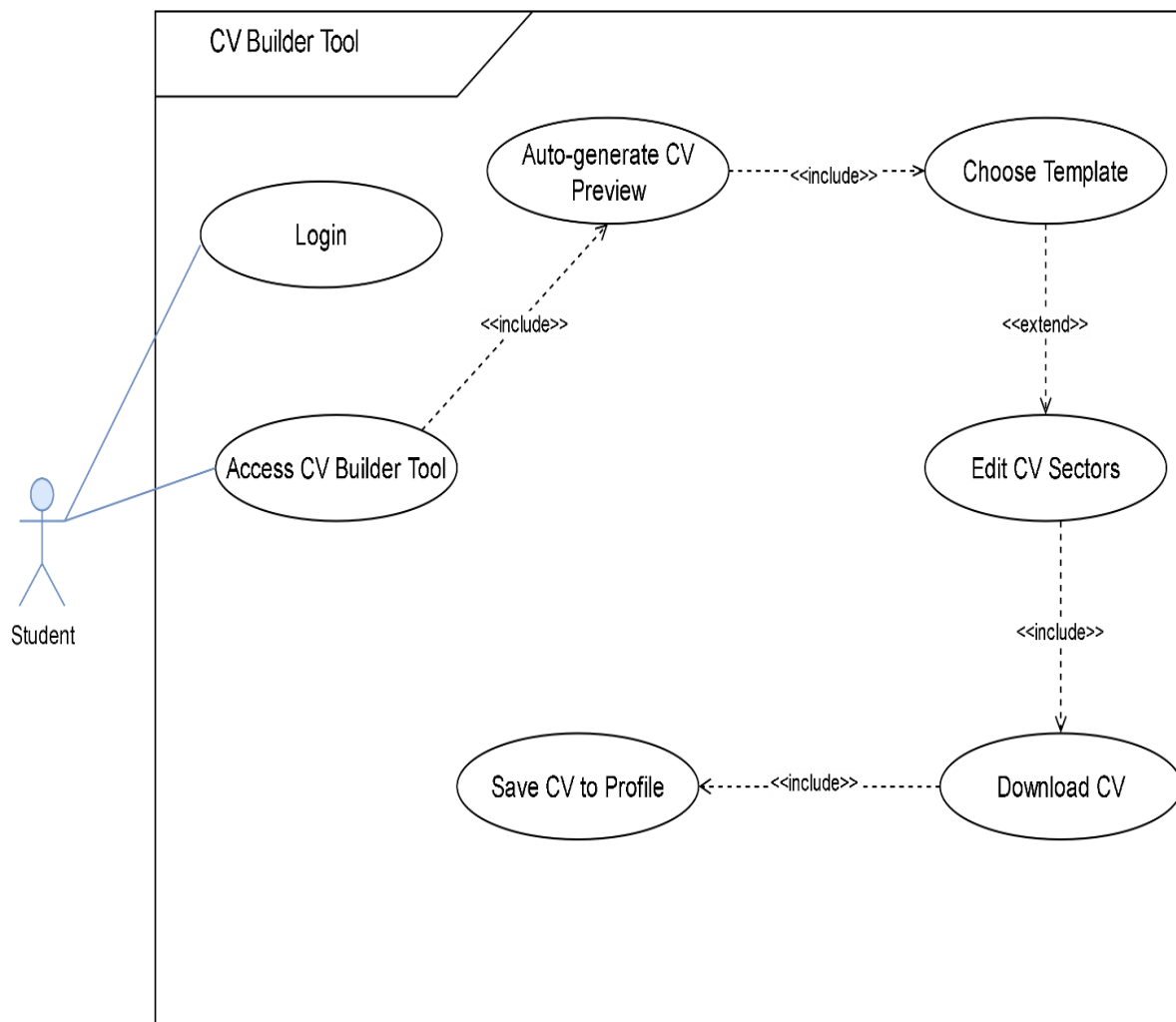


Figure 17: Use Case UC07 Use CV Builder Tool

Step-by-Step:

- 1 Student from their dashboard, clicks on the "CV Builder" tool in the main navigation menu.
- 2 System opens the CV Builder interface, displaying a live preview of their auto-generated CV on the right and an editing panel on the left.
- 3 System the editing panel has tabs for "Templates", "Personal Info", "Education", etc. The "Education" section is auto populated from their UFS academic data.
- 4 Student clicks the "Templates" tab and browses through different designs (Modern, Classic, Compact). They click on the "Modern" template.
- 5 System instantly applies the new template to the live preview.
- 6 Student clicks on the "Personal Statement" section in the live preview and begins typing to edit the text directly.
- 7 Student clicks the "Download as PDF" button located in the top toolbar of the preview pane.
- 8 System generates a PDF file and triggers a download to the student's device. A message appears: "CV downloaded successfully. The latest version has been saved to your profile for applications."

Best-Case Scenario:

1. Thabo opens the CV Builder tool. The system auto-generates a clean, professional CV based on his profile.
2. He browses the templates and selects a modern design.
3. He quickly tweaks the "Personal Statement" section to better reflect his career goals.
4. He clicks "Download as PDF," and the file is saved perfectly to his device. The CV is also auto saved to his profile for quick applications.

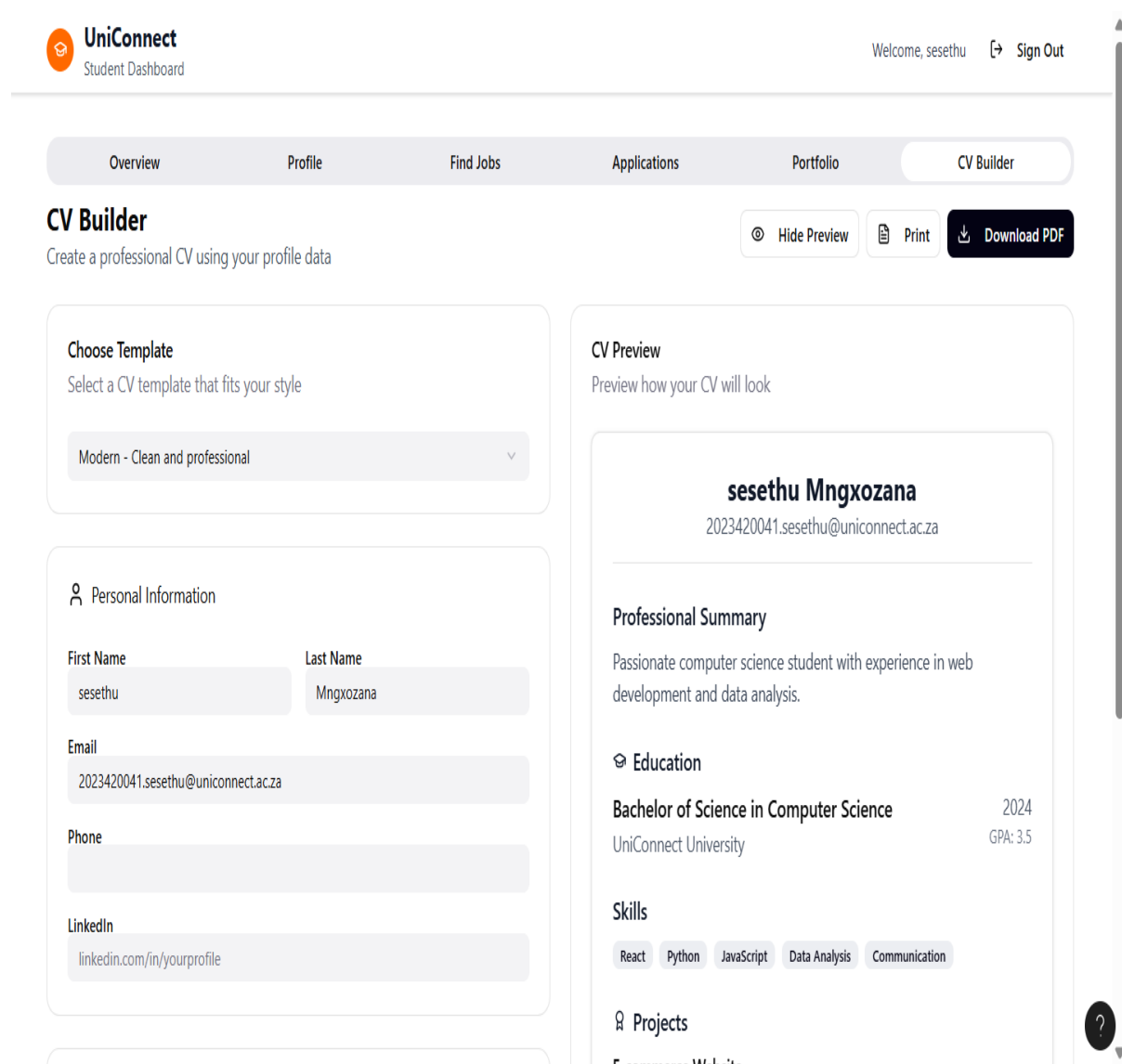
Worst-Case Scenario:

1. Thabo uses the CV Builder, but the system incorrectly formats his work experience dates.
2. When he tries to edit the text, the formatting tools are unresponsive, and the text becomes misaligned.
3. He attempts to download the CV, but the PDF generated has overlapping text boxes.
4. He must manually copy his information from the system into a separate Word document to create a presentable CV.

Alternative Scenario:

1. Thabo creates a general CV using the builder.
2. When applying for a specific data analyst role, he uses the "Save As" feature in the CV Builder to create a copy.
3. He tailors this copy, emphasizing his data analysis and Python skills more prominently, creating a targeted CV for that specific application.

Use CV Builder UI



The screenshot shows the UniConnect Student Dashboard with the CV Builder UI. The dashboard includes a header with the UniConnect logo, a user greeting, and a sign-out link. The main navigation bar has tabs for Overview, Profile, Find Jobs, Applications, Portfolio, and CV Builder. The CV Builder section is active, showing a 'Choose Template' dropdown set to 'Modern - Clean and professional'. Below this is a 'Personal Information' form with fields for First Name (sesethu), Last Name (Mngxozana), Email (2023420041.sesethu@uniconnect.ac.za), Phone, and LinkedIn (linkedin.com/in/yourprofile). To the right is a 'CV Preview' section showing a preview of the CV with the following details:

- CV Preview**: Preview how your CV will look
- sesethu Mngxozana**: 2023420041.sesethu@uniconnect.ac.za
- Professional Summary**: Passionate computer science student with experience in web development and data analysis.
- Education**: Bachelor of Science in Computer Science, UniConnect University, 2024, GPA: 3.5
- Skills**: React, Python, JavaScript, Data Analysis, Communication
- Projects**: E-commerce Website

Figure 18: Use CV Builder UI

Use Case UC08 Perform Skill Gap Analysis

Actor: Student

Description

This feature compares a student's current skills (as tagged in UC14 Manage Skill Tags & Badges) and academic records against the requirements of their desired job roles and suggests learning resources or workshops to bridge the gap.

UML



Figure 19: Use Case UC08 Perform Skill Gap Analysis

Step-by-Step:

1. Student from their dashboard, clicks on the "Skill Gap Analysis" tool.
2. System loads the Skill Gap Analysis page, featuring a search bar labelled "Search for a Job Role" and a button to "Upload Job Description".
3. Student types "Data Analyst" into the search bar and selects it from the dropdown of common roles.
4. System displays a two-column visual. The left column, "Your Skills," lists the student's current skills (e.g., Python, Statistics). The right column, "Required Skills," lists the typical requirements for a Data Analyst (e.g., Python, Statistics, SQL, Tableau).
5. System automatically matches "Python" and "Statistics," highlighting them in green. It identifies "SQL" and "Tableau" as "Missing Skills," highlighting them in red in the "Required Skills" column.
6. System below the visualization, a section titled "Recommended Learning" appears, suggesting resources like "UFS SQL Workshop (Starts Nov 10)" and a link to a "Coursera Tableau Course".
7. Student clicks the "+" icon next to the "UFS SQL Workshop" to add it to their "My Learning Plan".

Best-Case Scenario:

1. Thabo uses the tool and gets a clear, visual report showing he lacks SQL.
2. The system recommends a specific, upcoming university workshop.
3. Thabo adds it to his plan and registers, proactively closing a critical skill gap.

Worst-Case Scenario:

1. The system's data for the "Data Analyst" role is outdated and suggests "Adobe Flash" as a required skill.
2. Thabo is confused by the irrelevant recommendation and loses trust in the tool's usefulness.
3. He ignores all subsequent suggestions from the system.

Alternative Scenario:

1. Instead of searching for a role, Thabo copies and pastes the text of a real job description into the "**Upload Job Description**" text box.
2. The system parses the text, extracts the key skills, and runs a gap analysis against Thabo's profile specifically for that job.
3. This provides a highly personalized and relevant result.

Skills Gap Analysis UI

Skill Gap Analysis

Analyze skill gaps and get learning recommendations

Settings

Skill Gap Analysis

Compare your skills against job requirements and get personalized learning recommendations

Select Job RoleCustom Job DescriptionAnalysis Results

🎯 Choose a Job Role

Select from popular job roles to analyze your skill match

Select a job role to analyze

Analyze Skill Match

⭐ Your Current Skills

Skills from your profile, courses, and certifications

Python

Programming

Advanced

Statistics

Analytics

Intermediate

Communication

Soft Skills

Advanced

Teamwork

Soft Skills

Advanced

JavaScript

Programming

Beginner

Project Management

Management

Beginner

Figure 20: Skills Gap Analysis UI

49 | Page

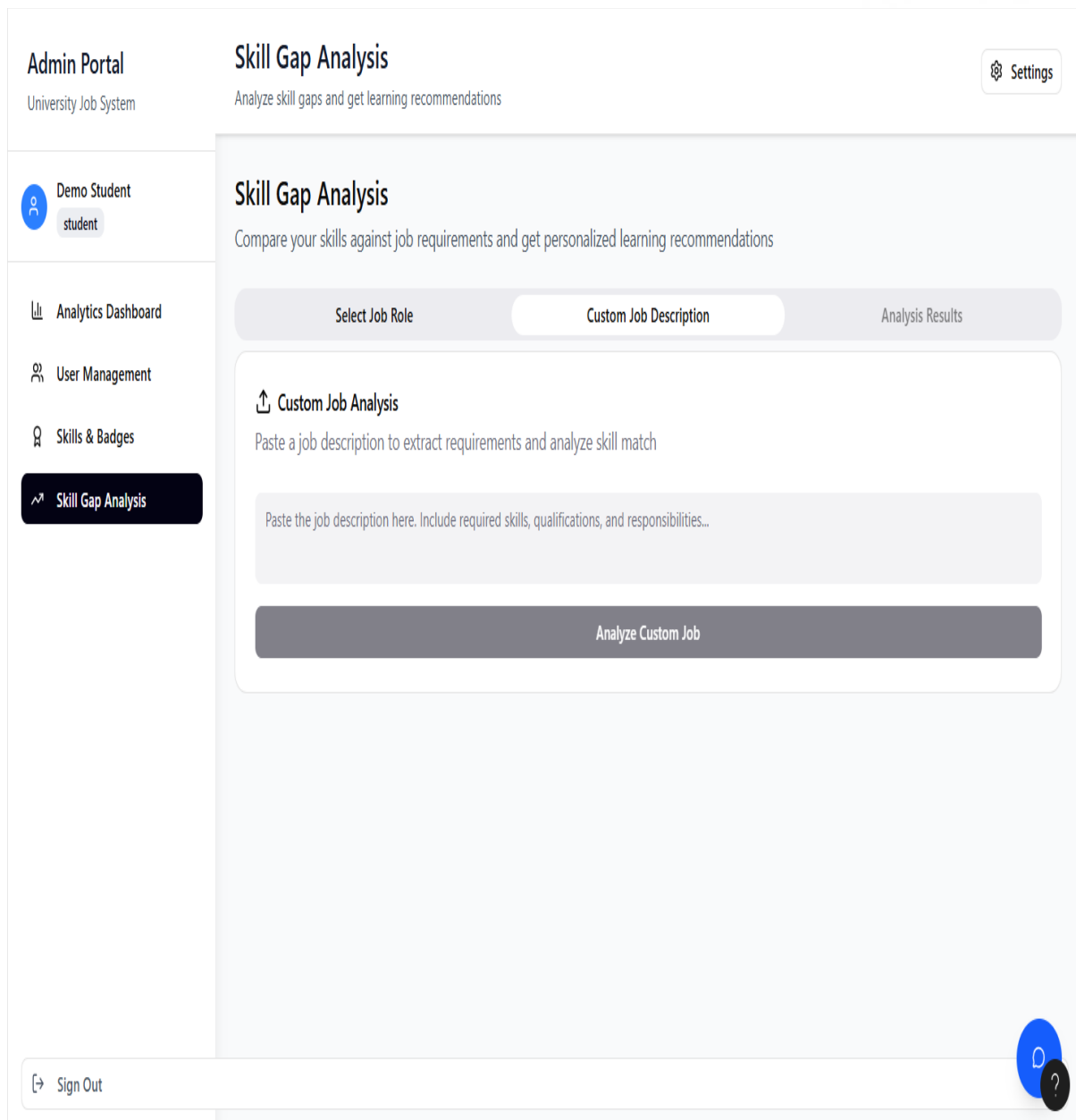


Figure 21: Skills Gap Analysis UI_2

Use Case UC09 Post a Job Opportunity

Actor Employer

Description

This functionality enables a verified employer to create and publish new job or internship opportunities on the platform. Employers can define the job title, description, requirements, type, and location, making the vacancy visible and searchable for eligible students.

UML

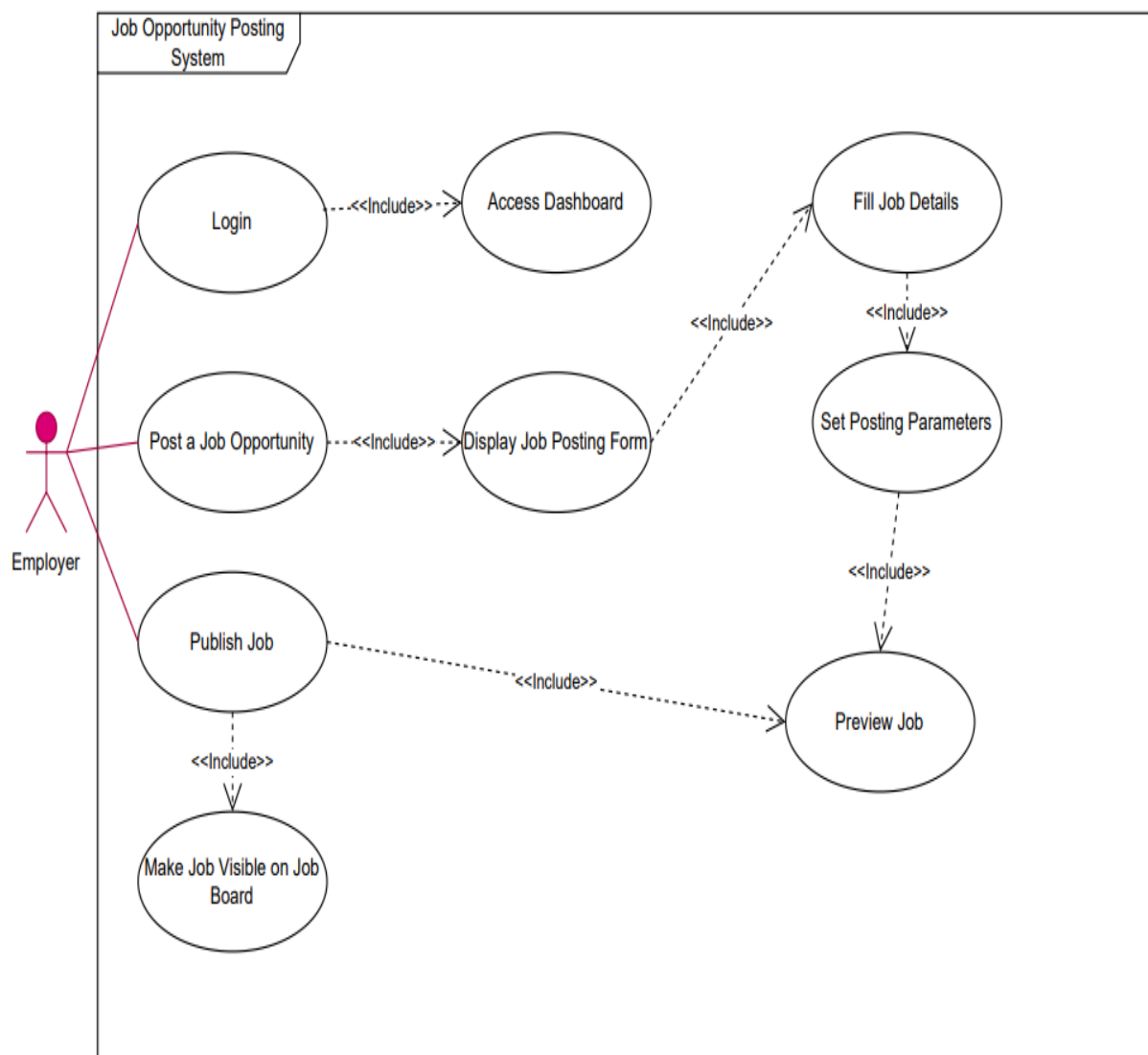


Figure 22: Use Case UC09 Post a Job Opportunity

Step-by-Step

1. Employer logs into the system and is directed to the Employer Dashboard.
2. Employer clicks the "Post a Job" button located in the dashboard's quick-action panel.
3. System redirects the employer to the "Create Job Posting" form.
4. Employer fills in the form fields: "Job Title", "Job Description" (using a rich text editor), "Key Responsibilities", and "Requirements".
5. Employer selects Full-time from the "Job Type" dropdown and enters Bloemfontein in the "Location" text box.
6. Employer sets a "Closing Date" using the date picker widget.
7. Employer clicks the "Preview" button at the bottom of the form.
8. System displays a preview of the job post exactly as candidates will see it, with two buttons: "Edit Post" and "Publish Job".
9. Employer reviews the preview and clicks "Publish Job".
10. System performs a final validation check.
 - 10.1 If successful the system publishes the job, displays a success message "Job Published! Your listing is now live on the job board.", and redirects the employer to their dashboard where the new job appears in their "Active Listings" section.
 - 10.2 If validation fails (e.g., a required field was cleared during preview) the system redirects the employer back to the form, displays an error message "Please complete all required fields before publishing." and highlights the missing fields in red.

Best-Case Scenario:

1. John (Employer) logs into the UFS Recruitment System successfully.
2. John navigates to the "Post a New Job" button in the employer dashboard and clicks it.
3. The system redirects John to the Job Posting Form, displaying fields for Job Title, Description, Responsibilities, Required Skills, Degree, Minimum GPA, Job Type, Location, and Closing Date.
4. John completes all fields correctly, ensuring the information is accurate and professional.
5. John clicks Preview Job Post, and the system displays a full preview of the job post exactly as candidates will see it.
6. John confirms the details are correct and clicks Publish Job.
7. The system successfully posts the job to the job board and sends notifications to eligible students.
8. The job is now visible to students in the job listings, and John can track applications in the employer dashboard.

Worst-Case Scenario:

1. Mark (Employer) logs into the system and clicks Post a New Job.
2. He fills in the form but leaves the Closing Date and Job Location fields empty.
3. He clicks Publish Job, but the system identifies missing required fields and displays error messages “Closing Date is required” and “Location is required.”
4. Mark ignores the warning and tries again without correcting the errors.
5. The system prevents the job from being published and highlights the problematic fields in red.
6. Mark becomes frustrated and abandons the posting process.

Alternative Scenario

1. Emily (Employer) logs in and navigates to Post a New Job.
2. She fills in the required fields but instead of directing candidates to apply via the platform, she provides an External Application Link to the company’s careers page.
3. The system accepts the URL and highlights it in blue and underlined in the preview.
4. Emily clicks Publish Job, and the system posts the job to the job board with an “Apply on External Site” button.
5. Students who click the button are redirected to Emily’s company website.

Post Job Opportunity UI

[← Back to Selector](#)

Create New Job Posting

Fill in the details below to create a compelling job posting

Preview

Basic Information

Job Title *

Job Type *

Work Location *

Office Location *

Job Description *

Key Responsibilities

Figure 23: Post Job Opportunity UI

Preview and Publish Job UI

[← Back to Selector](#)

\$ Compensation & Benefits

Minimum Salary	Maximum Salary	Currency	Period
15000	25000	ZAR	per month

Benefits & Perks

Add a benefit or perk...

📅 Important Dates & Contact

Application Closing Date *	Expected Start Date
yyyy/mm/dd	yyyy/mm/dd

Contact Email *

hr@company.com

☐
Mark as urgent hiring

Application Instructions

Provide specific instructions for applicants on how to apply, what documents to include, etc.

[← Back to Dashboard](#)
[Save as Draft](#)
[Preview & Publish](#)

Figure 24: Preview and Publish Job UI

Use Case UC10 Student Job Search & Application

Actor: Student

Description

This functionality enables a registered student to search and filter through available job postings, view detailed job descriptions, and submit applications directly through the platform using their pre-populated profile data.

UML

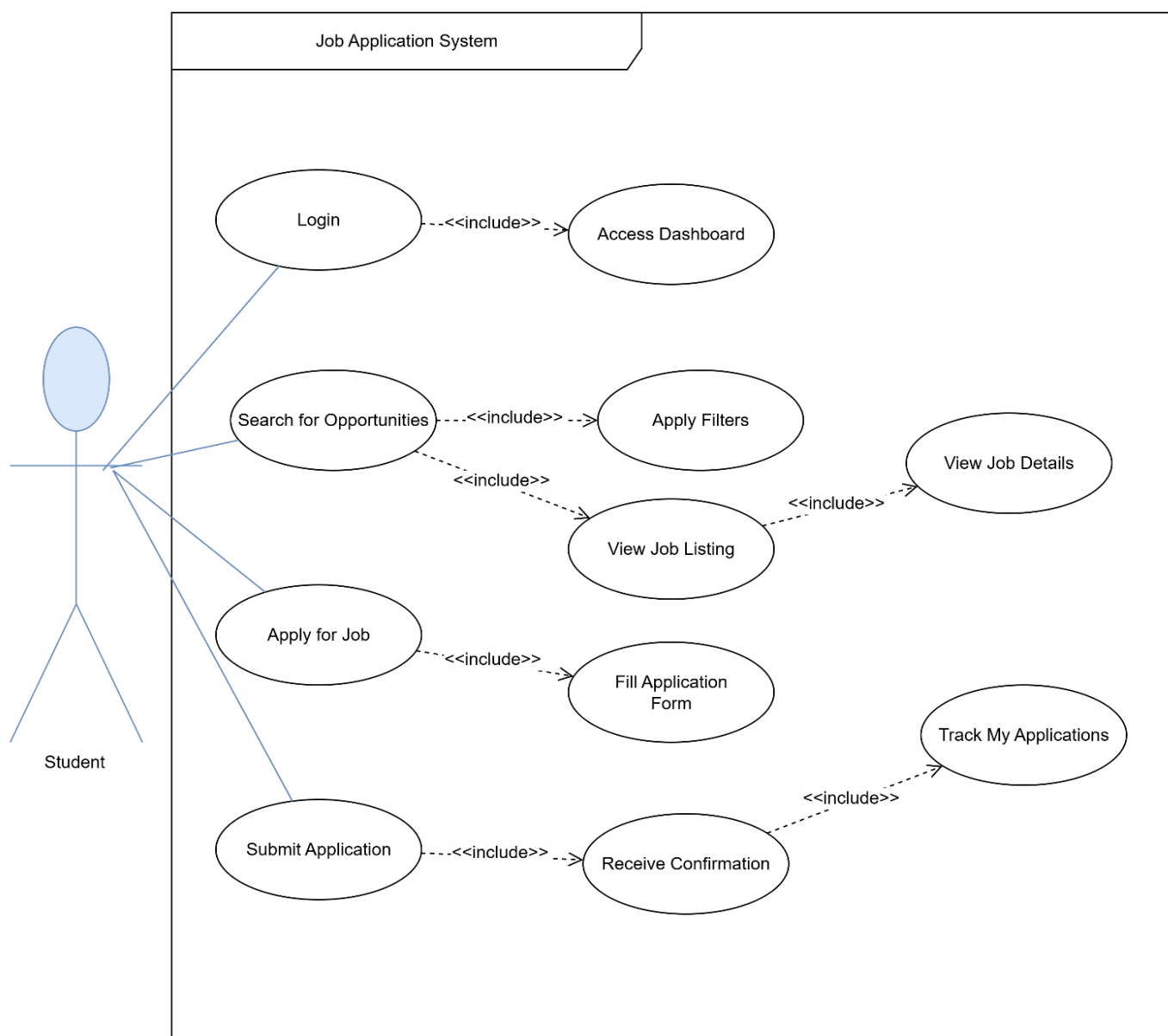


Figure 25: Use Case UC10 Student Job Search & Application

Step-by-Step:

1. Student: Logs into the system and is directed to the Student Dashboard.
2. Student: Clicks the "Job Board" tab in the main navigation menu.
3. System: Loads the Job Board page, displaying a search bar and filter panel (Job Type, Location, Field, etc.).
4. Student: Uses the filter panel, selecting Internship from the "Job Type" dropdown and Information Technology from the "Field" dropdown.
5. Student: Clicks the "Apply Filters" button.
1.6 System: Refreshes the page, displaying a list of job postings that match the selected criteria.
6. Student: Clicks on a specific Job Card titled "Software Developer Intern".
7. System: Opens a detailed view of the job posting in a new modal or page, showing the full description, requirements, and an "Apply Now" button.
8. Student: Clicks the "Apply Now" button.
9. System: Opens the "Application Form" modal. The form is pre-populated with the student's name, email, and the generated CV from their profile. A text area for a "Cover Letter" is empty.
10. Student: Writes a brief, tailored cover letter in the "Cover Letter" text area.
11. Student: Clicks the "Submit Application" button at the bottom of the modal.
12. System: Validates the application.
 - 12.1 If successful: The system displays a confirmation message "Application Submitted Successfully!", closes the modal, and adds the job to the student's "My Applications" tab with a status of "Submitted".
 - 12.2 If a network error occurs: The system displays an error message "Submission failed. Please check your connection and try again." The application form remains open with the data preserved.

Best-Case Scenario:

1. Thabo logs in and goes to the Job Board.
2. He filters for "IT Internships" in "Bloemfontein."
3. The system displays a list of relevant positions. He clicks on one from a leading tech company.

4. Impressed, he clicks "Apply Now." The system pre-populates the application form with his profile and generated CV.
5. Thabo adds a brief, tailored cover letter and clicks "Submit."
6. The system confirms the application was successful and adds it to his "My Applications" tab.

Worst-Case Scenario:

1. Thabo applies for a job, but his internet connection is unstable.
2. After clicking "Submit," the page hangs and then times out.
3. Thabo is unsure if the application went through. He checks his "My Applications" tab and doesn't see the job listed.
4. He must restart the application process, re-select the job, and hope the connection remains stable during the second attempt.

Alternative Scenario:

1. Thabo finds a perfect job but wants to update his CV first using the integrated CV Builder tool before applying.
2. He navigates to the CV Builder, selects a new template, and makes some edits.
3. He saves the new CV to his profile and then returns to the job posting.
4. He applies successfully, confident that he is presenting his best possible application.



Student Job Search UI

UniConnect
Student Dashboard

Welcome, sesethu [Sign Out](#)

Overview

Profile

Find Jobs

Applications

Portfolio

CV Builder

 Find Job Opportunities

Search and filter available positions

Job Type

All types

Location

e.g., Bloemfontein

Field/Industry

e.g., IT, Engineering

 Search Jobs

Software Developer Intern

Internship

 Bloemfontein

 Tech Solutions Inc.

Join our development team and work on exciting projects using modern technologies like React, Node.js, and MongoDB.

 Apply by 2024/03/15

View Details

Apply Now

Marketing Assistant

Part-time

 Remote

 Digital Marketing Pro

Support our marketing team with digital campaigns and content creation across multiple platforms.

 Apply by 2024/03/20

View Details

Apply Now

Data Analyst

Full-time

 Cape Town



Figure 26: Student Job Search UI

59 | Page

Use Case UC11- Employer Search for Candidates

Actors: Employer, Student (secondary actor)

Brief Description

This functionality provides verified employers with a powerful search tool to find potential candidates based on specific, detailed criteria such as skills, GPA, field of study, and graduation year. Employers view anonymized profiles initially and must request permission to access a student's full academic record.

UML

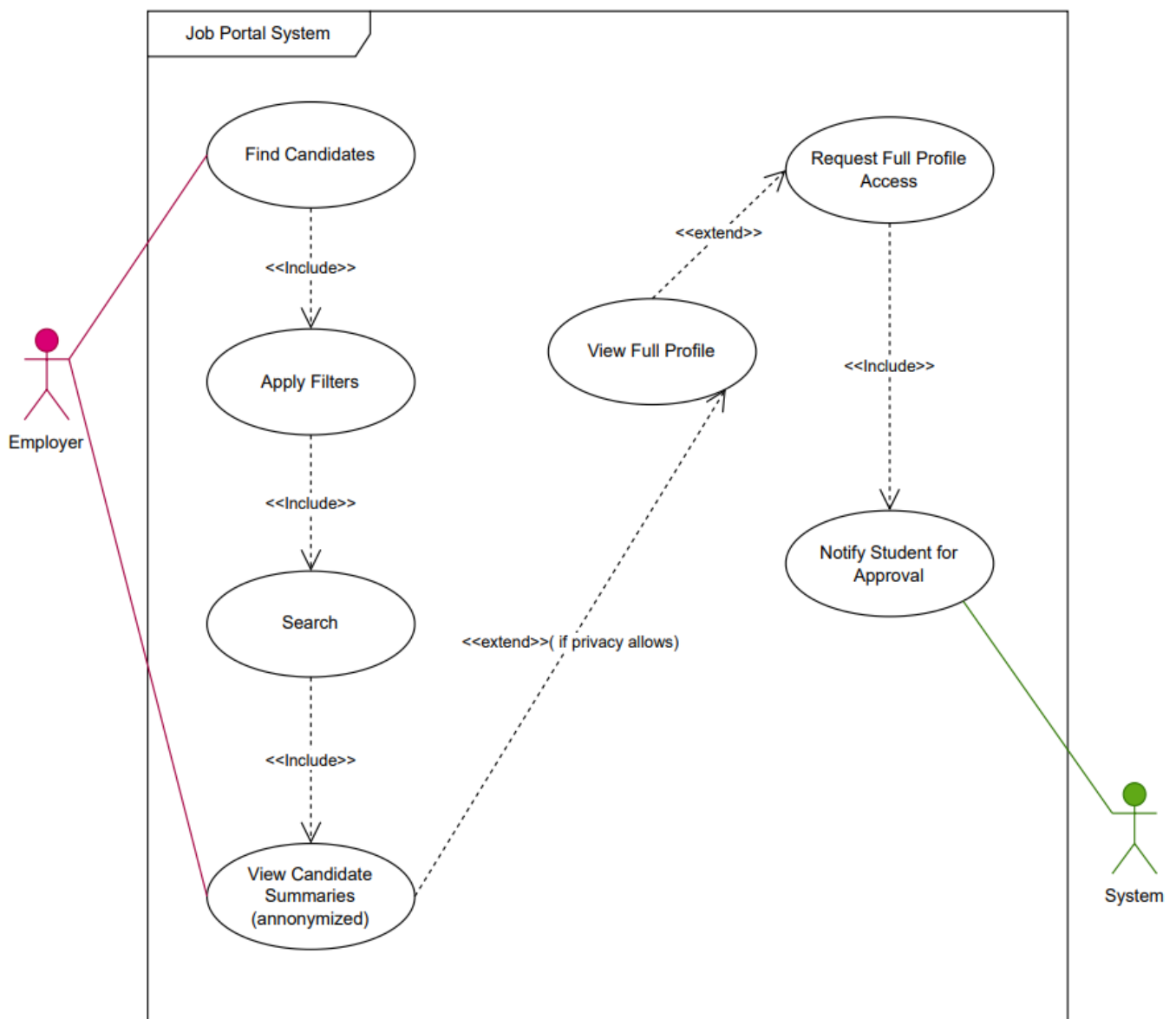


Figure 27: Use Case UC11 - Employer Search for Candidates

Step-by-Step:

1. Employer logs in and from their dashboard, clicks the "Find Candidates" tab.
2. System loads the candidate search page, featuring a prominent filter panel on the left.
3. Employer uses the filter panel, selecting Computer Science from the "Field of Study" dropdown, entering 3.0 in the "Minimum GPA" field, and typing Python into the "Skills" tag input.
4. Employer clicks the blue "Search" button.
5. System displays a list of anonymized candidate profile cards in the main content area. Each card shows skills, course, and graduation year, but no name or personal contact info.
6. Employer clicks on a promising candidate's profile card to expand it and view more details (e.g., bio, project links from SkillStack).
7. Employer clicks the "Request Full Profile" button on the candidate's expanded view.
8. System displays a pop-up: "A request to view [Candidate's] full academic record has been sent. They will be notified and must grant permission." The employer clicks "OK".

Best-Case Scenario:

1. Alice (Employer) logs into the system and clicks Find Candidates.
2. The system displays the candidate search page with filter options such as Degree, GPA, Skills, and Year of Study.
3. Alice selects Bachelor of Commerce, Minimum GPA 70%, and Skills Excel, Data Analysis.
4. She clicks Search, and the system returns a list of anonymised candidate profiles matching the criteria.
5. Alice clicks on three promising candidates and clicks Request Full Profile.
6. The system sends notifications to the students requesting permission.
7. Two students approve the request, and Alice is granted access to their full profiles.
8. Alice then clicks Schedule Interview, and the system generates a calendar invite for the selected candidates.

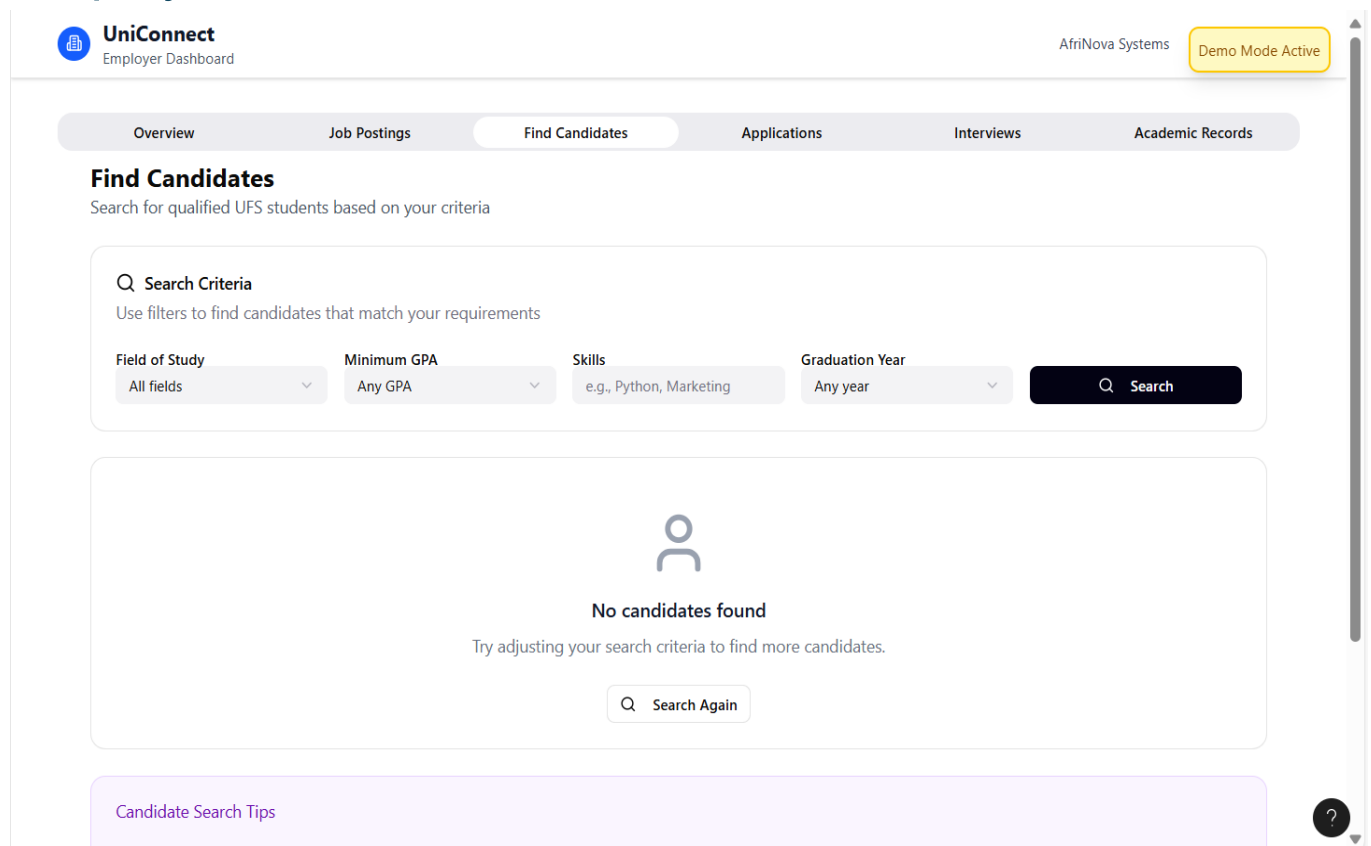
Worst-Case Scenario:

1. Sophia (Employer) logs into the platform and clicks Find Candidates.
2. She sets only one filter Degree = Any.
3. . She clicks Search, and the system attempts to display thousands of student profiles at once.
4. The page becomes slow to load, and Sophia receives an error message “Timeout Error – Too many results returned. Please refine your search criteria.”
5. Sophia is unable to find candidates effectively.

Alternative scenario:

1. David (Employer) searches for Engineering Students with minimum GPA 65%
2. The system displays anonymized candidate profiles.
3. Instead of requesting full profiles, David clicks Send Screening Questionnaire
4. The candidates receive a short questionnaire about availability and interest
5. Five students complete the questionnaire, and David then requests full profiles only for those who responded positively.

Employer Search for Candidates UI



UniConnect
Employer Dashboard

AfriNova Systems Demo Mode Active

Overview Job Postings **Find Candidates** Applications Interviews Academic Records

Find Candidates

Search for qualified UFS students based on your criteria

Search Criteria
Use filters to find candidates that match your requirements

Field of Study: All fields Minimum GPA: Any GPA Skills: e.g., Python, Marketing Graduation Year: Any year Search

No candidates found
Try adjusting your search criteria to find more candidates.

Search Again

[Candidate Search Tips](#)

Figure 28: Employer Search for Candidates UI

Use Case UC12 - Schedule Interview

Actors: Employer, Student

Brief Description

This functionality allows an employer to propose a formal interview time to a student directly from the application or candidate profile view. It facilitates the coordination of the interview schedule between both parties, with options for the student to accept, decline, or propose a new time.

UML

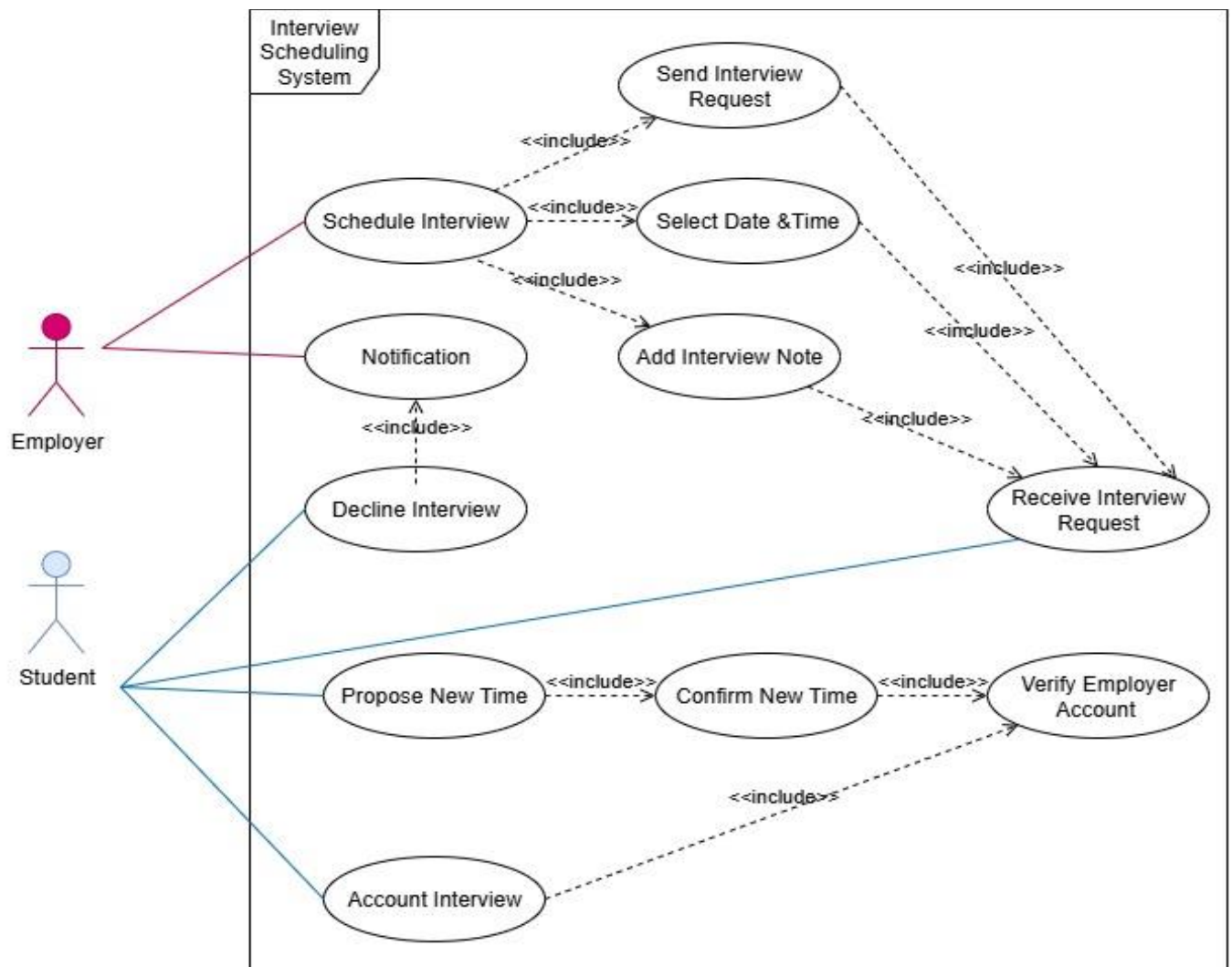


Figure 29: Use Case UC12 - Schedule Interview

Step-by-Step:

1. From an application or candidate profile, Employer clicks "Schedule Interview".
2. Employer selects a proposed date and time from their integrated calendar.
3. Employer adds a note (e.g., "Video call via Zoom, In-person").
4. System sends a notification to the student with the interview request.
5. Student logs in, sees the request, and can
 - **Accept** The event is added to both parties' calendars.
 - **Decline** The Employer is notified.
 - **Propose New Time** Student selects from their available times. System sends the new proposal to the Employer for confirmation.
6. Once confirmed, both parties receive a calendar invite and confirmation notification.

Best-Case Scenario:

1. Sarah (Employer) is reviewing applications for a Software Developer position and finds an impressive candidate, Michael (Student).
2. Sarah clicks the "Schedule Interview" button from Michael's application page.
3. Sarah's integrated calendar opens, showing her availability. She selects Tuesday, October 15th at 200 PM and adds the note "Technical interview via Zoom - 45 minutes duration."
4. Sarah clicks "Send Interview Request" and the system successfully processes the request.
5. Michael receives an instant notification on his dashboard and via email about the interview request.
6. Michael logs in, reviews the interview details, and finds the proposed time works perfectly with his schedule.
7. Michael clicks "Accept", and the system immediately adds the interview to both calendars.
8. Both Sarah and Michael receive calendar invites with Zoom meeting details and confirmation emails within minutes.
9. The interview is successfully scheduled with all details clearly communicated to both parties.

Worst-Case Scenario:

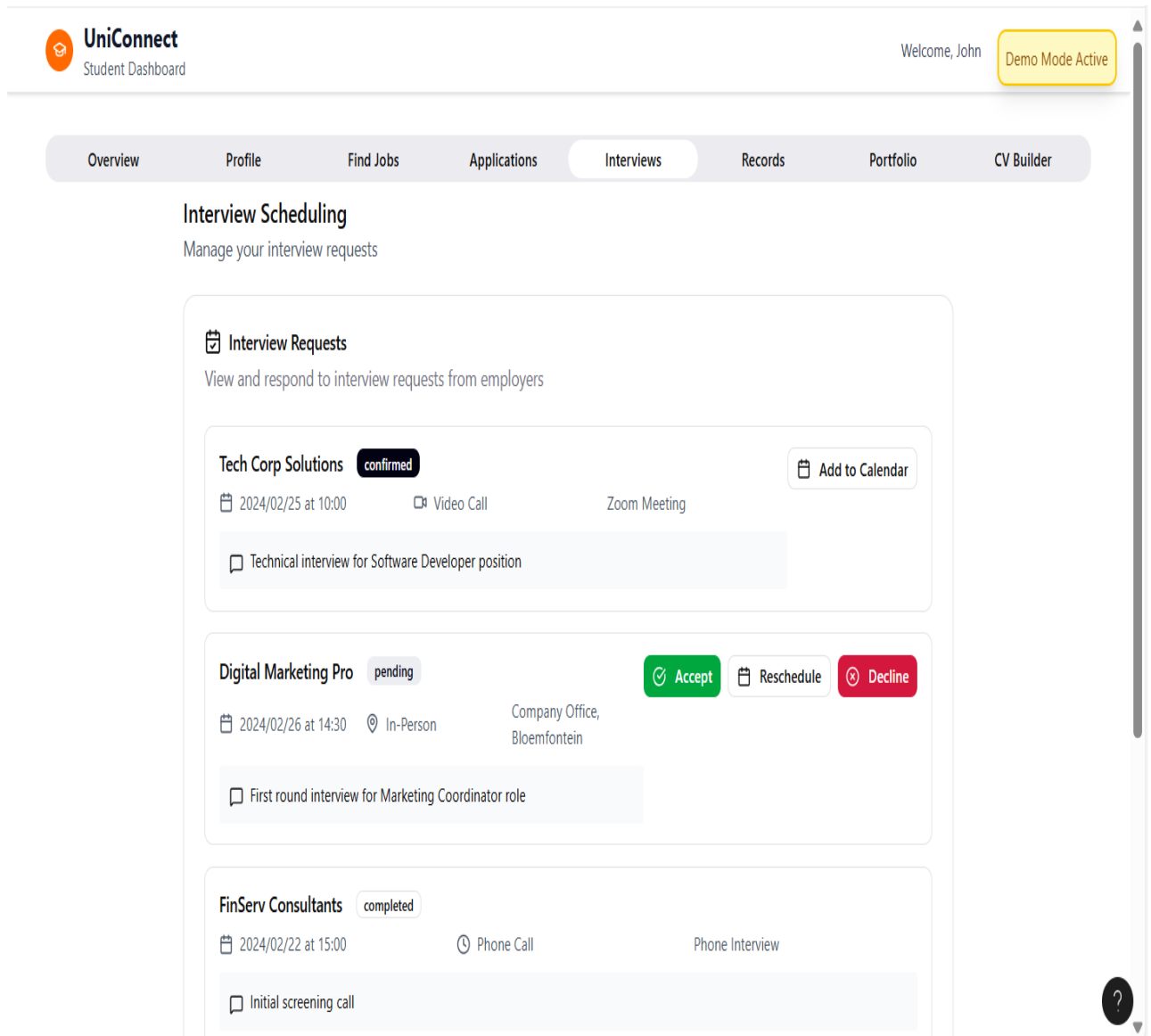
1. Robert (Employer) is rushing to schedule multiple interviews and clicks "Schedule Interview" for candidate Lisa (Student).
2. Robert hastily selects a date and time (Wednesday, October 16th at 1000 AM) without properly checking his calendar and adds a vague note "Interview."
3. The system sends the notification to Lisa, but due to a server issue, the email notification fails to deliver.
4. Lisa doesn't check the platform for several days and misses the interview request entirely.
5. Meanwhile, Robert realizes he has a conflict at the proposed time but cannot modify the sent request.
6. When Lisa finally logs in three days later, she sees the overdue interview request and clicks "Decline" because the time has already passed.
7. The system sends a decline notification to Robert, but he's confused about why it was declined so late.
8. Robert tries to send another interview request, but discovers the platform has a 48-hour cooldown period after a declined request.
9. Lisa, feeling the opportunity has been mishandled, loses interest in the position and accepts another job offer.
10. Both parties experience frustration, and the potential employment opportunity is lost due to poor communication and system limitations.

Alternative Scenario:

1. Amanda (Employer) reviews David's (Student) portfolio and decides to schedule an interview.
2. Amanda clicks "Schedule Interview" and proposes Friday, October 18th at 900 AM for an "In-person interview at company headquarters."
3. David receives the notification but has a university exam at that time.
4. David logs in and clicks "Propose New Time," selecting Friday, October 18th at 200 PM from his available slots.
5. The system sends David's counterproposal to Amanda with a note "Morning exam conflict - afternoon works better."
6. Amanda checks her calendar, sees she's free at 200 PM, and clicks "Accept Counterproposal."
7. The system finalizes the interview for Friday, October 18th at 200 PM.
8. Both parties receive updated calendar invites and confirmation notifications with the new time.

9. The interview is successfully rescheduled to accommodate both parties' availability.

Interview Scheduling UI



The screenshot displays the 'UniConnect Student Dashboard' with a navigation bar containing links for Overview, Profile, Find Jobs, Applications, Interviews (selected), Records, Portfolio, and CV Builder. The 'Interview Scheduling' section is active, showing a list of interview requests. Each request card includes the company name, status, date, time, location, and a description of the interview.

Company	Status	Date & Time	Location	Interview Type	Action
Tech Corp Solutions	confirmed	2024/02/25 at 10:00	Video Call	Zoom Meeting	Add to Calendar
Technical interview for Software Developer position					
Digital Marketing Pro	pending	2024/02/26 at 14:30	In-Person	Company Office, Bloemfontein	Accept, Reschedule, Decline
First round interview for Marketing Coordinator role					
FinServ Consultants	completed	2024/02/22 at 15:00	Phone Call	Phone Interview	
Initial screening call					

Figure 30: Interview Scheduling UI

Use Case UC013 Student Manage Interview

Actors: Student, Employer (secondary actor)

Description

This functionality allows a student to view, manage, and respond to interview requests sent by employers. The student can accept the proposed time, decline the request, or propose an alternative time directly through an integrated calendar system, ensuring a seamless and coordinated scheduling process.

UML

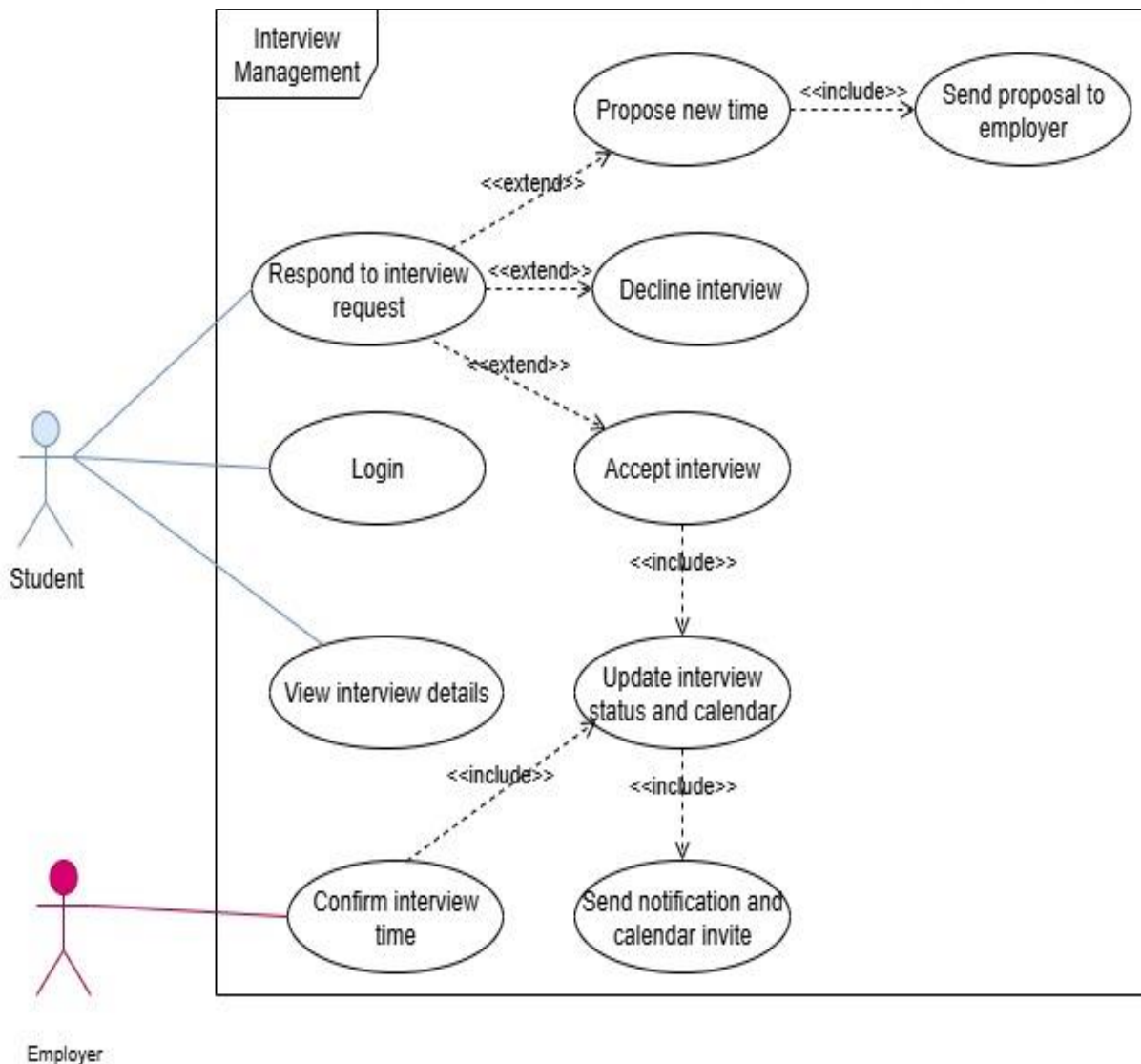


Figure 31: Use Case UC13 Student Manage Interview

Step-by-Step:

1. Student receives an email and an in-app notification: "You have a new interview request from TechCorp for 'Software Developer Intern'."
2. Student logs into the system and clicks the "Notifications" bell icon on the dashboard.
3. System displays a dropdown list of notifications. The student clicks on the interview request notification.
4. System redirects the student to the "Interview Details" page, showing the employer's name, job title, and the proposed date and time. Buttons for "Accept", "Decline", and "Propose New Time" are displayed.
5. Student finds the proposed time acceptable and clicks the "Accept" button.

6. System displays a confirmation pop-up: "Are you sure you want to accept this interview time?" with "Confirm" and "Cancel" buttons.
7. Student clicks "Confirm".
8. System updates the interview status to "Confirmed", sends a confirmation email and calendar invite to both parties, and displays a success message: "Interview Confirmed! A calendar invite has been sent to your email."

Best Scenario:

This flow uses the primary action button to confirm the interview immediately.

1. System student receives an email/mobile notification for a new interview request.
2. Student logs in and clicks the "Notifications" icon (or navigates to the "Interviews" section on the main dashboard).
3. Student clicks on the new request card to view the "Interview Details" screen, showing the proposed Date/Time.
4. Student Finds the time acceptable and clicks the prominent "Accept Interview Time" button.
5. System Displays a confirmation pop-up "Success! Interview Confirmed."
6. System The system sends a synchronized calendar invite (.ics file) to the student's linked email and updates the interview status on the UI to "Confirmed".

Worst Scenario:

This flow uses the UI elements for proposing a new time, but ultimately leads to a failure outcome.

1. System Student receives an interview request notification.
2. Student Navigates to the "Interviews" screen and clicks "Suggest Alternative".
3. Student Uses the integrated calendar picker UI to select an available time slot and clicks "Send Proposal".
4. System (Friction) Student receives an in-app alert "Employer Rejected Proposal."
5. Student Returns to the interview request screen, seeing the initial offer is now expired, and clicks the "Decline Interview" button.
6. Student Fills in the optional "Reason for Declining" text box (e.g., Scheduling Conflict) and clicks "Finalize Decline".
7. System The interview card disappears from the "Interviews" section, and the system updates the status to "Declined/Closed".

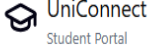
Alternative Scenario:

This flow successfully schedules the interview after using the calendar feature to negotiate a new time.

1. System Student receives an interview request notification.
2. Student Navigates to the "Interviews" screen and clicks "Suggest Alternative".
3. Student Uses the integrated calendar picker UI to select an available time slot and clicks "Send Proposal".
4. System Student receives a new notification "Alternative Time Confirmed by Employer."
5. Student Clicks the notification, and the system automatically updates the "Interview Details" screen with the newly agreed-upon time.

System The system sends a final calendar invite to both parties for the confirmed time, and the UI status changes to "Confirmed".

Student Manage Interview UI


Dashboard
Find Jobs
Applications
Interviews ²
Profile
5
AJ

Interview Management

Manage your interview requests and scheduled interviews

Pending Requests


2

Awaiting your response

Upcoming


0

Confirmed interviews

Completed


1

This month

Total Interviews


5

All time

 You have 2 interview requests awaiting your response. Please review and respond promptly.

Filter Interviews

All Interviews (5)
Pending (2) ²
Upcoming (0)
Completed (1)

All Statuses


All Statuses
 Pending
 Accepted
 Rescheduled
 Completed
 Declined

Software Developer Intern
Urgent


 TechCorp
  Video Call

 2024/01/22
  14:00 (60min)

 Awaiting Response

Round 1

Data Analyst

 DataSoft
  Video Call

 2024/01/25
  10:30 (45min)

 Confirmed

Round 1

Figure 32: Student Manage Interview UI

Figure 33: Use Case UC14 Request Academic Record Access

Step-by-Step:

1. Employer is viewing a student's profile (e.g., after the student has applied to their job). They see a section titled "Academic Record" with a status of "Available upon Request".
2. Employer clicks the "Request Access" button within the Academic Record section.
3. System opens a pop-up modal titled "Request Academic Information". It contains checkboxes for "Transcript", "GPA Verification", and "Course-Specific Grades", and a text area for "Message to Student (Optional)".
4. Employer selects the "Transcript" and "GPA Verification" checkboxes.
5. Employer types a polite message: "Hello, as part of our shortlisting process for the Data Analyst role, we would like to verify your academic record. Thank you."
6. Employer clicks the "Send Request" button.
7. System sends a secure notification to the student and changes the button on the employer's view to "Request Pending", which is disabled.

Best-Case Scenario:

1. Patricia (Employer) from TechCorp is reviewing final candidates for a Software Engineering position and accesses Kevin's (Student) profile.
2. Patricia notices Kevin's academic details show "Available upon Request" and clicks "Request Academic Record" to verify his Computer Science degree claims.
3. The system presents academic record options, and Patricia selects "Transcript" and "GPA Verification" as these are required for her company's hiring policy.
4. Patricia adds a professional message "Hello Kevin, we're moving forward with your application and need to verify your academic credentials as part of our standard hiring process. Thank you for your understanding."
5. Patricia clicks "Send Request" and the system successfully processes and sends the notification to Kevin.
6. Kevin receives the notification within minutes and logs into the platform to review the request.
7. Kevin reads Patricia's message, appreciates the transparency about why the information is needed, and feels comfortable with TechCorp's professionalism.
8. Kevin clicks "Grant Access" and confirms his decision when prompted by the system.

9. Patricia immediately receives a notification that access has been granted and can now view Kevin's official transcript and GPA verification.
10. Patricia reviews the academic information, confirms Kevin's qualifications meet their requirements, and proceeds with the final hiring decision.

Worst-Case Scenario:

1. Richard (Employer) from QuestionableCorp is browsing student profiles and decides to request academic records from Maria (Student) without a clear legitimate need.
2. Richard clicks "Request Academic Record" and selects all available options including "Transcript," "GPA Verification," and "Course-specific grades" without considering what's necessary.
3. Richard adds a vague and unprofessional message "Need to see your grades. Send everything."
4. The system sends the request to Maria, but due to a server delay, the notification arrives 6 hours late.
5. Maria receives the request and is immediately suspicious of the unprofessional tone and excessive information being requested.
6. Maria researches Questionable Corp online and discovers negative reviews about their hiring practices and data handling.
7. Maria clicks "Deny Access" and adds a reason "Request seems excessive and unprofessional for the position applied."
8. Richard receives the denial notification but becomes frustrated and attempts to send multiple additional requests.
9. The system flags Richard's repeated requests as potential harassment and suspends his ability to request academic records for 48 hours.
10. Maria reports the inappropriate behaviour to the platform administrators, and Richard's employer account receives a warning about proper platform usage.
11. The potential employment opportunity is lost due to unprofessional handling of sensitive information requests.

Alternative Scenario:

1. Susan (Employer) from Healthcare Plus is interested in hiring **Ahmad (Student)** for a Medical Research Assistant position.
2. Susan views Ahmad's profile and clicks "Request Academic Record" to verify his biology degree and research experience.
3. Susan selects "Transcript" and "Academic standing/honours" and adds a message "We'd like to verify your biology coursework and any academic honours for our research position."

4. Ahmad receives the request but feels uncomfortable sharing his full transcript due to one poor grade in an unrelated elective course.
5. Ahmad contacts Susan through the platform's messaging system asking if GPA verification alone would be sufficient instead of the full transcript.
6. Susan responds positively, appreciating Ahmad's proactive communication, and agrees that GPA verification would be adequate.
7. Ahmad modifies his response to grant access only to "GPA Verification" and "Academic standing/honours" while politely declining transcript access.
8. The system processes the partial approval, and Susan receives access to the agreed-upon information.
9. Susan reviews Ahmad's GPA and academic honours, finds them satisfactory, and proceeds with the interview process.
10. Both parties maintain a positive professional relationship through respectful communication and flexibility.

Academic Record Access UI


UniConnect
Employer Dashboard

AfriNova Systems

Demo Mode Active

Overview
Job Postings
Find Candidates
Applications
Interviews
Academic Records

Academic Record Access

Request access to student academic records for verification


Request Academic Record Access
Request specific academic information from a student

Select Information to Request
Choose the specific academic records you need for verification

☐ **Official Transcript**
Complete academic record with grades

☐ **GPA Verification**
Verified Grade Point Average

☐ **Enrollment Status**
Current/past enrollment verification

☐ **Degree Verification**
Degree completion status

☐ **Course List**
List of completed courses

☐ **Honors & Awards**
Academic achievements and recognitions

Request Message

Hi! I'm interested in verifying your academic background for a position at AfriNova Systems. This information will help us better understand your qualifications for the role.

This message will be sent to the student along with your request

Figure 34: Academic Record Access UI

Grant/Deny Access UI

Your Access Requests

Track the status of your academic record requests

John Smith pending

 Requested: 2024/02/20

Requested Records:

 Official Transcript  GPA Verification

Request Message:

We would like to verify your academic credentials for the Software Developer position.

Jane Doe approved  Access Granted

 View Records

 Requested: 2024/02/18

 Responded: 2024/02/19

Requested Records:

 Degree Verification  Course List

Request Message:

For the Marketing Coordinator role, we need to verify your business administration background.

Student Response:

Happy to share my academic records. Looking forward to the opportunity!

Mike Johnson denied

 Requested: 2024/02/15

 Responded: 2024/02/16

Requested Records:

 Official Transcript  Honors & Awards

Figure 35: Grant/Deny Access UI

Use Case UC15 - Administrator Views Analytics Dashboard

Actor: Administrator

Brief Description

This functionality provides administrators with a comprehensive, system-wide analytics dashboard. It displays key metrics and trends related to user engagement, job placement rates, popular skills, and application statistics, offering valuable insights for reporting and strategic decision-making.

UML

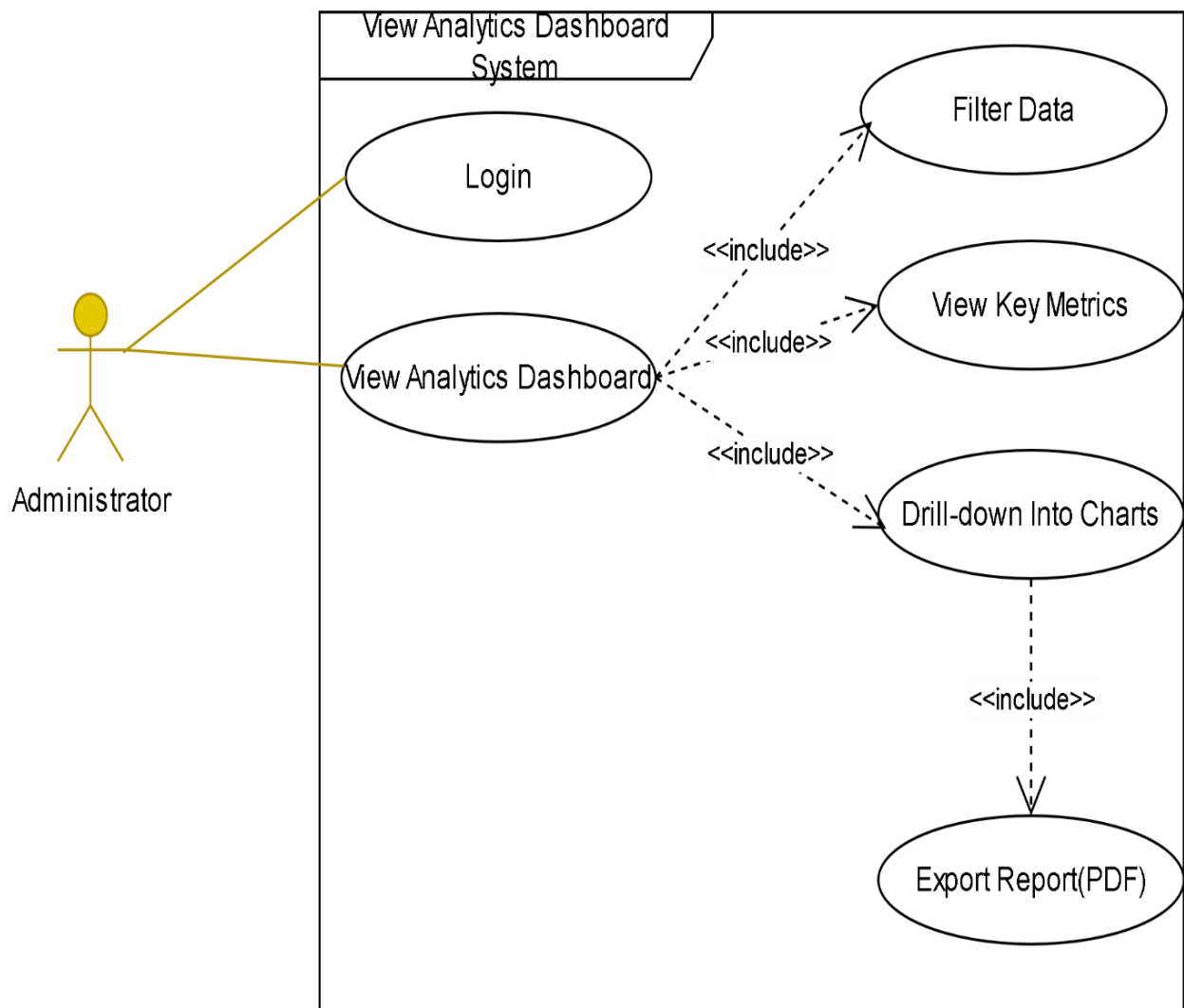


Figure 36: Use Case UC15 - Administrator Views Analytics Dashboard

Step-by-Step:

1. Administrator logs into the system and is automatically directed to the Admin Dashboard.
2. System the dashboard loads, displaying several key metrics widgets:
 - Total Active Users (showing separate counts for Students and Employers)
 - Job Placement Rate (%)
 - Top In-Demand Skills (as a horizontal bar chart)
 - Monthly Applications (as a line graph)
3. Administrator wants to view data for a specific faculty. They click the "Filter" button in the top-right of the dashboard.
4. System expands a filter panel with dropdowns for "Faculty", "Department", and "Time Period".
5. Administrator selects the faculty from the "Faculty" dropdown.
6. All widgets on the dashboard instantly refresh to display data filtered for the selected faculty. The Top In-Demand Skills chart now shows skills relevant to that faculty.
7. Administrator notices a high number of applications in the Monthly Applications graph. They click directly on the highest bar in the graph (representing the current month).
8. System drills down to a new page showing a detailed breakdown of applications for that month, including a list of the most popular job postings and applicant demographics.
9. Administrator clicks the "Export Report" button on the detailed view page.
10. System generates and downloads a PDF report containing the filtered analytics data for the selected faculty and time period.

Best-Case Scenario:

1. The admin logs into the portal and is immediately directed to the Analytics Dashboard.
2. The system loads quickly, displaying all key metrics widgets Admin Routing (2,847), Job Placement Rate (78.4%), Application This Mode (1,256), and the Monthly Application & Placements chart.
3. The admin wants to see data specifically for the Engineering faculty. They use the Filter section, selects "Faculty," and chooses "Engineering."
4. The dashboard updates instantly, showing all the metrics and charts filtered for Engineering students and employers.
5. Satisfied with the data, the admin logs out.

Worst-Case Scenario:

1. The admin logs in and is directed to the Analytics Dashboard.
2. Instead of the metrics, the dashboard displays loading spinners indefinitely. Some widgets show an error message "Data Unavailable."
3. The admin tries to use the Filter to see if it triggers a refresh, but the filter dropdowns are unresponsive.
4. The Recent System Activity panel shows a critical error notification from "2 days ago" about a database connection failure.
5. The admin cannot view any analytics and must contact technical support to resolve the backend issue.

Alternative Scenario:

1. The admin logs in and views the main dashboard. They notice the Job Placement Rate is 78.4%.
2. Instead of filtering, the admin clicks directly on the **78.4%** widget to "drill down."
3. The system navigates to a new, detailed view showing a breakdown of placement rates by department (e.g., Computer Science 20%, Engineering 40% as seen in the UI).
4. From this detailed view, the admin can see which departments are performing well and which may need more support.
5. The admin then uses the "Back to Dashboard" button to return to the main overview.

Admin View Analytics Dashboard UI

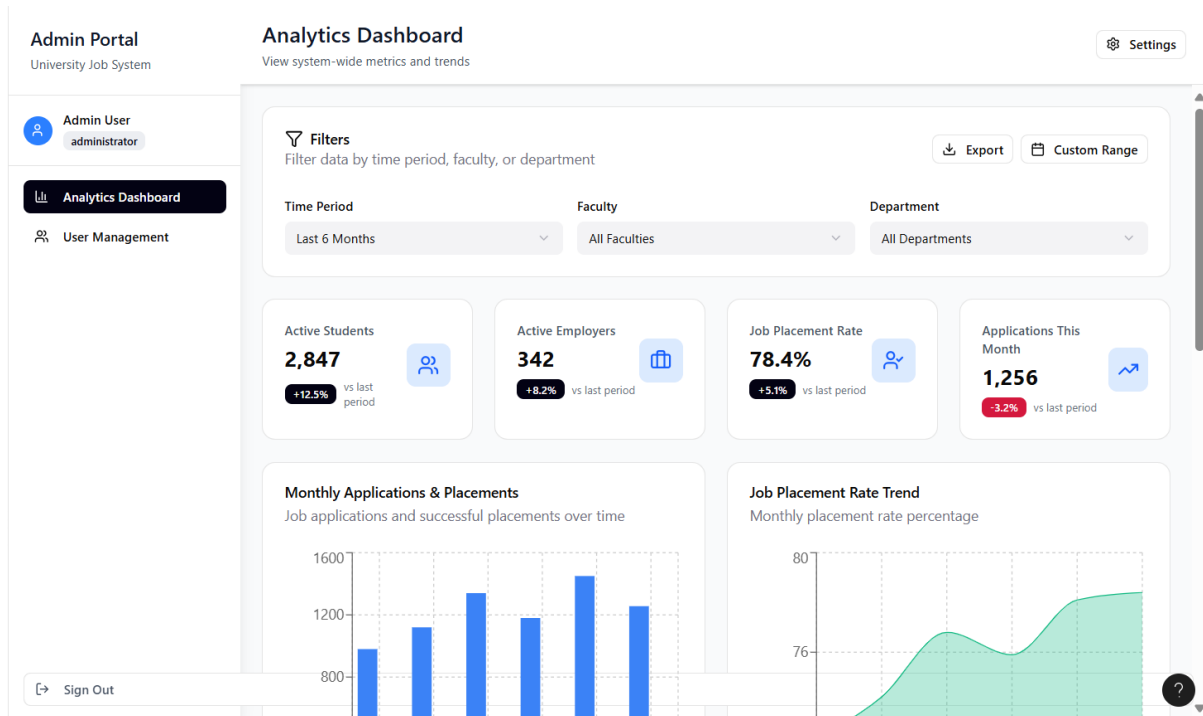


Figure 37: Admin View Analytics Dashboard UI

Admin Analytics sign out UI

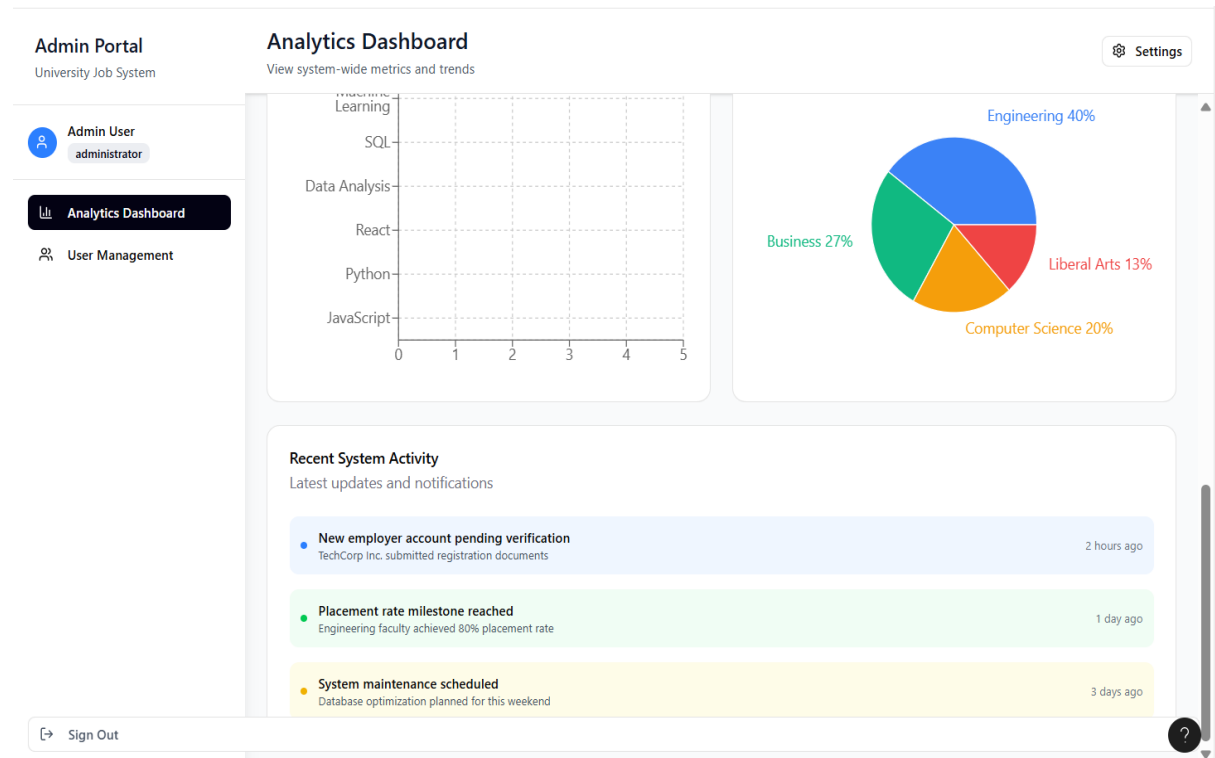


Figure 38: Admin Analytics sign out UI

Use Case UC16 Use Support Chatbot

Actors: Student, Employer, Administrator

Description

This functionality provides contextual help for all system features documented in UC01-UC14, including profile setup (UC01), CV generation (UC05), and job applications (UC02).

Precondition: User is on a dashboard (logged in)

Trigger: User clicks **Help** on their dashboard

Postcondition: User receives an answer (or an escalated support ticket is created and admin notified)

UML

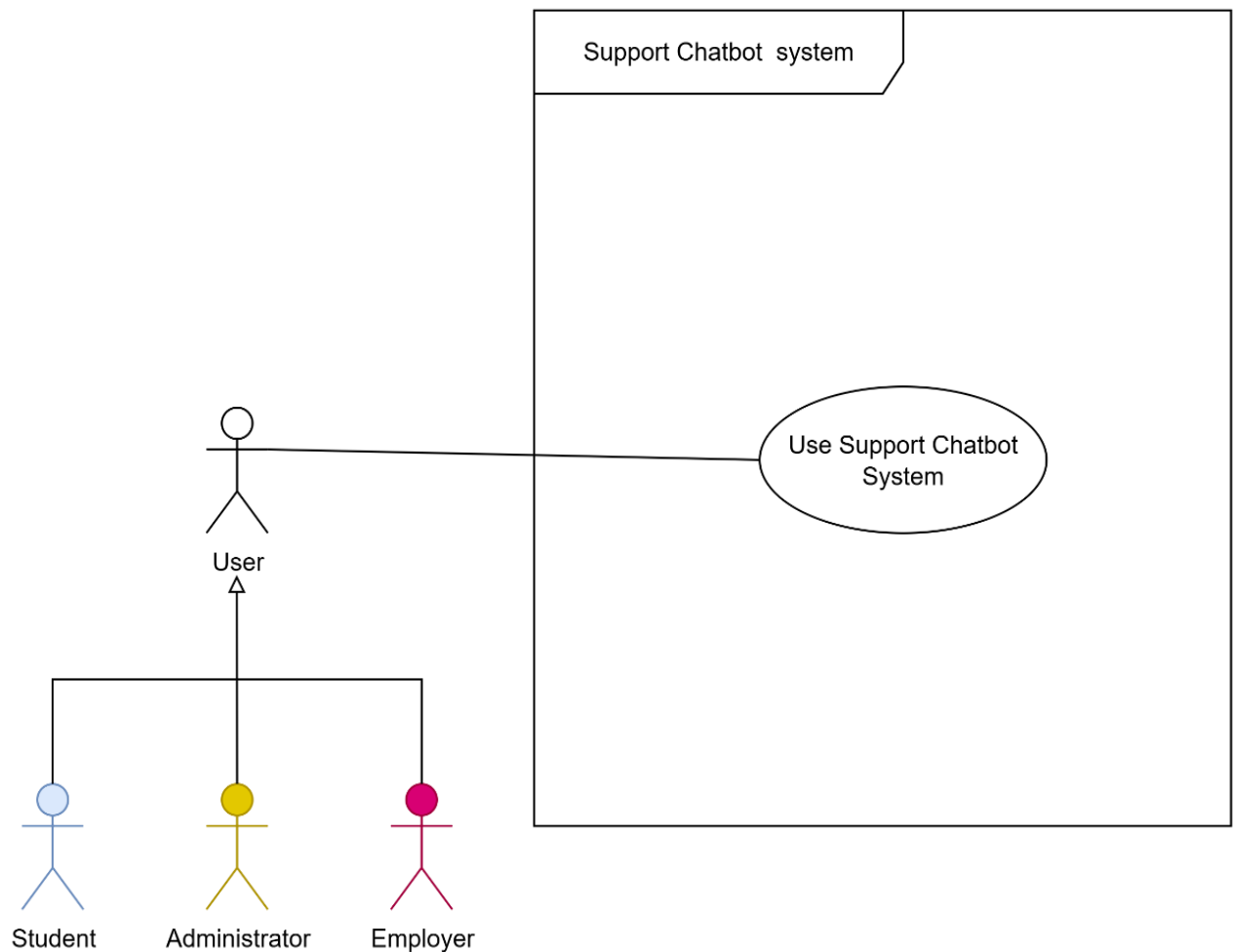


Figure 39: Use Case UC16 Use Support Chatbot

Step-by-Step:

1. The user on any page within the system, they click the floating "Help" icon (a chat bubble) in the bottom-right corner of the screen.
2. The system opens the "Support Assistant" chat window with a greeting message "Hello! I'm here to help. You can ask me about using the platform or choose a common topic below." It displays quick-action buttons like "CV Builder Help," "Job Application Process," and "Employer Verification."
3. The student clicks the "CV Builder Help" button.
4. The system (Chatbot) responds with a step-by-step guide "To use the CV Builder 1. Go to your Dashboard. 2. Click 'CV Builder' in the main menu. 3. Select a template... Would you like me to open the CV Builder for you?" It provides a "Take me there" button.
5. The student clicks "Take me there".



6. System Closes the chat window and redirects the student to the CV Builder tool.

Best-Case Scenario:

1. A student is confused about how to apply for a job.
2. They ask the chatbot, which provides a clear, step-by-step guide and a direct link to the job board.
3. The student completes the application without needing to wait for human support.

Worst-Case Scenario:

1. A student asks a complex, poorly phrased question *"My thing won't upload."*
2. The chatbot misunderstands and provides irrelevant answers about profile pictures instead of project uploads.
3. After three failed attempts, the chatbot displays *"I'm having trouble. Let me connect you with our support team."* and creates a support ticket, forcing the student to wait for an email response.

Alternative Scenario:

1. An employer asks the chatbot *"What documents are needed for verification?"*
2. The chatbot provides a definitive list and a link to the document upload section of their settings.
3. The employer successfully uploads the correct documents on the first try.

Disclaimer: UC15 (Support Chatbot) is a cross-cutting feature accessible from all pages in the system. While documented as UC15, it can be invoked at any point during user interaction with UC01-UC14.

Support Chatbot UI

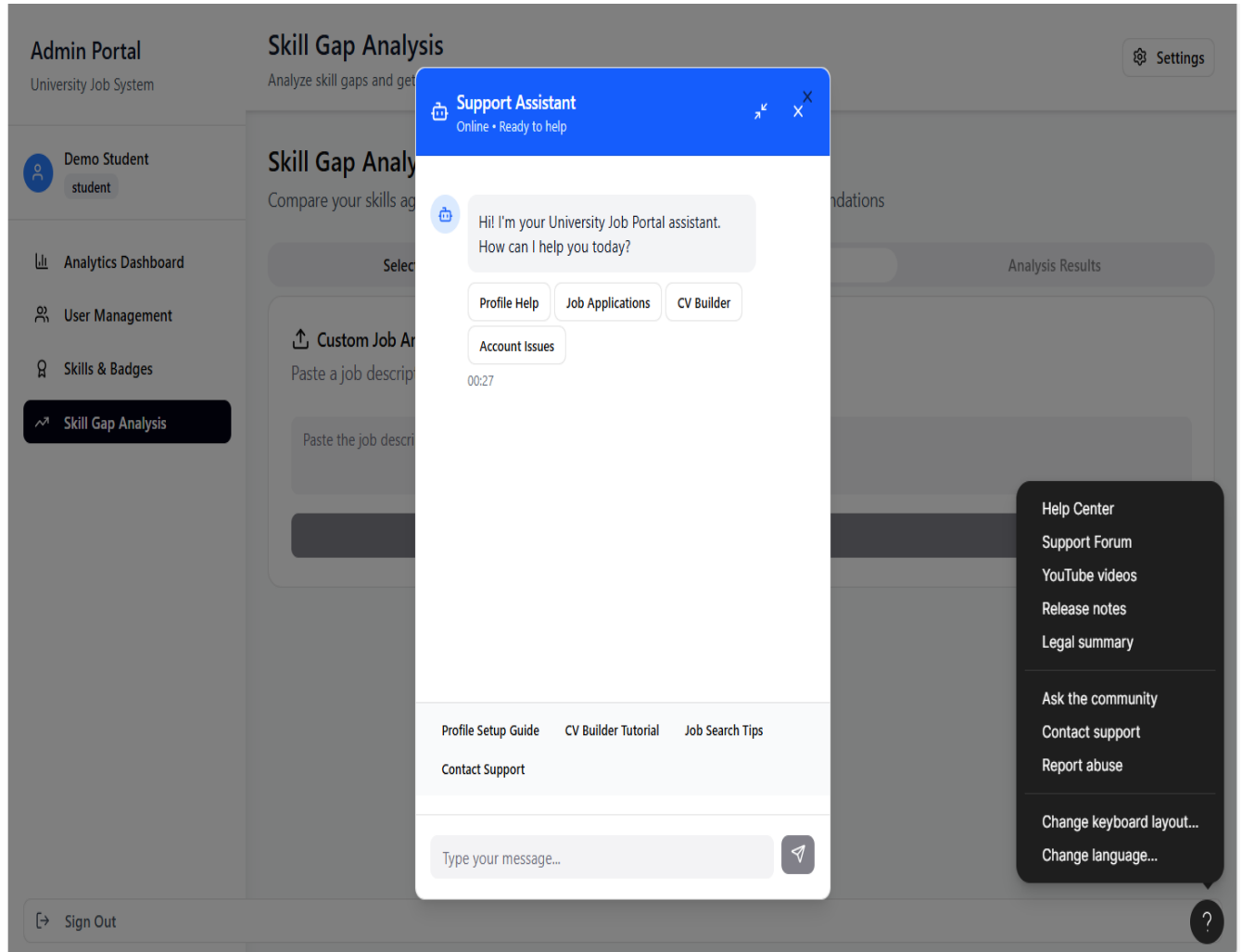


Figure 40: Support Chatbot UI

Accessibility and User Interface

To ensure inclusivity, all web and mobile interfaces will adhere to the W3C Web Content Accessibility Guidelines (WCAG) 2.1 (W3C, 2018).

Conclusion for Chapter 2

In conclusion, the Specification Document provides a structured blueprint for the UFS Recruitment System, emphasising usability, security, and role-based permissions. The detailed use cases ensure that all stakeholders—students, employers, and administrators—can navigate the system efficiently to achieve their goals. This chapter guides the platform's development to fully meet the project's objectives of creating a seamless bridge between students and potential employers.

Chapter 3: Software Project Management Plan

3.1 Introduction

This Software Project Management Plan (SPMP) serves as the definitive master document for the development, deployment, and initial maintenance of the UFS Recruitment System, a comprehensive web-based platform engineered by AfriNova Systems for the University of the Free State (UFS). The system is a strategic initiative designed to bridge the critical gap between academic achievement and successful career placement by creating a dynamic, secure, and integrated digital ecosystem for students, employers, and university administrators. This plan formally establishes the project's governance, defining its objectives, scope, deliverables, schedules, resource allocations, risk mitigation strategies, and quality standards. Its primary function is to provide a structured and actionable framework that will guide the project team through a disciplined development lifecycle, ensuring the on-time, in-budget delivery of a robust, scalable, and production-ready system that fully meets the functional and non-functional requirements as specified in the Project Description and Specification Document.

3.2 Purpose

The purpose of this SPMP is to articulate a clear and comprehensive roadmap for the successful execution of the UFS Recruitment System project. It is designed to:

- **Define Vision and Objectives:** Establish the project's strategic vision, concrete objectives, and measurable success criteria that align with UFS's goal of enhancing student employability and employer engagement.
- **Delineate Scope and Deliverables:** Provide a precise and bounded statement of the project's scope, explicitly listing all in-scope and out-of-scope items to serve as a formal baseline and prevent scope creep.
- **Establish Organizational Structure:** Define the project's organizational structure, clearly outlining the roles, responsibilities, and lines of authority for all team members and stakeholders.
- **Detail Management and Technical Processes:** Specify the managerial control plans (for requirements, schedule, budget, and quality) and the technical process plans (including the development methodology, tools, and techniques).
- **Provide a Execution Framework:** Present a detailed project schedule, resource allocation plan, and budget, enabling effective tracking and control throughout the project lifecycle.
- **Implement Proactive Safeguards:** Outline a robust risk management plan to proactively identify, assess, and mitigate potential project threats, and a rigorous quality assurance plan to ensure the delivery of a high-fidelity, reliable product.
- **Ensure Compliance and Sustainability:** Guarantee that the system is developed with best practices in security and data privacy, adhering to regulations such as POPIA, and is designed for maintainability and future scalability.

3.3 Project Scope

The scope of this project is strictly confined to the design, development, testing, and deployment of the web-based application for the UFS Recruitment System. The following functionalities are included and excluded for this release.

In-Scope (Core Web Application Features):

- **Student Module:**

- Registration with UFS student number and email verification via secure API integration.
 - Comprehensive professional profile setup with automated academic data import (subject to student consent).
 - SkillStack Portfolio management for uploading and showcasing projects, achievements, and associated media/files.
 - AI-assisted CV Builder Tool with multiple professional templates and PDF export functionality.
 - Intelligent Job Search & Application system with advanced filtering and one-click application using profile data.
 - Skill Gap Analysis Tool that compares student skills with market roles and suggests learning resources.
 - Interview Management interface to view, accept, decline, or propose new times for employer-scheduled interviews.
 - Privacy-controlled management and sharing of academic records with employers upon request.
- **Employer Module:**
 - Verified registration process requiring company documentation and administrative approval.
 - Company profile creation and management, including the interactive "Culture Compass" showcase.
 - Job Posting functionality with rich text editors, preview capabilities, and publishing to the job board.
 - Advanced Candidate Search with filters for skills, GPA, field of study, and graduation year, initially showing anonymized profiles.
 - Ability to Request Access to a student's full academic record, pending student approval.
 - Integrated Interview Scheduling system with calendar synchronization and notification alerts.
 - Dashboard for tracking and managing received applications.

- **Administrator Module:**

- Centralised User Management for overseeing all Student, Employer, and Administrator accounts.
- Employer Verification Workflow for reviewing and approving new employer registrations.
- Comprehensive Analytics Dashboard with visual data on placement rates, in-demand skills, user engagement, and application trends, with faculty/department-level filtering.
- System oversight, log monitoring, and functionality for sending broadcast notifications.

- **Cross-Cutting System Features:**

- Universal Login & Authentication with security measures including account lockout after multiple failed attempts.
- Contextual Support Chatbot for in-app user assistance, with escalation to a support ticket system.
- Integrated Notification System delivering alerts via both in-app notifications and email.
- Secure File Management for user-uploaded documents including CVs, portfolio files, and employer verification documents.

Out of Scope (Excluded from Current Project Release):

- The development of a native iOS or Android mobile application. (*Planned for a future phase.*)
- A direct, free-form messaging system between students and employers outside of the interview scheduling context.
- Real-time integration with external social or professional platforms such as LinkedIn or Indeed.
- Any functionality related to processing salaries, negotiating employment contracts, or conducting formal background checks.
- A fully autonomous AI-based career path prediction engine.

3.3.1 Definitions and acronyms.

3.3.1.1 Definitions

- **Client:** The entity or individual responsible for commissioning and overseeing the UFS Recruitment System project. The client (Dr. R. Fouché representing the University of the Free State) provides project requirements, feedback, and approves the final product.
- **UFS Recruitment System:** A comprehensive web-based platform designed to bridge the gap between academic achievement and successful career placement by connecting UFS students with potential employers through an integrated digital ecosystem.
- **Student:** A registered UFS student who uses the platform to create professional profiles, build CVs, showcase portfolios, search for jobs, and apply for employment opportunities.
- **Employer:** A verified company representative who uses the platform to post job opportunities, search for qualified candidates, schedule interviews, and manage applications.
- **Administrator:** A system administrator responsible for managing user roles, verifying employer accounts, overseeing system configurations, and monitoring analytics within the UFS Recruitment System.
- **SkillStack Portfolio:** A digital showcase feature allowing students to upload and display their projects, achievements, coursework, and associated media files to demonstrate practical skills to potential employers.
- **Academic Record Access:** A privacy-controlled feature that allows employers to request access to a student's full academic transcript and records, pending explicit student approval.
- **Job Board:** The central platform interface displaying all active job postings and internship opportunities posted by verified employers.

3.3.1.2 Acronyms

- **SPMP:** Software Project Management Plan
- **UFS:** University of the Free State
- **POPIA:** Protection of Personal Information Act
- **CV:** Curriculum Vitae
- **GPA:** Grade Point Average
- **API:** Application Programming Interface
- **UI/UX:** User Interface/User Experience
- **RBAC:** Role-Based Access Control
- **UAT:** User Acceptance Testing
- **CI/CD:** Continuous Integration/Continuous Deployment

- **OWASP:** Open Web Application Security Project
- **WCAG:** Web Content Accessibility Guidelines
- **JWT:** JSON Web Tokens
- **VPS:** Virtual Private Server
- **IDE:** Integrated Development Environment
- **SRS:** Software Requirements Specification
- **QA:** Quality Assurance
- **PDF:** Portable Document Format
- **SMS:** Short Message Service
- **ACID:** Atomicity, Consistency, Isolation, Durability

3.4 Project Organisation

3.4.1 Organisational Structure & Responsibilities

The project will be executed by the AfriNova Systems team, with each member assuming critical, well-defined roles. The structure is designed to ensure clear accountability and efficient collaboration.

Role	Lead	Key Responsibilities
Project Manager	Mngxozana Sesethu	Overall project planning, scheduling, and tracking for all documentation deliverables. Risk management, stakeholder communication (primary liaison with Dr. Fouché), and ensuring the final documentation package is complete and submitted on time. Facilitates team coordination and progress reviews.
Systems Analyst & QA Lead	Vuyolwethu Bera	Eliciting, analysing, and documenting all functional and non-functional requirements. Authoring the core Software Requirements Specification (SRS), detailed use cases, UML diagrams (use case, class, sequence), and ensuring requirements traceability throughout the design documentation.
Lead Developer	Velenkosini Darrell Nkosi	Leading the technical design specification. Defining the system architecture, technology stack, and high-level component design for the documentation. Creating the software architecture document and technical specifications that would guide future implementation.
UI/UX Designer	Likhona Neku	Conducting user research, creating wireframes, and developing high-fidelity interactive prototypes. Designing the visual interface to ensure an intuitive and engaging user experience. Guaranteeing all designs comply with WCAG 2.1 AA accessibility standards.
Database Architect	Katleho Diseko	Designing the logical and physical database schema. Documenting all entities, attributes, relationships, and defining data integrity, security policies, and backup strategies within the system design documentation.
DevOps & Configuration Manager	Omphah Ramaru	Defining the development and deployment environment specifications. Documenting the version control strategy (using GitHub as the proposed platform), the CI/CD pipeline plan,

		and the configuration management processes that would be used during implementation. Creating the infrastructure and deployment plan documentation.
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3.4.2 Stakeholder Register

Stakeholder	Role	Interest / Influence
Dr. R. Fouché (UFS)	Client & Sponsor	High Interest, High Influence. Provides strategic guidance, approves key deliverables and scope changes, and acts as the primary university liaison.
UFS Students	End-User	High Interest, Low Influence. The primary beneficiaries as documented in the system's scope; their needs are analysed and designed for, but they are not directly involved in the documentation phase.
UFS Employers	End-User	High Interest, Low Influence. Key platform users as defined in the requirements; their engagement needs are captured in the design, but they are not active participants in documentation creation.
UFS Administrators	End-User	High Interest, Low Influence. Their operational requirements are documented within the admin module specifications, but they do not provide direct input on the project documentation.
AfriNova Development Team	Project Team	High Interest, High Influence. Directly responsible for building, testing, and delivering the system; their performance determines the project's outcome.

3.4.3 Communication Plan

Purpose	Audience	Frequency	Medium / Tool
Team Progress Sync	Documentation Team	Weekly / As Needed	WhatsApp / In-Person Meeting
Document Collaboration	Documentation Team	Continuous	Shared Google Drive
Deliverable Submission & Review	Dr. R. Fouché	At Milestones	Formal Email
Version Control for Documents	Documentation Team	Continuous	Google Drive (Version History)
Issue/Task Tracking	Documentation Team	As Needed	Shared Task List / WhatsApp

3.5 Project Management Approach

3.5.1 Development Methodology: Hybrid (Agile-Waterfall)

We will employ a hybrid methodology to balance structure with flexibility.

- **Phases (Waterfall):** The project will follow sequential high-level phases: Planning, Requirements, Design, Implementation, Testing, Deployment, and Closure. This ensures disciplined milestone tracking and deliverable sign-off.
- **Sprints (Agile):** The Implementation phase will be divided into two-week Agile sprints. Each sprint will focus on delivering specific, testable modules from the product backlog, allowing for regular feedback and adaptive planning.

3.5.2 Work Breakdown Structure (WBS) & Master Schedule – REVISED

CURRENT STATUS: Documentation Phase

- **Phase 1: Project Initiation & Planning (Weeks 1-2) ☒ COMPLETED**
 - Deliverable: Approved SPMP Document
 - Status: Delivered
- **Phase 2: Requirements Analysis (Weeks 3-4) ☒ COMPLETED**

- Deliverable: Signed-off Software Requirements Specification (SRS) with Detailed Use Cases
- Status: Delivered
- **Phase 3: System Design (Weeks 5-6) ✓ COMPLETED**
 - Deliverable: Complete System Design Document (Architecture Diagrams, UI/UX Mock-ups, Database Schema, CRC Cards, Class Diagrams, Sequence Diagrams (**in progress**))
 - Status: Delivered

POST-SUBMISSION PHASES (Planned Implementation):

- **Phase 4: Iterative Development & Unit Testing (Weeks 7-12) - PLANNED**
 - Sprint 1 (2 weeks): User Authentication & Registration Modules
 - Sprint 2 (2 weeks): Student & Employer Profile Management
 - Sprint 3 (2 weeks): Job Posting & Search/Application Modules
 - Sprint 4 (2 weeks): Candidate Search & Interview Scheduling
 - Sprint 5 (2 weeks): Analytics Dashboard & Admin Controls
 - Sprint 6 (2 weeks): Integration, Security Hardening, and Final Polish
 - Deliverable: Fully coded and unit-tested system modules
- **Phase 5: System & User Acceptance Testing (UAT) (Weeks 13-14) - PLANNED**
 - Deliverable: Test Summary Report & Resolved Bug Log
- **Phase 6: Deployment & Closure (Week 15-16) - PLANNED**
 - Deliverable: Live Production System, Final Project Report, and Handover Documentation

3.5.3 Resource Management

- **Human Resources:** The team of 6 members, as defined in Section 3.4.1.
- **Tools & Software:**

- **Design:** Figma, [Draw.io](#).
- **Development:** Visual Studio, Node.js, React, Microsoft SQL Server, Git.
- **Project Management:** GitHub Projects, WhatsApp, Google Drive, MS Teams.
- **Testing:** Jest, Postman, Selenium.

3.6 Technical Process Plan

3.6.1 Technology Stack

Layer	Technology Choices	Justification
Frontend	React.js, HTML5, CSS3 (with Sass), JavaScript	Component-based architecture for reusability, strong ecosystem, and excellent performance for a dynamic UI.
Backend	Node.js with Express.js	Unified JavaScript stack (full-stack efficiency), non-blocking I/O for handling concurrent requests, vast npm ecosystem
Database	Microsoft SQL Server	Relational database for ACID compliance, data integrity for academic records, and proven reliability.
Authentication	JWT (JSON Web Tokens), bcrypt for password hashing	Stateless, scalable authentication. Bcrypt provides robust security against brute-force attacks.

Hosting	VPS (e.g., DigitalOcean, AWS EC2) or UFS-provided server	Full control over environment, suitable for handling sensitive student data within South Africa (POPIA).
Version Control	GitHub	Industry standard for collaboration, code review, and CI/CD integration.

3.6.2 Development Standards & Conventions

- **Coding Standards:** ESLint for JavaScript/React, Prettier for code formatting.
- **Git Workflow:** Feature Branch Workflow. All changes must be made via pull requests and require at least one code review before merging into the main branch.
- **Documentation:** Inline code comments (JSDoc style), a dedicated README.md for setup, and API documentation.

3.6.3 Testing Strategy

A multi-layered testing approach will be implemented post-documentation delivery to ensure system quality and reliability:

1. Unit Testing

- **Tool:** Jest (for React components) and NUnit (for Node.js/C# backend)
- **Objective:** Validate individual functions, methods, and React components
- **Coverage Target:** >80% code coverage
- **Responsibility:** Each developer tests their own modules
- **Timeline:** Concurrent with development (Sprints 1-6)

2. Integration Testing

- **Tool:** Supertest (API testing), Postman

- **Objective:** Verify interactions between system modules (e.g., frontend ↔ backend ↔ database)
- **Focus Areas:**
 - Authentication flow (Login → Session Management)
 - Data flow (Profile Creation → Database Storage → Retrieval)
 - File uploads (Resources → Server Storage → Database Metadata)
- **Timeline:** End of each Sprint (Weeks 7-12)

3. System Testing (End-to-End)

- **Tool:** Selenium WebDriver
- **Objective:** Validate complete user workflows from start to finish
- **Key Workflows to Test:**
 - Student registration → Profile setup → Job application
 - Employer registration → Job posting → Candidate search
 - Admin approval workflows
- **Timeline:** Week 13

4. User Acceptance Testing (UAT)

- **Participants:**
 - 5 UFS students (representing different faculties)
 - 3 employers (from partnered organizations)
 - 2 UFS administrators
- **Environment:** Staging server (mirroring production)
- **Test Scenarios:** Based on use cases from Chapter 2
- **Acceptance Criteria:**
 - 95% task success rate for core workflows
 - Average System Usability Scale (SUS) score > 70

- **Timeline:** Week 14
- **Deliverable:** UAT Sign-off Document

5. Performance & Load Testing

- **Tool:** Apache JMeter
- **Objective:** Ensure system handles expected user load
- **Test Scenarios:**
 - 50 concurrent users (normal load)
 - 100+ concurrent users (peak load during registration periods)
 - Database query response times (<2 seconds)
- **Timeline:** Week 13

6. Security Testing

- **Tool:** OWASP ZAP (Zed Attack Proxy)
- **Objective:** Identify vulnerabilities (SQL injection, XSS, CSRF)
- **Compliance Check:** POPIA data handling requirements
- **Deliverable:** Security Audit Report with remediation plan
- **Timeline:** Week 14

7. Regression Testing

- **Objective:** Ensure new features don't break existing functionality
- **Approach:** Automated test suite runs after each code merge
- **Timeline:** Ongoing (Weeks 7-14)

Defect Management:

- **Tool:** Jira (integrated with GitHub)
- **Severity Levels:**
 - **Critical:** System crash, data loss (Fix within 24 hours)
 - **High:** Major feature broken (Fix within 3 days)

- **Medium:** Minor bug affecting usability (Fix within 1 week)
- **Low:** Cosmetic issues (Fix in next sprint)

Test Deliverables:

1. Test Plan Document (Week 7)
2. Test Case Repository (Ongoing)
3. Test Execution Reports (Per sprint)
4. UAT Sign-off Document (Week 14)
5. Final Test Summary Report (Week 14)

3.6.4 Implementation Plan (POST-DOCUMENTATION)

This section outlines the detailed approach for system implementation, which will commence after the documentation delivery and client approval.

3.6.4.1 Implementation Phases

Phase 4A: Sprint 1 - Authentication & Core Infrastructure (Weeks 7-8)

Objectives:

- Establish secure user authentication system
- Implement role-based access control (RBAC)
- Set up database schema and initial API endpoints

Key Deliverables:

1. Login Module:

- User authentication with JWT tokens
- Password encryption using bcrypt
- Account lockout mechanism (5 failed attempts)
- Forgot password functionality with SMS/email verification

2. Registration Module:

- Student/Employer registration forms

- UFS API integration for academic data import
- Admin approval workflow for employer accounts
- Username/password generation system

3. Database Setup:

- Microsoft SQL Server deployment
- Tables: Users, Profiles, Authentication, Roles
- Stored procedures for authentication logic
- Initial data seeding (test users, system admins)

Dependencies:

- GitHub repository setup complete
- Development environment configured (Visual Studio, Node.js 18+)
- SQL Server instance provisioned

Phase 4B: Sprint 2 - Profile Management (Weeks 9-10)

Objectives:

- Develop student and employer profile creation/editing
- Implement SkillStack Portfolio module
- Build CV Builder Tool

Key Deliverables:

1. Student Profile:

- Academic information display
- Skills and experience sections
- Privacy controls for data sharing
- Profile completeness indicator

2. Employer Profile:

- Company information management
- Culture Compass interactive showcase
- Team member management

3. SkillStack Portfolio:

- Project upload system (DOCX, PDF, images)
- Achievement timeline view
- Media file management (up to 12MB per file)

4. CV Builder Tool:

- Template selection system (3+ professional templates)
- Dynamic form with auto-population from profile data
- PDF export functionality (using libraries like jsPDF or PDFKit)

Phase 4C: Sprint 3 - Job Board & Applications (Weeks 11-12)

Objectives:

- Create job posting system for employers
- Develop job search and filtering for students
- Implement application submission workflow

Key Deliverables:

1. Job Posting Module:

- Rich text editor integration (CKEditor or similar)
- Job preview before publishing
- Draft/publish workflow
- Job expiration dates

2. Job Search & Filtering:

- Advanced search by skills, location, GPA, field of study
- Real-time filtering using React state management

- Bookmark/save jobs feature
- One-click application using profile data

3. Application Management:

- Application submission with cover letter attachment
- Application status tracking (Submitted, Viewed, Shortlisted, Rejected)
- Employer dashboard to manage applications

Phase 4D: Sprint 4 - Candidate Search & Interviews (Weeks 13-14)

Objectives:

- Build employer candidate search with anonymized profiles
- Implement interview scheduling system
- Create academic record access request workflow

Key Deliverables:

1. Candidate Search:

- Anonymized profile views (no names initially)
- Filtering by GPA, skills, graduation year, faculty
- "Request Full Profile" button
- Student consent notification system

2. Interview Scheduling:

- Calendar integration (using libraries like FullCalendar)
- Propose/accept/decline interview times
- Email and in-app notifications
- Time zone handling for remote interviews

3. Academic Record Access:

- Request workflow (Employer → System → Student)

- Student approval/denial interface
- Secure PDF generation of academic transcripts
- Access expiration after 30 days

Phase 4E: Sprint 5 - Analytics & Admin Tools (Weeks 15-16)

Objectives:

- Develop analytics dashboard with visual data representations
- Build admin user management interface
- Implement employer verification workflow

Key Deliverables:

1. Analytics Dashboard:

- Placement rate charts (by faculty/department)
- In-demand skills visualizations
- User engagement metrics (logins, applications, profile views)
- Filterable data (date ranges, user types)
- Export reports to CSV/PDF

2. Admin Module:

- User management (create, edit, deactivate accounts)
- Employer verification queue
- System log viewer
- Broadcast notification tool

3. Data Visualization:

- Charts using Recharts library
- Real-time data updates via WebSocket (optional)

Implementation Tools:

- **Recharts** for React-based charts

- **Lodash** for data manipulation
- **PapaParse** for CSV export

Phase 4F: Sprint 6 - Integration & Polish (Weeks 17-18)

Objectives:

- Integrate all modules into cohesive system
- Security hardening
- UI/UX refinements
- Performance optimization

Key Activities:

1. System Integration:

- End-to-end workflow testing
- API endpoint consolidation
- Error handling standardization
- Loading states and user feedback improvements

2. Security Hardening:

- OWASP Top 10 vulnerability checks
- Input sanitization across all forms
- HTTPS enforcement
- Rate limiting on API endpoints (e.g., login attempts)

3. Performance Optimization:

- Database query optimization (indexing)
- Frontend lazy loading (code splitting)
- CDN setup for static assets (images, CSS, JS)
- Caching strategy (Redis for session management)

4. UI/UX Polish:

- Responsive design testing (mobile, tablet, desktop)
- Accessibility audit (WCAG 2.1 AA compliance)
- User onboarding tooltips
- Error message clarity

Deliverable: Feature-complete system ready for Phase 5 (Testing)

3.6.4.2 Development Standards & Best Practices

Coding Standards:

- **JavaScript/React:**
 - ESLint configuration enforced
 - Prettier for code formatting
 - Functional components with React Hooks (no class components)
 - PropTypes or TypeScript for type checking
- **Node.js/Backend:**
 - RESTful API design principles
 - Async/wait for asynchronous operations
 - Error handling middleware
 - JSDoc comments for all functions

Git Workflow:

1. Feature branches: feature/module-name
2. Pull request required for all merges to main
3. Minimum 1 code review approval
4. Automated tests must pass before merge
5. Squash commits on merge

Code Review Checklist:

- Code follows ESLint rules

- No hardcoded credentials or sensitive data
- Error handling implemented
- Unit tests written and passing
- No console.log statements in production code
- Comments explain "why," not "what"

Database Management:

- All schema changes via migration scripts
- No direct production database edits
- Backup before major migrations
- Rollback scripts prepared for each migration

3.6.4.3 Development Environment Setup

Required Software (Per Developer):

1. Visual Studio Code (latest version)
2. Node.js v18+ & npm v9+
3. Git v2.40+
4. Microsoft SQL Server Management Studio
5. Postman (API testing)
6. Figma Desktop App (for design reference)

3.6.4.4 Integration Points

External System Integrations:

1. **UFS Database API Integration:**
 - **Purpose:** Import student academic data
 - **Authentication:** API key provided by UFS IT
 - **Endpoints:**

- GET /api/students/{student-number}/academic-record
- **Fallback:** Manual CSV upload if API unavailable
- **Timeline:** Sprint 2 (Weeks 9-10)

2. SMS Gateway (for notifications):

- **Provider:** TBD (Twilio or local provider)
- **Use Cases:**
 - Account verification
 - Password reset codes
 - Emergency announcements
- **Timeline:** Sprint 1 (Week 8)

3. Email Service (SMTP):

- **Provider:** Gmail SMTP or SendGrid
- **Use Cases:**
 - Welcome emails
 - Interview invitations
 - Application status updates
- **Timeline:** Sprint 1 (Week 8)

4. Cloud Storage (Optional):

- **Provider:** DigitalOcean Spaces or AWS S3
- **Use Cases:**
 - CV file storage
 - Portfolio media files
- **Timeline:** Sprint 2 (Week 10)

3.6.4.5 Deployment Strategy (Week 19)

Pre-Deployment Checklist:

- All Phase 5 tests passed (UAT sign-off received)
- Security audit completed (no critical vulnerabilities)
- Database migration scripts tested on staging
- SSL certificates installed and verified
- Backup and rollback procedures documented
- Monitoring tools configured (e.g., New Relic, Sentry)
- Production environment variables configured
- Client training materials prepared

Deployment Steps:

1. Database Deployment:

- Run migration scripts on production SQL Server
- Verify schema integrity
- Seed initial data (admin accounts, system settings)

2. Backend Deployment:

- Deploy Node.js application to VPS (DigitalOcean/AWS EC2)
- Configure Nginx reverse proxy
- Setup PM2 for process management
- Enable HTTPS (Let's Encrypt SSL)

3. Frontend Deployment:

- Build React production bundle (npm run build)
- Deploy to CDN or serve via Nginx
- Configure domain DNS records

4. Post-Deployment Verification:

- Smoke tests on production
- Monitor error logs for 24 hours
- Client walkthrough and acceptance

Rollback Plan:

- Database snapshot taken before migration
- Previous version code tagged in Git (v1.0-stable)
- Rollback can be executed within 15 minutes if critical issues arise

3.7 Schedule and Budget

3.7.1 Schedule

Workflow	Hours Allocated	Status
Requirements	82	☒ Complete
S. Mngxozana	17 Hrs	
V. Bera	16 Hrs	
O. Ramaru	14 Hrs	
L.Neku	12 Hrs	
K. Diseko	11Hrs 30 min	
D.Nkosi	11 Hrs 30 min	
Analysis	77	☒ Complete

V. Bera	14 Hrs	
O. Ramaru	13 Hrs	
S. Mngxozana	13 Hrs	
L.Neku	12 Hrs 20 min	
K. Diseko	12 Hrs 20 min	
D.Nkosi	12 Hrs 20 min	
Design	95	☒ Complete
S. Mngxozana	17 Hrs	
O. Ramaru	16 Hrs	
K. Diseko	16 Hrs	
L. Neku	16 Hrs	
V. Bera	15 Hrs	
D. Nkosi	15 Hrs	
SPMP & Documentation	120	☒ Complete (additional)
S. Mngxozana	27 Hrs	
V. Bera	26 Hrs	
O. Ramaru	20 Hrs	
L.Neku	19 Hrs	
K. Diseko	14 Hrs	
D. Nkosi	14 Hrs	
TOTAL COMPLETED	374 hours	

Planned Phases (**Post-Submission**):

Workflow	Planned Hours	Timeline post-submission
Implementation	321	Weeks 1-6 (Phase 4)
Testing	450	Weeks 7-8 (Phase 5)
Deployment	40	Week 9 (Phase 6)
TOTAL PLANNED	811 hours	

Grand Total Project: 374 + 811 = 1,185 hours

Note: The additional 160 hours (from 1,025 to 1,185) accounts for:

- Enhanced testing procedures
- Deployment activities
- Post-deployment support planning
- Documentation finalization

3.7.2 Cost and Quotation (TWO-PHASE MODEL)

This cost estimation is based on a professional services model, accounting for the full labour cost of the development team and all associated business expenses required to deliver a production-grade system over the 16-week project timeline as defined in our SPMP. The hourly rate of R350 is a conservative industry-standard rate for specialized software development services in South Africa.

Phase 1: Documentation Delivery (Current Submission)

Breakdown of Completed Work:

Workflow	Hours	Hourly Rate (R)	Estimated Cost (R)
Project Planning (Weeks 1-2)	82	350	28,700
Requirements Analysis (Weeks 3-4)	77	350	26,950
System Design (Weeks 5-6)	95	350	33,250
SPMP & Documentation	120	350	42,000
Total Development Hours	374	350	130,900

Calculation of Phase 1 Labour Cost:

- **Total Development Time (entire team):** 374 Hours
- **Fully Loaded Hourly Rate (per person):** R 350
- **Number of Developers:** 6
- **Total Labour Cost:** $374 \times (R350 \times 6) = \text{R } 786,600$

General Expenses (Phase 1 - 4 months 1 week prorated):

Expense Category	Cost (R)	Calculation	Estimated Cost (R)
Software Licenses			65,000
Visual Studio Enterprise	4,750 p/m	$R4,750 \times 4.25 \text{ months} \times 6 \text{ users}$	121,125 (prorated to 40,000)
Microsoft SQL Server Standard	16,813 p/a	One-time for development	16,813 (prorated to 5,000)

GitHub Team	168 p/m	$R168 \times 4.25 \text{ months} \times 6 \text{ users}$	4,284
Figma Professional	105 p/m	$R105 \times 4.25 \text{ months}$	446
Domain Registration (.co.za)	1,200	One-time for 2 years	1,200
SSL Certificate	2,400	One-time for 2 years	2,400
Operational Overheads			47,000
Office Rent & Utilities	18,500 p/m	$R18,500 \times 4.25 \text{ months}$	78,625 (prorated to 30,000)
Travel & Client Meetings	8,500 p/m	$R8,500 \times 4.25 \text{ months}$	36,125 (prorated to 12,000)
Professional Insurance	3,500 p/m	$R3,500 \times 4.25 \text{ months}$	14,875 (prorated to 5,000)
Total Phase 1 Expenses			112,000

Phase 1 Summary:

Item	Cost (R)
Total Development Labour	786,600
General Expenses	112,000
Subtotal	898,600

Profit Margin (25%)	224,650
Total Phase 1 Quotation (Excl. VAT)	R 1,123,250

Phase 2: Implementation & Deployment (Post-Documentation)

Breakdown of Planned Work:

Workflow	Hours	Hourly Rate (R)	Estimated Cost (R)
Implementation (Weeks 7-18)	321	350	112,350
Testing & QA (Weeks 19-20)	450	350	157,500
Deployment & Support (Weeks 21-22)	40	350	14,000
Total Development Hours	811	350	283,850

Calculation of Phase 2 Labour Cost:

- **Total Development Time (entire team):** 811 Hours
- **Fully Loaded Hourly Rate (per person):** R 350
- **Number of Developers:** 6
- **Total Labour Cost:** $811 \times (R350 \times 6) = \text{R } 1,703,100$

General Expenses (Phase 2 - estimated 5.5 months):

Expense Category	Cost (R)	Calculation	Estimated Cost (R)
Software Licenses (continued)			59,000
Visual Studio Enterprise	4,750 p/m	$R4,750 \times 5.5 \text{ months} \times 6 \text{ users}$	156,750 (prorated to 40,000)
DigitalOcean VPS	680 p/m	$R680 \times 5.5 \text{ months}$	3,740
Postman Pro	420 p/m	$R420 \times 5.5 \text{ months}$	2,310
Apache JMeter (Free)	0	Open source	0
Operational Overheads			103,905
Office Rent & Utilities	18,500 p/m	$R18,500 \times 5.5 \text{ months}$	101,750
Team Equipment (Testing Devices)	12,000	One-time	12,000 (prorated to 8,000)
Travel & UAT Sessions	8,500 p/m	$R8,500 \times 5.5 \text{ months}$	46,750
Professional Insurance	3,500 p/m	$R3,500 \times 5.5 \text{ months}$	19,250
Miscellaneous & Contingency	75,000	Buffer for unforeseen costs	75,000
Total Phase 2 Expenses			162,905

Phase 2 Summary:

Item	Cost (R)
Total Development Labour	1,703,100
General Expenses	162,905
Subtotal	1,866,005
Profit Margin (25%)	466,501
Total Phase 2 Quotation (Excl. VAT)	R 2,332,506

Complete Project Pricing:

Phase	Cost (Excl. VAT)
Phase 1: Documentation	R 1,123,250
Phase 2: Implementation & Deployment	R 2,332,506
TOTAL PROJECT COST	R 3,455,756

Payment Terms:

- Phase 1: 30% upon documentation approval (R 336,975)
- Phase 1: 70% upon Phase 1 completion (R 786,275)
- Phase 2: Payable upon implementation commencement (milestone-based)

Quotation Validity: 60 days from documentation delivery date

Phase 2 Trigger: Client written approval of Phase 1 deliverables

3.8 Risk Management

A comprehensive risk management strategy will be implemented throughout the project lifecycle to identify, analyse, and mitigate potential threats to project success. The following sections detail the risk management approach, categorized by risk type with corresponding mitigation and contingency plans.

3.3.101 Technical Risks

Data Security and POPIA Compliance Breach

- **Probability:** Medium
- **Impact:** High
- **Mitigation Strategy:** Implement end-to-end encryptions for all sensitive data, enforce strict role-based access control (RBAC), and adhere to the principle of least privilege. All candidate data in initial search results will be anonymized. A comprehensive security audit will be conducted prior to deployment, and all team members will undergo data privacy training.
- **Contingency Plan:** In the event of a breach, an immediate incident response plan will be activated. This includes legal consultation, notification of the Information Regulator as per POPIA requirements, and transparent communication with affected users.

Integration Failure with UFS Database API

- **Probability:** Medium
- **Impact:** High
- **Mitigation Strategy:** Early and continuous engagement with UFS IT services to secure API specifications and test environments. A mock API will be developed and used during the development phase to ensure progress is not hindered by external dependencies.
- **Contingency Plan:** If integration fails or is delayed, a fallback manual data entry and document upload process will be implemented for student academic information, ensuring core functionality remains available.

Performance and Scalability Issues Under Load

- **Probability:** Low
- **Impact:** High

- **Mitigation Strategy:** The system will be architected for scalability from the outset, utilizing stateless APIs, efficient database indexing, and a content delivery network (CDN) for static assets. Rigorous load testing simulating 100+ concurrent users will be performed during the testing phase.
- **Contingency Plan:** Should performance issues arise post-launch, server resources will be scaled vertically, and advanced optimization techniques such as query caching with Redis will be implemented.

3.3.102 Project Management Risks

Scope Creep

- **Probability:** High
- **Impact:** Medium
- **Mitigation Strategy:** A formal change control process will be enforced. Any request for new features or significant alterations to approved requirements must be submitted in writing, analysed for impact on timeline and budget, and require formal sign-off from the client (Dr. Fouché) before implementation. A prioritized product backlog will be maintained and reviewed weekly.
- **Contingency Plan:** Minor, non-critical changes will be deferred to a "Phase 1.1" post-launch update. For major changes that threaten project deliverables, a formal contract re-negotiation will be initiated to adjust timelines and costs.

Key Team Member Unavailability

- **Probability:** Medium
- **Impact:** High
- **Mitigation Strategy:** Comprehensive and up-to-date documentation of all systems, code, and processes will be maintained in a shared wiki. Pair programming will be practiced on critical modules to ensure knowledge redundancy. Weekly knowledge-sharing sessions will be mandated for all team members.
- **Contingency Plan:** Tasks will be dynamically re-allocated based on skill overlap within the team. If necessary and budget allow, a temporary contractor will be brought in to fill critical skill gaps.

3.3.103 Business and External Risks

Poor User Adoption by Students and Employers

- **Probability:** Medium
- **Impact:** Medium
- **Mitigation Strategy:** The UI/UX will be designed with a strong emphasis on usability and intuitive navigation, incorporating UFS branding to foster familiarity. A series of video tutorials and a comprehensive FAQ knowledge base will be created. UFS Career Services will be engaged as champions to actively promote the platform.
- **Contingency Plan:** Post-launch feedback sessions will be conducted, and usage analytics will be monitored closely. An iterative improvement plan will be executed to address usability concerns and enhance features based on real user feedback.

Vendor or Technology Obsolescence

- **Probability:** Low
- **Impact:** Medium
- **Mitigation Strategy:** The technology stack has been selected based on current industry standards, strong community support, and long-term viability. A clear technology roadmap will be established at the project's outset.
- **Contingency Plan:** The modular architecture of the system ensures that specific technologies can be replaced with minimal impact on the overall system. A technology refresh plan will be part of the long-term maintenance strategy.

Budget Overruns

- **Probability:** Medium
- **Impact:** High
- **Mitigation Strategy:** A detailed budget with a 15% contingency allocation has been established. Financial tracking against the budget will be a standing item in weekly progress meetings. The formal change control process will directly address the financial impact of any scope changes.
- **Contingency Plan:** If overruns occur, non-essential features will be descope in consultation with the client, with a focus on delivering the Minimum Viable Product (MVP) as defined in the SRS.

This proactive risk management approach, with clear ownership assigned to relevant team leads, will be integral to guiding the project to a successful and timely

completion, ensuring that potential issues are identified early and addressed effectively.

3.9 Quality Assurance

Quality is built into every phase of the project, not just tested at the end.

- **Process Standards:** Adherence to ISO/IEC 25010 standards for system and software quality models.
- **Quality Metrics:**
 - **Code Quality:** >80% unit test coverage; zero critical/open bugs at release.
 - **Performance:** Page load times <3 seconds; system supports 50+ concurrent users without degradation.
 - **Usability:** Task success rate of >95% in UAT for core workflows (Registration, Job Application).
 - **Security:** Zero critical/high vulnerabilities in OWASP ZAP scan.
- **Code Quality:** Mandatory peer code reviews for all pull requests. Automated linting and formatting checks in the CI pipeline. SonarQube code analysis.
- **Testing Rigor:** As defined in Section 3.6.3. A "Definition of Done" for every user story includes code written, unit tests passed, code reviewed, integrated successfully, and documented.
- **Usability Testing:** Formal sessions with at least 5 representative users per user group (students, employers) to gather feedback on navigation, clarity, and task completion rates

3.10 Additional plans

3.10.1 Training Plan:

AfriNova Systems is committed to ensuring that all stakeholders can effectively use and maintain the UFS Recruitment System. This training plan will be executed **during Phase 2 (Implementation)** to prepare users for the system launch.

Training Needs Analysis

Pre-Training Activities (Week 17-18 of Implementation):

1. User Profiling:

- Conduct surveys to assess current technical literacy levels
- Identify specific training needs per user group
- Determine preferred learning formats (video, hands-on, written guides)

2. Training Environment Setup:

- Configure dedicated training server (staging environment)
- Create test accounts for each user type
- Populate with sample data for realistic scenarios

3. Training Materials Development:

- Video tutorials (5-15 minutes each)
- User manuals (PDF format, 30-50 pages per user type)
- Quick reference guides (2-page cheat sheets)
- Interactive help tooltips (embedded in application)

3.10.1.1. End-User Training:

3.10.1.1.1 Student User Training:

Target Audience: UFS students who will use the platform to create profiles, search jobs, and apply for opportunities.

Training Objectives:

- Navigate the student dashboard confidently
- Create and optimize professional profiles
- Utilize the CV Builder tool effectively
- Search, filter, and apply for job opportunities
- Manage interview schedules
- Use the SkillStack Portfolio to showcase projects
- Understand privacy controls for academic data sharing

Training Modules:

Module	Duration	Delivery Method	Content Covered
1. Getting Started	10 min	Video + Written	Account activation, first login, password change
2. Profile Setup	15 min	Video + Hands-on	Academic info import, skills section, photo upload
3. CV Builder Masterclass	20 min	Interactive	Template selection, data autofill, PDF export
4. SkillStack Portfolio	15 min	Video + Written	Project uploads, achievement timelines, media management
5. Job Search & Applications	20 min	Video + Hands-on	Advanced filters, bookmarking jobs, one-click apply
6. Interview Management	10 min	Video + Written	Viewing invitations, proposing times, calendar sync
7. Privacy & Security	10 min	Video + Written	Data sharing controls, academic record requests

Training Delivery:

- **Self-Paced Learning:** All modules available on-demand through the platform's "Help Center"
- **Live Webinars:** Two optional 1-hour sessions during orientation week
 - Session 1: Profile setup and CV building
 - Session 2: Job search strategies and application tips
- **In-App Onboarding:** Guided tooltips appear on first login (using libraries like Intro.js)
- **Support Resources:** FAQ section, searchable knowledge base, support chatbot

Success Metrics:

- 80% of students complete profile setup within first week
- 90% successfully submit at least one job application within first month
- Average profile completeness score > 75%

Timeline:

- Week 20: Training materials finalized
- Week 21: Pre-launch webinar promotion begins
- Week 22: Launch week - webinars conducted, students onboard

3.10.1.1.2 Employer User Training:

Target Audience: Verified company representatives who will post jobs and search for candidates.

Training Objectives:

- Create compelling company profiles
- Post job opportunities with best practices
- Utilize advanced candidate search effectively
- Manage applications efficiently
- Schedule and coordinate interviews
- Request and review academic records appropriately

Training Modules:

Module	Duration	Delivery Method	Content Covered
1. Employer Onboarding	15 min	Personalized video call	System overview, account setup, verification process
2. Company Profile Excellence	20 min	Video + Written	Profile creation, Culture Compass showcase, branding
3. Job Posting Best Practices	25 min	Interactive tutorial	Rich text editor, job descriptions, preview, publishing
4. Candidate Search Strategies	30 min	Video + Hands-on	Search filters, anonymized profiles, skill matching
5. Application Management	15 min	Video + Written	Dashboard navigation, candidate tracking, shortlisting
6. Interview Coordination	15 min	Video + Written	Scheduling invitations, calendar integration, reminders
7. Academic Record Access	10 min	Video + Written	Request process, student consent, compliance

Training Delivery:

- **Personalized Onboarding:** Each employer receives a 1-hour virtual orientation session via MS Teams
 - Conducted by AfriNova project team member
 - Customized to employer's specific needs
 - Q&A session included
- **Comprehensive Employer Manual:** 40-page PDF guide with screenshots
- **Video Library:** All modules available on-demand

- **Dedicated Support Channel:** Email (employers@ufs-recruitment.co.za) with 4-hour response time

Success Metrics:

- 100% of employers complete onboarding within 5 business days of approval
- 85% post first job within 2 weeks of onboarding
- Average job posts quality score > 4/5 (based on completeness)

Timeline:

- Week 20: Employer manual finalized
- Week 21: Pilot onboarding with 3 partnered companies
- Week 22+: Rolling onboarding for new employer registrations

3.10.1.1.3 Administrator Training:

Target Audience: UFS Career Services staff responsible for system oversight, user management, and analytics.

Training Objectives:

- Perform comprehensive user management tasks
- Verify and approve employer registrations
- Monitor system logs and performance
- Generate and interpret analytics reports
- Handle escalated support issues
- Send broadcast notifications effectively

Training Program:

Day 1: System Administration Fundamentals (8 hours - On-site at UFS or Virtual)

Session	Duration	Content
Morning Session (3 hours)		
- System Architecture Overview	45 min	Backend structure, database schema, security model
- Admin Dashboard Navigation	30 min	Menu structure, quick actions, customization
- User Management Deep Dive	90 min	Creating accounts, role assignments, deactivation, password resets
- Break	15 min	
Afternoon Session (4 hours)		
- Employer Verification Workflow	60 min	Document review, approval criteria, rejection handling

- Analytics Dashboard Mastery	90 min	Report generation, filters, data interpretation, export
- System Logs & Monitoring	45 min	Error logs, user activity tracking, security alerts
- Break	15 min	
- Support Escalation Procedures	30 min	Ticket management, issue categorization, contact protocols

Day 2: Hands-On Practice & Advanced Features (4 hours - Virtual)

Session	Duration	Content
Morning Session (2 hours)		
- Practice Exercises in Staging	90 min	Simulated scenarios: user issues, employer verification, report generation
- Broadcast Notifications	30 min	Creating announcements, targeting user groups, scheduling
Afternoon Session (2 hours)		
- Troubleshooting Common Issues	60 min	Login problems, data sync errors, missing profiles
- Q&A and Certification	60 min	Open forum, best practices, completion certificate

Training Materials Provided:

- **Administrator Manual:** 60-page comprehensive guide
 - Quick reference sections
 - Step-by-step procedures with screenshots
 - Troubleshooting flowcharts
 - Emergency contact information
- **Practice Scenarios Document:** 20 realistic cases to solve
- **Video Recordings:** All training sessions recorded for future reference
- **Direct Support Line:** Dedicated WhatsApp group with AfriNova team

Success Metrics:

- 100% of admins pass certification assessment (80% score required)
- Admins independently resolve 90% of Tier-1 support issues
- Average employer verification turnaround < 24 hours

Timeline:

- Week 19: Administrator manual finalized
- Week 21: Day 1 training conducted
- Week 22: Day 2 training + certification

3.10.1.2 Technical Support Training:

3.10.1.2.1 UFS IT Support Staff Training:

Target Audience: UFS IT department staff who will provide first-line technical support and system maintenance.

Training Objectives:

- Understand system architecture and infrastructure
- Troubleshoot common technical issues independently
- Perform routine maintenance tasks (backups, updates)
- Monitor system health and performance
- Escalate complex issues appropriately to AfriNova

Training Program Structure:

Session 1: Technical Architecture (Half-day - 4 hours)

Topic	Duration	Content
System Overview	30 min	Frontend (React), Backend (Node.js), Database (SQL Server)
Server Infrastructure	45 min	VPS configuration, Nginx setup, SSL certificates
Database Architecture	60 min	Schema walkthrough, relationships, backup procedures
API Documentation	45 min	Endpoint overview, authentication flow, rate limiting
Break	15 min	
Security Configuration	45 min	Firewall rules, encryption, POPIA compliance measures

Session 2: Operations & Troubleshooting (Half-day - 4 hours)

Topic	Duration	Content
System Monitoring Tools	45 min	Server logs, application logs, error tracking (Sentry)
Common Issues & Fixes	90 min	Login failures, slow performance, database connection errors
Backup & Recovery	45 min	Automated backup verification, manual backup procedures, restoration testing

Break	15 min	
Escalation Protocols	30 min	Issue categorization, SLA response times, contact procedures
Hands-On Lab	45 min	Simulated technical issues to resolve in test environment

Training Materials Provided:

- **Technical Documentation Package:**
 - System architecture diagrams (network, application layers)
 - Database schema and ER diagrams
 - API documentation (Swagger/OpenAPI format)
 - Server configuration files (commented)
 - Troubleshooting decision trees
- **Runbook:** 40-page operational guide
 - Startup/shutdown procedures
 - Backup verification checklist
 - Common error codes and resolutions
 - Emergency contact information
- **Access to Test Environment:** Replica of production for practice

Success Metrics:

- IT staff independently resolve 85% of technical issues
- Average response time to critical issues < 2 hours
- Zero unplanned downtime due to preventable maintenance issues

Timeline:

- Week 20: Technical documentation finalized
- Week 21: Session 1 conducted
- Week 22: Session 2 conducted + knowledge transfer

3.10.1.3 Training Materials Development Timeline

Week 17-18 (During Implementation Phase):

Deliverable	Status
Student video tutorials (7 modules)	To be created
Student user manual (30 pages)	To be created
Employer video tutorials (7 modules)	To be created
Employer manual (40 pages)	To be created
Administrator manual (60 pages)	To be created
Technical documentation (40 pages)	To be created

Quick reference guides (8 cheat sheets)	To be created
In-app tooltips (50+ elements)	To be created
FAQ knowledge base (100+ articles)	To be created

Week 19-20 (Testing Phase):

- Beta user testing of training materials
- Revisions based on feedback
- Final proofreading and formatting

Week 21-22 (Deployment):

- Training delivery (as per schedules above)
- Real-time adjustments based on participant feedback

3.10.1.4 Post-Launch Continuous Training

Ongoing Training Support:

- **Monthly Webinars:** Advanced tips, new features, Q&A sessions
- **Help Center Updates:** New articles added based on common support queries
- **In-App Announcements:** Tips and tricks pushed via notification system
- **Refresher Training:** Quarterly sessions for administrators (optional)
- **Version Update Training:** Whenever major features are released

Feedback Mechanism:

- Training effectiveness surveys (after each session)
- User satisfaction metrics (monthly)
- Support ticket analysis (identify training gaps)

3.10.2 Comprehensive Post-Deployment Maintenance Strategy

AfriNova Systems commits to a robust maintenance framework ensuring the UFS Recruitment System remains operational, secure, and aligned with evolving user needs.

3.10.2.1 Corrective Maintenance

Objective: Rapidly identify and resolve defects, bugs, or errors discovered in the production environment.

Bug Severity Classification:

Severity	Definition	Response Time	Resolution Target	Examples
Critical	System unavailable, data loss, security breach	1 hour	4 hours	Database connection failure, authentication bypass
High	Major feature broken, affects many users	4 hours	24 hours	Job application submission failing, CV builder not generating PDFs
Medium	Minor feature issue, workaround available	1 business day	5 business days	Search filters not working correctly, email notifications delayed
Low	Cosmetic issues, minor inconvenience	2 business days	Next release cycle	Button misalignment, spelling errors

Bug Resolution Workflow:

User Reports Bug (via Support Chatbot/Email/Phone)



Support Team Logs in Jira (includes description, steps to reproduce, user impact)



QA Lead Assigns Severity Level



Developer Assigned Based on Module Expertise



Fix Implemented in Development Branch



Code Review (Minimum 1 Peer Review)



Testing in Staging Environment



Client Approval (for High/Critical bugs)



Deployment to Production (with rollback plan)



Verification & User Notification



Bug Status: CLOSED

Monitoring Tools:

- **Error Tracking:** Sentry (real-time error monitoring)
- **Performance Monitoring:** New Relic or Azure Monitor
- **User-Reported Issues:** Integrated support chatbot + ticketing system (Freshdesk or Zendesk)

24/7 Critical Issue Response:

- On-call rotation: Each team member takes 1-week on-call duty
- Response SLA: 1 hour acknowledgment, 4 hours resolution for Critical bugs
- Escalation path: Developer → Lead Developer (B. Kasumba) → Project Manager (L. Malaza) → Client

Monthly Bug Report:

- Total bugs reported
- Bugs resolved by severity
- Average resolution time per severity level
- Top 5 recurring issues (for proactive fixes)

3.10.2.2 Adaptive Maintenance

Objective: Ensure the system remains compatible with external changes (OS updates, browser versions, regulatory changes, third-party API updates).

Monitoring & Response Strategy:

1. Operating System & Browser Compatibility:

Update Type	Monitoring Frequency	Action Plan
Browser updates (Chrome, Firefox, Safari, Edge)	Quarterly	Test system on beta versions before public release, update polyfills/CSS if needed
Windows Server updates	Monthly	Review update notes, test in staging, schedule maintenance window
Mobile OS updates (Android, iOS)	Upon release	Test responsive design, update app if developed

2. POPI Act Compliance:

- **Annual Review:** Legal consultant reviews system compliance
- **Audit Trail:** All data access logged and retained for 5 years
- **Consent Management:** Periodic review of consent forms and user opt-in flows
- **Data Breach Response:** Documented 72-hour notification protocol to Information Regulator

3. UFS Database API Changes:

- **Quarterly Sync:** Meet with UFS IT to review any upcoming API changes
- **Versioning Strategy:** Support previous API version for 6 months after new version release
- **Fallback Mechanism:** CSV import always available if API temporarily unavailable

4. Third-Party Service Updates:

Service	Impact Area	Monitoring	Adaptation Strategy
Email SMTP (Gmail/SendGrid)	Notifications	API changelog subscription	Update authentication if protocols change
SMS Gateway (Twilio)	Announcements, verification	Version monitoring	Test new endpoints in staging before migration

Cloud Hosting (DigitalOcean/AWS)	Infrastructure	Service health dashboard	Schedule migrations during low-traffic periods
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Adaptive Maintenance Timeline:

- **Quarterly:** Browser compatibility testing
- **Bi-annually:** Security certificate renewal
- **Annually:** POPIA compliance audit, third-party service contract reviews

Examples of Adaptive Changes:

- Updating system to support new POPI amendments (e.g., extended data subject rights)
- Ensuring compatibility with Chrome 120+ (when released)
- Migrating to new UFS API v2.0 when UFS IT upgrades their systems
- Supporting new authentication methods (e.g., UFS single sign-on if implemented)

3.10.2.3 Perfective Maintenance

Objective: Improve system performance, usability, and functionality based on user feedback and analytics insights.

User Feedback Collection Mechanisms:

1. **In-App Surveys:**
 - Pop-up surveys after key actions (e.g., "How was your job application experience?")
 - Net Promoter Score (NPS) surveys quarterly
 - Target: 500+ responses per quarter
2. **Support Ticket Analysis:**
 - Monthly review of top 10 support issues
 - Identify patterns indicating usability problems
 - Feature requests logged and prioritized
3. **Analytics Dashboard Insights:**
 - Track user behavior: time spent on pages, drop-off points, feature usage
 - Identify underutilized features (e.g., if <20% use CV Builder, investigate why)
 - A/B testing for UI improvements (e.g., button placements, form layouts)

4. Stakeholder Feedback Sessions:

- Quarterly meetings with UFS Career Services
- Semi-annual employer focus groups
- Annual student surveys

Enhancement Prioritization Framework:

Priority	Criteria	Examples	Release Cycle
High	Requested by >50% users, high business impact	Add new CV templates, improve search algorithm	Next quarterly release
Medium	Quality-of-life improvements, moderate impact	Add dark mode, keyboard shortcuts	Within 6 months
Low	Nice-to-have features, minimal impact	New dashboard widget colors	Backlog (as capacity allows)

Enhancement Request Workflow:

User Submits Request (via Support/Surveys)



Request Logged in Jira (tagged: "Enhancement")



Project Manager Reviews (L. Malaza)



Feasibility Analysis (Technical team estimates effort)



Client Prioritization Meeting (if significant cost/time)



Approved Enhancements Added to Development Backlog



Scheduled for Quarterly Release



Development → Testing → Deployment



User Notification (via Announcements)

Quarterly Release Cycle:

- **Q1 (Jan-Mar):** Minor enhancements, bug fixes, UI polish
- **Q2 (Apr-Jun):** Medium enhancements (e.g., new features based on feedback)
- **Q3 (Jul-Sep):** Performance optimization, major enhancements
- **Q4 (Oct-Dec):** Year-end features, preparing for new academic year

Examples of Perfective Changes:

- **Based on Employer Feedback:** "Add bulk job posting feature" (requested by 15 employers)
- **Based on Student Analytics:** "Improve job search algorithm - students spend 10+ minutes filtering, can we make it faster?"
- **Based on Support Tickets:** "Add 'Save Draft' for job applications" (65 requests)
- **Performance:** "Database queries taking >3 seconds on Analytics page - optimize with indexing"
- **UI/UX:** "Mobile users struggle with CV upload - redesign for better mobile experience"

3.10.2.4 Preventive Maintenance

Objective: Prevent future problems through proactive monitoring, optimization, and maintenance.

Scheduled Preventive Activities:

Daily:

- **Automated System Health Checks:**
 - Server uptime monitoring (Pingdom or UptimeRobot)
 - Database connection tests
 - API endpoint availability checks
- **Log Monitoring:**
 - Review error logs for anomalies
 - Check for unusual spikes in failed login attempts (potential security threat)
- **Backup Verification:**

- Automated database backup runs at 2:00 AM daily
- Verification script confirms backup integrity

Weekly:

- **Support Ticket Review:**
 - Team meeting to discuss trends in user issues
 - Identify recurring problems for proactive fixes
- **Performance Metrics Review:**
 - Page load times (target: <3 seconds)
 - API response times (target: <500ms)
 - Database query performance
- **Security Scan:**
 - Automated vulnerability scan (OWASP ZAP or similar)
 - Review of suspicious activity logs

Monthly:

- **Database Optimization:**
 - Index rebuilding to maintain query performance
 - Archive old data (e.g., job posts >2 years old moved to archive table)
 - Analyse slow queries and optimize
- **Backup Restoration Test:**
 - Full database restores to test environment
 - Verify data integrity and application functionality
- **Dependency Updates:**
 - Review npm packages for security updates
 - Test updates in staging before production deployment
- **Capacity Planning:**
 - Review server resource usage (CPU, RAM, disk space)
 - Forecast growth and plan for scaling if needed

Quarterly:

- **Comprehensive Security Audit:**
 - Penetration testing by independent security consultant
 - Review of user access logs for anomalies
 - Update security policies and procedures
- **Performance Benchmarking:**
 - Load testing with simulated 200+ concurrent users
 - Identify bottlenecks and optimize
- **Code Quality Review:**
 - Static code analysis (SonarQube)
 - Refactor technical debt areas
- **Disaster Recovery Drill:**
 - Simulate server failure
 - Test backup restoration procedure
 - Measure Recovery Time Objective (RTO) - Target: 4 hours

Annually:

- **Technology Stack Review:**
 - Assess if technologies are still optimal
 - Plan major upgrades (e.g., Node.js version, SQL Server version)
- **Comprehensive System Review:**
 - Architecture assessment
 - Scalability planning for next 2-3 years
- **Infrastructure Assessment:**
 - Server performance review
 - Cost optimization (e.g., migrate to better hosting if needed)

Preventive Maintenance Calendar:

Activity	Daily	Weekly	Monthly	Quarterly	Annually
System health checks	✓				
Log monitoring	✓				
Backup verification	✓				
Support ticket review		✓			
Performance metrics		✓			
Security scan		✓			
Database optimization			✓		
Backup restoration test			✓		
Dependency updates			✓		
Security audit				✓	
Performance benchmarking				✓	
Disaster recovery drill				✓	
Technology review					✓

3.10.2.5 Maintenance Schedule Matrix

Comprehensive Maintenance Activity Schedule:

Maintenance Type	Activity	Frequency	Estimated Time
Corrective	Bug fixing	As reported	Varies by severity
Corrective	Emergency hotfixes	As needed	1-4 hours
Adaptive	Browser compatibility testing	Quarterly	4 hours
Adaptive	POPI compliance review	Annually	8 hours
Adaptive	Third-party API updates	As released	2-8 hours
Perfective	Feature enhancements	Quarterly	40-80 hours
Perfective	UI/UX improvements	Bi-annually	20-40 hours
Perfective	Performance optimization	Quarterly	8-16 hours
Preventive	Database maintenance	Monthly	2 hours
Preventive	Security audits	Quarterly	8 hours
Preventive	Backup testing	Monthly	2 hours
Preventive	Disaster recovery drill	Quarterly	4 hours

3.10.2.6 Support Structure During Maintenance Period

Tiered Support Model:

Tier 1: UFS IT Support (Basic User Assistance)

- **Handles:**
 - Password resets
 - Account lockout issues
 - Basic navigation questions
 - File upload problems
- **Tools:** Admin panel access for password resets
- **Response Time:** Within 2 hours during business hours (8 AM - 5 PM)
- **Escalation:** To Tier 2 if issue requires code-level investigation

Tier 2: AfriNova Technical Support

- **Handles:**
 - System bugs
 - Feature malfunctions
 - Database issues
 - Integration problems (e.g., UFS API not syncing)
- **Tools:** Access to logs, database, staging environment
- **Response Time:**
 - Critical: 1 hour
 - High: 4 hours
 - Medium: 1 business day
- **Team:** Rotating on-call developer
- **Escalation:** To Tier 3 for architectural issues

Tier 3: AfriNova Development Team (Complex Issues)

- **Handles:**
 - System architecture modifications
 - Major security incidents
 - Performance optimization
 - Customization requests
- **Team:** Project Manager + Lead Developer + relevant specialists
- **Response Time:** Within 24 hours for assessment, timeline provided based on complexity

Support Channels:

Channel	Purpose	Availability	Response Time
In-App Support Chatbot	Instant answers to FAQs	24/7	Immediate (automated)
Email: support@ufs-recruitment.co.za	General inquiries, bug reports	Business hours	4 hours
Phone Hotline	Critical issues only	Business hours	1 hour
Support Ticket System	Structured issue tracking	24/7 submission	Based on severity
Admin Dashboard	For UFS administrators	24/7	Self-service

Response Time Commitments (SLA):

Severity	Acknowledgment	Resolution Target	Example
Critical	1 hour	4 hours	System down, no users can login
High	4 hours	24 hours	Job applications not submitting

Medium	1 business day	5 business days	Search results not accurate
Low	2 business days	Next release	Button color preference

Emergency After-Hours Protocol:

- **Critical Issues Only:** System completely unavailable, data breach
- **Contact:** Emergency hotline (number provided to client)
- **Response:** On-call developer responds within 1 hour
- **Escalation:** If unable to resolve in 2 hours, full team mobilized

3.10.3 Security Plan

The security of the UFS Recruitment System is paramount given the sensitivity of student academic records and employer information. AfriNova Systems will implement comprehensive security measures across all system layers.

3.10.3.1 Security Measures:

3.10.3.1.1 Authentication and Access Control:

- Role-Based Access Control (RBAC) for all user types
- Secure password policies (minimum 12 characters, complexity requirements)
- Account lockout after 3 failed login attempts
- JWT (JSON Web Tokens) for stateless, secure session management
- Bcrypt password hashing (industry-standard encryption)
- Multi-factor authentication capability for administrator accounts
- Automatic session timeout after 30 minutes of inactivity

3.10.3.1.2 Data Protection:

- End-to-end encryption for all sensitive data transmission (TLS 1.3)
- Encryption at rest for database storage (AES-256)
- Anonymization of student profiles in initial candidate searches
- Privacy-controlled sharing of academic records (student consent required)
- Principle of least privilege for all data access
- Secure file upload validation (type, size, content scanning)
- Regular data backups with encrypted storage

3.10.3.1.3 Application Security:

- Input validation and sanitization to prevent injection attacks
- Protection against OWASP Top 10 vulnerabilities:
- SQL Injection prevention (parameterized queries)
- Cross-Site Scripting (XSS) protection
- Cross-Site Request Forgery (CSRF) tokens
- Security misconfiguration hardening
- Broken authentication prevention
- API rate limiting to prevent abuse
- Secure file upload mechanisms with malware scanning
- Content Security Policy (CSP) headers
- HTTP security headers (HSTS, X-Frame-Options, etc.)

3.10.3.1.4 Infrastructure Security:

- Firewall configuration and network segmentation
- Regular security patches and updates
- Intrusion Detection System (IDS) monitoring
- DDoS protection (if using cloud hosting)
- Secure server configuration (hardened OS)
- VPN access for administrative functions
- Regular vulnerability scanning

3.10.3.2 POPIA Compliance

The system will adhere to South Africa's Protection of Personal Information Act (POPIA) requirements:

3.10.3.2.1 Data Minimization:

- Collect only information necessary for system functionality
- Students control visibility of their academic information
- Clear data retention policies and procedures

3.10.3.2.2 Consent Management:

- Explicit consent for academic data import from UFS database
- Employers must request (and receive approval for) access to academic records
- Users can withdraw consent and request data deletion

3.10.3.2.3 Right to Access and Correction:

- Students can view and edit all their profile information
- Users can download their data (data portability)
- Clear procedures for data correction requests

3.10.3.2.4 Security Safeguards:

- Technical measures as described in Section 3.103.1
- Organizational measures (staff training, access policies)
- Incident response plan for data breaches

3.10.3.2.4 Breach Notification:

- Immediate incident response activation
- Notification to Information Regulator within 72 hours (as required)
- Transparent communication with affected users
- Legal consultation and remediation

3.10.3.3 Security Audits and Testing

3.10.3.3.1 Pre-Deployment Security Audit:

- Comprehensive security assessment before go-live
- OWASP ZAP security scanning
- Penetration testing by independent security consultant
- Vulnerability assessment report
- Remediation of all high and critical vulnerabilities

3.10.3.3.2 Ongoing Security Monitoring:

- Continuous automated security scanning
- Log monitoring for suspicious activities
- Regular manual security reviews
- Quarterly penetration testing
- Annual comprehensive security audit

3.10.3.3.3 Security Incident Response Plan:

1. Detection and identification of security incident
2. Containment (isolate affected systems)
3. Eradication (remove threat, patch vulnerabilities)
4. Recovery (restore systems to normal operation)
5. Post-incident analysis (lessons learned)

6. Notification (authorities, affected users as required)

3.10.3.4 Team Security Training

All development team members will undergo:

- POPIA compliance training
- Secure coding practices workshops
- OWASP Top 10 awareness training
- Regular security awareness updates

3.10.4 Installation Plan

The deployment of the UFS Recruitment System will follow a structured, phased approach to minimize risks and ensure a smooth transition to production.

3.10.4.1 Pre-Deployment Phase (Staging - Week 15)

1. Environment Setup:

- Provision and configure staging server environment
- Mirror production specifications:
 - Operating System (Ubuntu 22.04 LTS or Windows Server)
 - Web Server (Nginx or IIS)
 - Database Server (Microsoft SQL Server 2022)
 - Node.js runtime environment
- Configure network settings, firewall rules, and security groups
- Set up SSL/TLS certificates for HTTPS
- Verify all required ports are accessible

2. Database Migration:

- Export finalized database schema from development environment
- Create migration scripts for:
 - Table creation
 - Index generation
 - Stored procedure deployment
 - Initial data seeding (faculties, departments, system settings)
- Test migration scripts in staging environment
- Verify database integrity and referential constraints

- Create database backup and restoration procedures
- Document rollback procedures

3. Application Deployment to Staging:

- Build release candidate from main branch in GitHub
- Deploy backend API services
- Deploy frontend application
- Configure environment variables and secrets
- Set up automated CI/CD pipeline (GitHub Actions or Jenkins)
- Verify all third-party integrations (UFS Database API)
- Test API endpoints using Postman

4. Staging Testing:

- Execute full smoke test suite
- Run critical end-to-end test scenarios
- Perform load testing (simulate 100+ concurrent users)
- Security scan with OWASP ZAP
- Conduct final User Acceptance Testing (UAT) with selected users
- Document and resolve any identified issues
- Obtain final UAT sign-off from client

3.104.2 Production Deployment (Week 16)

1. Final Client Sign-Off:

- Present UAT results to Dr. R. Fouché
- Demonstrate successful completion of all acceptance criteria
- Obtain formal written approval to proceed with production deployment
- Confirm deployment date and time (preferably weekend or low-traffic period)

2. Pre-Deployment Checklist:

- ☐ All UAT issues resolved and verified
- ☐ Production server environment prepared and tested
- ☐ SSL certificates installed and validated
- ☐ Database backup strategy implemented

- ☐ Monitoring tools configured (application performance, error logging)
- ☐ Support team briefed and on standby
- ☐ Rollback plan documented and tested
- ☐ Communication plan for users prepared

3. Production Deployment Execution:

- Schedule deployment during low-usage window (e.g., Saturday 02:00 AM)
- Deploy using blue-green deployment strategy:
 - Maintain current system (blue) as fallback
 - Deploy new system to separate environment (green)
 - Perform final verification tests on green environment
 - Switch traffic from blue to green via load balancer/DNS
 - Monitor closely for 2 hours
 - If issues arise, instant rollback to blue environment

Detailed Steps:

- a) Place maintenance notice on old system (if applicable)
- b) Deploy database migrations to production database
- c) Verify database migration success
- d) Deploy backend application services
- e) Deploy frontend application
- f) Configure environment-specific settings
- g) Verify API connectivity and database connections
- h) Switch DNS/load balancer to point to new system
- i) Remove maintenance notice

4. Post-Deployment Verification:

- Execute rapid smoke test on live production site:
 - Student login and profile creation
 - Employer login and job posting
 - Job search and application process
 - Administrator dashboard access
 - CV Builder functionality
 - Academic data integration

- Verify all integrations with UFS Database API
- Check SSL certificate validity
- Test email notification delivery
- Confirm file upload/download functionality

5. Go-Live Monitoring (First 48 Hours):

- Real-time application performance monitoring (New Relic or Azure Monitor)
- Error log monitoring (Sentry or custom logging)
- Database performance tracking
- Server resource utilization (CPU, memory, disk)
- Network traffic analysis
- User feedback collection
- Support ticket monitoring
- AfriNova technical team on call for immediate response

6. Rollback Plan (If Critical Issues Arise):

If critical issues are detected that cannot be resolved quickly:

- a) Immediately switch DNS/load balancer back to blue (old) environment
- b) Notify all stakeholders of rollback
- c) Analyse root cause of failure
- d) Implement fixes in development environment
- e) Re-test thoroughly in staging
- f) Schedule new deployment window

3.10.4.3 Post-Deployment Phase (Week 17+)

1. User Notification and Communication:

- Announce system launch via multiple channels:
 - UFS official email to all students
 - Announcement on UFS student portal
 - Social media posts (UFS Facebook, Twitter, Instagram)
 - UFS Career Services office notifications
 - Posters on campus bulletin boards
- Provide links to user guides and video tutorials
- Highlight key features and benefits

- Communicate support channels

2. Support Handover to UFS:

- Conduct knowledge transfer session with UFS IT staff
- Provide comprehensive documentation:
 - System architecture diagrams
 - Database schema and ER diagrams
 - API documentation
 - Troubleshooting guide
 - Common issues and solutions
 - Administrator manual
- Establish support contact hierarchy:
 - Tier 1: UFS IT Support (basic issues)
 - Tier 2: AfriNova Support Email (technical issues)
 - Tier 3: AfriNova Development Team (critical system failures)
- Set up communication channels (email, WhatsApp group, emergency phone)

3. Monitoring and Optimization (First Month):

- Daily review of system logs and error reports
- Weekly performance analysis and optimization
- Collection and analysis of user feedback
- Identification of usability issues
- Minor bug fixes and patches as needed
- Continuous improvement based on real-world usage patterns

3.10.5 Post-Delivery Maintenance Plan

A comprehensive 6-month post-deployment support and maintenance period is included in the project scope. After this period, ongoing maintenance agreements can be established based on UFS requirements.

3.10.5.1 Initial 6-Month Support Period (Included in Project Cost):

- Scope of Support:
 - Bug fixes and error corrections
 - Performance optimization
 - Security patches and updates
 - User support via email and ticketing system
 - Minor enhancements based on user feedback (subject to effort estimation)
 - System monitoring and maintenance

- Response Time Commitments:
 - Critical issues (system unavailable): 1 hour response, 4-hour resolution target
 - High-priority bugs: 4 hours response, 24-hour resolution target
 - Medium-priority issues: 1 business day response, 5 business day resolution
 - Low-priority issues: 2 business days response, scheduled for future update
- Included Services:
 - Monthly system health reports
 - Quarterly performance reviews
 - Security vulnerability monitoring
 - Database backup monitoring
 - User support ticket resolution
 - Documentation updates as needed

3.10.5.2 Long-Term Maintenance (Post 6-Month Period)

After the initial 6-month support period, AfriNova Systems offers flexible maintenance agreements tailored to UFS needs:

- Option 1: Standard Maintenance Agreement
 - Monthly retainer: R15,000
 - Includes: 10 hours of maintenance/support per month
 - Bug fixes and minor updates
 - Security patches
 - Email/ticket support during business hours
 - Monthly system health reports
- Option 2: Comprehensive Maintenance Agreement
 - Monthly retainer: R35,000
 - Includes: 25 hours of maintenance/support/enhancements per month
 - All services in Standard plan
 - Priority support (faster response times)
 - Quarterly feature enhancements
 - Proactive system optimization
 - Extended support hours (8AM-8PM weekdays)
- Option 3: Pay-Per-Service Model
 - No monthly retainer
 - Services billed at R500/hour
 - Suitable for minimal maintenance needs
 - Response times not guaranteed

- Minimum 4-hour billing increments
- Additional Services (Available in All Plans):
 - Major feature development (quoted separately)
 - System upgrades and migrations (quoted separately)
 - Custom integration development (quoted separately)
 - On-site training sessions (R8,000 per day)
 - Emergency support outside business hours (R1,500/hour)

3.10.5.3 Maintenance Agreement Terms

- Service Level Agreement (SLA):
 - Uptime guarantee: 99.5% (excluding scheduled maintenance)
 - Scheduled maintenance windows: Last Sunday of each month, 02:00-06:00 AM
 - Advance notice: 7 days for scheduled maintenance
- Backup and Disaster Recovery:
 - Automated daily incremental backups
 - Weekly full system backups
 - Off-site backup storage (30-day retention)
 - Quarterly disaster recovery drills
 - Recovery Time Objective (RTO): 4 hours
 - Recovery Point Objective (RPO): 24 hours
- Performance Monitoring:
 - 24/7 automated system monitoring
 - Real-time alerts for critical issues
 - Monthly performance reports
 - Capacity planning recommendations
- Communication and Reporting:
 - Monthly maintenance reports
 - Quarterly business reviews
 - Annual system health assessment
 - Dedicated account manager
 - Regular check-in meetings

3.10.5.4 Transition to Maintenance Phase

At the end of the 16-week development period:

1. Final project deliverables handed over
2. All documentation transferred to UFS
3. Support responsibility transitions to maintenance model
4. Support email (ufs-recruitment-support@afriNova.co.za) becomes active

5. Maintenance SLA takes effect
6. Monthly reporting begins

The maintenance agreement ensures the UFS Recruitment System remains secure, performant, and aligned with evolving university needs long after the initial deployment.

3.11 Comprehensive Project Close-Out Strategy

The Project Close-Out Plan outlines the steps to be taken once the UFS Recruitment System project is completed, ensuring a smooth transition and proper closure of all project activities.

3.11.1 Pre-Closure Activities (Week 21-22)

1. Deliverable Verification:

Checklist of All Deliverables:

Category	Deliverable	Status	Sign-Off Required
Documentation			
	Software Requirements Specification (SRS)	✓ Delivered	Client
	Software Project Management Plan (SPMP)	✓ Delivered	Client
	System Design Document	✓ Delivered	Client
	User Manuals (Student, Employer, Admin)	In Progress	Client
	Technical Documentation	In Progress	UFS IT
	API Documentation	In Progress	UFS IT
Software			
	Production-ready system (deployed)	Planned	Client
	Source code repository (GitHub)	Planned	Client
	Database schema and scripts	Planned	UFS IT
Testing			
	Unit test suite	Planned	QA Lead
	Integration test results	Planned	QA Lead
	UAT sign-off document	Planned	Client
	Security audit report	Planned	Client
Training			
	Training materials (videos, manuals)	Planned	Client
	Training session completion certificates	Planned	Client

2. Final Quality Review:

Quality Gates:

- ✓ All acceptance criteria met (from SRS)
- ✓ Zero Critical or High-severity bugs in production
- ✓ Performance benchmarks achieved (page load <3s, 50+ concurrent users)
- ✓ Security audit passed (no Critical/High vulnerabilities)
- ✓ UAT sign-off received (95% task success rate)

- ☒ Accessibility compliance (WCAG 2.1 AA)
- ☒ All documentation reviewed and approved

Final Quality Report:

- Test coverage: >80% (target achieved)
- Defect density: <2 defects per 1000 lines of code
- System uptime during test period: >99.5%
- User satisfaction (UAT): Average SUS score 75/100

3.11.2. Final Project Review and Sign-Off:

A formal project review will be conducted with the client (Dr. R. Fouché) to ensure all deliverables have been met according to the agreed-upon specifications. Once the client approves the final product, a formal sign-off document will be completed to officially close the project. This sign-off will confirm that:

- All functional and non-functional requirements have been met
- The system has passed User Acceptance Testing (UAT)
- All documentation has been delivered
- Training materials have been provided
- The 6-month post-launch support period has been acknowledged

3.11.3. Archiving of Artifacts:

All project-related documents, including source code, design documents, test results, project management files, and communication logs, will be archived in a secure repository (GitHub and cloud backup storage). This ensures that future teams can access these artifacts if needed for reference, maintenance, or future enhancements. The archived materials will include:

- Complete Software Requirements Specification (SRS)
- System Design Documentation
- Source code repository with version history
- Test plans, test cases, and test results
- User manuals and technical documentation
- Project management plans and schedules
- All correspondence with stakeholders
- Risk management and quality assurance records

3.11.4. Financial Close-Out

A final review of the budget will be conducted, ensuring that all expenses are accounted for and any remaining funds are reconciled. A final budget report will be submitted to stakeholders, detailing:

- Total actual costs versus budgeted costs
- Variance analysis and explanations
- Final invoice and payment confirmation
- Closure of all vendor contracts and subscriptions

3.11.5. Lessons Learned Documentation

The team will document lessons learned during the project, identifying areas for improvement in future projects. This documentation will capture:

- Successes and best practices that worked well
- Challenges encountered and how they were resolved
- Technical difficulties and their solutions
- Process improvements for future projects
- Recommendations for similar future initiatives
- Team feedback on collaboration and tools used

These lessons learned document will be shared with AfriNova Systems management and archived for organizational learning.

3.11.6. Knowledge Transfer:

A comprehensive knowledge transfer session will be conducted with UFS technical staff and administrators to ensure they can maintain and support the system post-deployment. This includes:

- System architecture walkthrough
- Database administration training
- User management procedures
- Troubleshooting common issues
- Access to technical documentation
- Emergency support contacts

3.11.7. Formal Project Closure:

Mngxozana Sesethu will submit a formal project closure report to the client and internal stakeholders, officially marking the end of the project. The closure report will provide:

- Summary of project achievements and outcomes
- Confirmation of deliverable completion

- Final metrics and performance indicators
- Client satisfaction assessment
- Transition to maintenance and support phase
- Formal release of project resources

The project will be officially closed upon receipt of final payment and client acceptance, at which point the 6-month post-launch support and maintenance period will commence. The team adopted an Iterative-and-Incremental Model consistent with IEEE 1058-1998 recommendations for structured project management (IEEE, 1998).

According to Pressman and Maxim (2020), iterative approaches improve adaptability and allow for early user feedback during system development.

Conclusion for Chapter 3

This SPMP defines a robust and professional framework for the successful delivery of the UFS Recruitment System. Through careful planning, a hybrid management approach, a modern technology stack, and a relentless focus on quality and security, AfriNova Systems is committed to delivering a platform that will significantly enhance student employability and employer engagement at the University of the Free State, ready for real-world production use.

Chapter 4: Quotation

Introduction

This chapter presents the formal commercial quotation for the development of the UFS Recruitment System, as prepared by AfriNova Systems for the University of the Free State. The quotation provides a detailed and transparent financial breakdown for the complete project lifecycle, from the initial documentation phase through to the final implementation and deployment of the fully functional platform. The costs detailed herein are derived directly from the resource allocations, timelines, and budgetary guidelines established in the Software Project Management Plan (SPMP). This document serves as a binding financial proposal, outlining the investment required to transform the outlined specifications and design into a secure, scalable, and operational system that will deliver significant value to the university, its students, and employer partners.



Chapter 4: Quotation

Quote Number: 024506

Quote Date: 2025-10-12

BILLED TO

Dr R. Fouché University of the Free State Street address Bloemfontein, Free State, SA 9301
Phone: 0501-401-9318

YOUR COMPANY

AfriNova Systems Suite 5, Silver Oak Business Park 42 Tech Road, Midrand Johannesburg,
Gauteng, SA 1685 Phone: 078-300-8888 Email: quotations@afriNova.co.za Website:
www.afriNova-systems.co.za

Disclaimer: A 25% profit margin has been added to labour and expenses as per the SPMP cost model.

QUOTATION DETAILS

Description	Unit Cost (R)	Qty	Amount (R)
Phase 1 — Documentation Delivery			
Total (Labour + Expenses, excl. VAT)	1,123,250.00	1	1,123,250.00
Phase 2 — Implementation & Deployment			
Total (Labour + Expenses, excl. VAT)	2,332,506.00	1	2,332,506.00
TOTAL PROJECT COST (Excl. VAT)	3,455,756.00	1	3,455,756.00

Subtotal (excl. VAT): R3,455,756.00

VAT (15%): R518,363.40

TOTAL (Incl. VAT): R3,974,119.40

PAYMENT TERMS

- 30% deposit on acceptance (payable within 7 days).
- Balance payable as milestones (detailed schedule to be attached to contract).
- Quotation valid for 60 days from quote date.

NOTES

- This quotation covers both Phase 1 (Documentation) and Phase 2 (Implementation & Deployment) as per the submitted SPMP.
 - Any changes in scope will be quoted separately and will affect timeline and cost.
 - The quoted price includes a 25% profit margin applied to labour and general expenses (as documented in the SPMP).
-

Prepared by: AfriNova Systems

Date: 2025-10-12

Conclusion for Chapter 4

The formal quotation of **R 3,974,119.40** (inclusive of VAT) represents the total investment required to bring the UFS Recruitment System from concept to reality. This comprehensive cost covers both the meticulous planning and design phase and the robust development and deployment phase, ensuring a high-quality, secure, and professionally managed project delivery. The quoted amount is a direct reflection of the scope, complexity, and strategic importance of the system as detailed in the preceding chapters. By accepting this quotation, the University of the Free State is not merely procuring software, but is making a strategic investment in a platform designed to enhance student employability, streamline employer engagement, and provide data-driven insights for years to come. AfriNova Systems is confident that this proposal offers exceptional value and looks forward to partnering with UFS to successfully execute this visionary project.

Chapter 5: Design Documentation

Introduction

The Design Documentation details the architecture and design elements of the UFS Recruitment System. It covers the system's core features, including user roles, profile management, job search and posting, interview scheduling, and analytics. This section provides **class diagrams**, **CRC cards**, **state charts** and **sequence diagrams** to outline system interactions, responsibilities, and key functionalities, focusing on the business logic layer as required.

Noun Extraction

Noun Extraction: Paragraph

Disclaimer:

Nouns are bold

Abstract nouns are underlined

The **UFS Recruitment System** provides a comprehensive collection of features to connect **Students**, **Employers**, and **Administrators**. **Login** enables all **users** to securely access the **platform**, while **Registration** captures **student numbers**, **university emails** and verifies them against the **UFS database**. **Student profiles** contain bios, skills, grades, projects, and privacy settings. The **CV Builder** and **Portfolio (SkillStack)** let **students** assemble **documents** and media for **employers** to review. The **job board** and **job postings** enable **employers** to advertise internships, part-time roles, and graduate positions. An **intelligent matching engine** recommends opportunities based on GPA, skills, and career interests. The **application portal** handles **applications**, status tracking, and optional academic access requests. The **interview scheduler** coordinates **calendar** slots and sends **notifications** and **reminders**. **Administrators** operate the **analytics dashboard** to view *placement rates*, *employer engagement*, and *skill trends*, and manage roles, **verifications**, and access control. Supporting components include **file management**, **chatbot support**, **notifications**, reporting, and secure API links to the **UFS database system**. Privacy, security, and scalability are cross-cutting concerns that govern all modules.

Noun Extraction: Table

Disclaimer:

All nouns listed represent potential concepts from the system description. Nouns identified as **Class Nouns** are candidates for becoming classes in the system design, as they represent core entities with data and behaviour. **Abstract Nouns** represent concepts, attributes, or properties that describe the state or quality of other nouns but do not typically become standalone classes.

Noun extraction		
Class Nouns	Nouns	Abstract Nouns
User	UFS Recruitment System/Platform	Privacy Settings
Student	Student number	Security
Employer	University emails	Bio
Administrator	Documents	Skills
User management	Application portal	Interest
Student Profile		Features
Registration		Project
Login		Media
Job Posting		Internship
Application		Part-time roles
CV Builder		Graduate positions

Skill Stack		Intelligent machine engine
Skill Gap Analysis		Opportunities
Interview Schedule		GPA
Calendar		Career interest
Notification		Status tracking
Dashboard		Academic access request
Database Manager		roles
File Management		Access control
Chatbot		reporting
Job Search		API
Employer Profile		Scalability
Verification/Authentication		
Notification		
Calendar		

CRC Cards

Login: CRC Card

Class Name: Login
RESPONSIBILITY
-Authenticate user credentials (email and password) -Validate password strength and format requirements -Manage user sessions and maintain login state -Implement security measures (account lockout after 3 failed attempts) -Redirect users to role-specific dashboards (Student/Employer/Admin) -Handle "Forgot Password" functionality and password reset -Maintain login audit trails and security logs -Verify user role permissions and access rights -Manage session timeouts and automatic logout -Provide multi-factor authentication options
COLLABORATER
- Student -Employer -Administrator -Authentication Controller -Email Service -Security Service -Session Manager -Logging Service

Student Registration: CRC Card

Class Name: Student Registration
RESPONSIBILITY
- Provide an interface for students to input student number and official UFS email for identity validation. - Verify student credentials against the UFS Database API to confirm active enrolment. - Prompt creation of a secure password and enforce password strength rules with live feedback indicators. - Request student consent before fetching academic data from UFS Database. - Import verified student academic details such as modules completed, GPA, and degree program. - Automatically generate preliminary skill tags (e.g., 'Python', 'Data Analysis') from academic results and allow editing. - Allow students to add profile details: personal bio, career interests, links to GitHub/portfolio projects. - Offer granular privacy settings so students can choose visibility of their academic details to employers. - Handle database errors or missing course data by allowing manual entry and displaying fallback messages. - Confirm successful registration by sending a welcome message and enabling dashboard access.
COLLABORATER
- Student - UFS Database - Profile Management

Employer Registration: CRC Card

Class Name: Employer Registration
RESPONSIBILITY
- Allow employers to initiate registration by submitting company details (legal name, address, website). - Capture contact person details including name, designation, email, and phone number for correspondence. - Provide upload feature for required documentation (e.g., company registration certificate, proof of legitimacy). - Submit registration to the Administrator for verification workflow. - Send automated acknowledgment email confirming receipt of application. - Enable draft saving so employers can resume incomplete registrations later. - Notify administrators when new employer accounts require approval. - Update employer account status based on admin review (approved, pending, rejected). - On approval, send activation link to employer and allow completion of company profile (mission, logo, recruitment needs). - Flag and reject suspicious or incomplete employer applications.
COLLABORATER
- Employer - Administrator - Verification System

User Management: CRC Card

Class Name: User Management
RESPONSIBILITY - Provide administrator with a searchable and filterable list of all users with details and statuses. - Allow admin to verify employer registrations by checking uploaded documents before activating accounts. - Support account maintenance actions: reset passwords, unlock accounts, deactivate accounts. - Enable privilege assignment: grant or revoke administrator-level access for certain users. - Log all account actions for audit purposes (lockouts, resets, role changes). - Notify users about important account changes via email or in-app messages. - Handle system errors or database issues gracefully with clear administrator feedback.
COLLABORATER - Administrator - Employer - Student

Portfolio Management: CRC Card

Class Name: Portfolio Management
RESPONSIBILITY
- Provide interface for students to create and maintain a digital portfolio (SkillStack). - Enable uploading of project details, descriptions, source code links, and attached files. - Allow editing and deletion of portfolio entries with version history. - Ensure file validation and secure storage for uploaded documents or multimedia files. - Integrate employer view of portfolio showcasing relevant skills and achievements. - Allow students to categorize projects by skill or academic module. - Provide option to mark certain projects as private or public. - Handle error cases like invalid file types or upload failures.
COLLABORATER
- Student - Employer

Skill Tags Management: CRC Card

Class Name: Skill Tags Management
RESPONSIBILITY
- Allow students to manually add or edit skill tags associated with their profiles. - Auto-generate skill tags from academic records and uploaded portfolio content. - Assign badges automatically when students reach milestones (e.g., completion of advanced modules). - Support skill endorsements by employers or peers. - Allow visibility toggling for specific skills (public/private). - Handle duplicates by merging or suggesting existing tags. - Ensure badge criteria are transparent and consistent. - Maintain system logs of skill modifications.
COLLABORATION
- Student - System - Employer

CV Builder: CRC Card

Class Name: CV Builder
RESPONSIBILITY
- Extract academic and portfolio data to auto-generate a structured CV. - Provide multiple CV templates with previews before download. - Allow customization of sections (skills, education, projects, references). - Validate formatting and ensure consistent professional style. - Enable export to PDF and save a copy to student profile. - Allow students to edit generated CVs before finalizing. - Handle errors gracefully if academic or portfolio data is missing.
COLLABORATER
- Student - Profile Management

Skill Gap Analysis: CRC Card

Class Name: Skill Gap Analysis
RESPONSIBILITY
- Accept job description or selected job role from employer database. - Compare student's current skills with job requirements. - Highlight missing skills in a structured report. - Recommend relevant courses, certifications, or learning resources. - Allow students to simulate gap analysis for desired job positions. - Ensure reports are exportable and stored in student dashboard. - Handle incomplete student profiles by providing disclaimers.
COLLABORATER
- Student - Skill Management - Learning Resource

Job Posting: CRC Card

Class Name: Job Posting
RESPONSIBILITY
- Provide form for employers to create job postings with mandatory fields (title, description, requirements, location, salary). - Validate all fields before posting (e.g., salary format, application deadline). - Allow employers to preview job before publishing. - Publish job on central job board accessible to students. - Notify eligible students based on skill matching algorithms. - Enable employers to edit or delete postings post-publication. - Track job post analytics (views, applications received).
COLLABORATER
- Employer - Student - Job Board

Job Search: CRC Card

Class Name: Job Search
RESPONSIBILITY
- Allow students to search jobs using filters (industry, location, salary, skills). - Display detailed job descriptions and employer information. - Enable direct application submission with one-click apply using stored CV. - Pre-fill applications with student data while allowing edits. - Send confirmation to student after successful application submission. - Store application history in student profile. - Handle cases where job is closed or withdrawn by employer.
COLLABORATER
- Student - Employer - Job Board

Notification: CRC Card

Class Name: Notification
RESPONSIBILITY -Alert users to events (interview requests, application updates, access requests). -Deliver notifications via in-app alerts and/or email. -Allow users to set preferences for notification types (email only, in-app only, both). -Ensure time-sensitive alerts (e.g., interview reminders) are delivered promptly. -Provide read/unread status for tracking engagement. - Support grouping of notifications (e.g., “3 new applications updates”). - Log notification history for auditing and reference. - Send push notifications (if integrated with mobile app). - Additional: Prioritize critical notifications (e.g., accepted interview request vs. general info). - Allow administrators to broadcast system-wide announcements.
COLLABORATER - Student (receives job/interview/application updates) -Employer (receives candidate responses, interview confirmations) -Administrator (system-wide updates, account alerts) -Interview (linked to scheduled or updated interview times) -Application (updates on submission, review, or status changes)

Application: CRC Card

Class Name: Application
RESPONSIBILITY - Link a student to a specific Job. - Store application data (submission date, cover letter, uploaded CV). - Track application status (Submitted, Under Review, Shortlisted, Rejected, Accepted). - Allow students to withdraw applications before deadlines. - Notify employers of new applications and students of status changes. - Maintain application history for students and employers. - Enable students to update documents (CV, cover letter) before review begins. - Support “Quick Apply” using stored profile/CV. - Provide analytics for students (number of applications sent, acceptance rate). - Allow employers to filter/sort applications. -Ensure data privacy and secure handling of sensitive files.
COLLABORATER -Student (submits application) -Job (posting being applied to) -CV (document attached to application) -Employer (reviews and updates status) -Notification (alerts about submission, review, or updates) -Administrator (audits application records if needed)

Candidate Search: CRC Card

Class Name: Candidate Search
RESPONSIBILITY
- Enable employers to search for candidates by skills, GPA, or keywords. - Display anonymized student profiles to protect sensitive data. - Allow employers to request access to full profiles or academic records. - Notify students of employer requests for approval. - Grant or deny academic data access based on student response. - Provide employers with candidate shortlisting tools. - Log employer search activity for auditing.
COLLABORATER
- Employer - Student - UFS Database

Interview Scheduling: CRC Card

Class Name: Interview Scheduling
RESPONSIBILITY
- Provide employers with calendar interface to propose interview slots. - Send interview invitations to selected students with multiple time options - Allow students to accept, decline, or propose new times. - Sync confirmed interviews to both student and employer calendars. - Send reminders and notifications before scheduled interviews. - Handle rescheduling and cancellation requests. - Ensure time zone and availability conflicts are resolved automatically.
COLLABORATER
- Employer - Student - Calendar System

Interview Management: CRC Card

Class Name: Interview Management
RESPONSIBILITY
- Allow students to view pending interview requests from employers. - Provide options to accept, decline, or request rescheduling. - Display employer details and job context for each interview. - Sync confirmed interview details into student calendar. - Send confirmation back to employer upon acceptance. - Provide interface for rescheduling in case of conflicts. - Handle expired or withdrawn interview requests.
COLLABORATER
- Employer - Student - Calendar System

Academic Record Access: CRC Card

Class Name: Academic Record Access
RESPONSIBILITY
- Allow employers to submit requests for access to specific academic records. - Notify students when an employer requests access to their data. - Provide students with option to approve, partially approve, or deny request. - Ensure privacy rules are respected (only approved data is shared). - Log all record access transactions for compliance auditing. - Notify employers of student decision and provide access if granted. - Handle expired requests automatically after set duration.
COLLABORATER
- Employer - Student - UFS Database

Analytics Dashboard: CRC Card

Class Name: Analytics Dashboard
RESPONSIBILITY - Collect and aggregate data across the recruitment system (jobs posted, applications, success rates). - Visualize trends such as top demanded skills, placement rates, and employer engagement. - Allow filtering by faculty, department, or academic year. - Provide drill-down analytics to track student or employer-specific performance. - Enable export of analytics reports in PDF or Excel formats. - Ensure data is updated in real time with minimal lag. - Handle large datasets without performance degradation.
COLLABORATER - Employer - Student - UFS Database

Support Chatbot: CRC Card

Class Name: Support Chatbot
RESPONSIBILITY
- Provide 24/7 automated assistance through contextual help dialogs. - Answer frequently asked questions with predefined responses. - Guide users step-by-step through processes (e.g., job application, CV building). - Escalate unresolved issues to live support or administrators. - Offer quick links to relevant sections of the system. - Log all interactions for future analysis and system improvement. - Handle errors gracefully if queries are out of scope.
COLLABORATER
- Student - Employer - Administrator - Support System

Calendar: CRC Card

Class Name: Calendar
RESPONSIBILITY - Schedule and manage interview appointments -Check and manage user availability across time zones -Send automated calendar invites and reminders -Handle interview rescheduling and cancellation requests -Prevent scheduling conflicts and double-booking -Sync with external calendar systems (.ics file generation) -Manage recurring availability patterns and working hours -Send notification reminders 24 hours before interviews -Coordinate time zone conversions for international participants -Maintain interview status (pending, confirmed, cancelled)
COLLABORATER - Employer -Student -Interview -Email Service -Notification -External Calendar API -Time zone Service -Conflict Detector

File management: CRC Card

Class Name: File management
RESPONSIBILITY
- Handle CV file uploads, storage, and retrieval -Manage portfolio project files and attachments -Process employer verification document uploads -Store and secure academic transcripts and records -Validate file types, sizes, and security compliance -Manage file permissions and privacy settings -Generate and optimize PDF files (CVs, reports) -Implement file version control and backup systems -Scan uploaded files for malware and security threats -Compress and optimize large files for storage
COLLABORATER
-Student -Employer -Portfolio -CV Builder -Academic Record -Security Service -Validation Service -Backup Service -Privacy Settings

Class Diagram

Disclaimer: This class diagram focuses on the business logic layer and core domain entities. Methods and attributes for persistence (e.g., save ()) or UI (e.g., display ()) are omitted for clarity



Figure 41: Support Chatbot UI

State Chart: Student

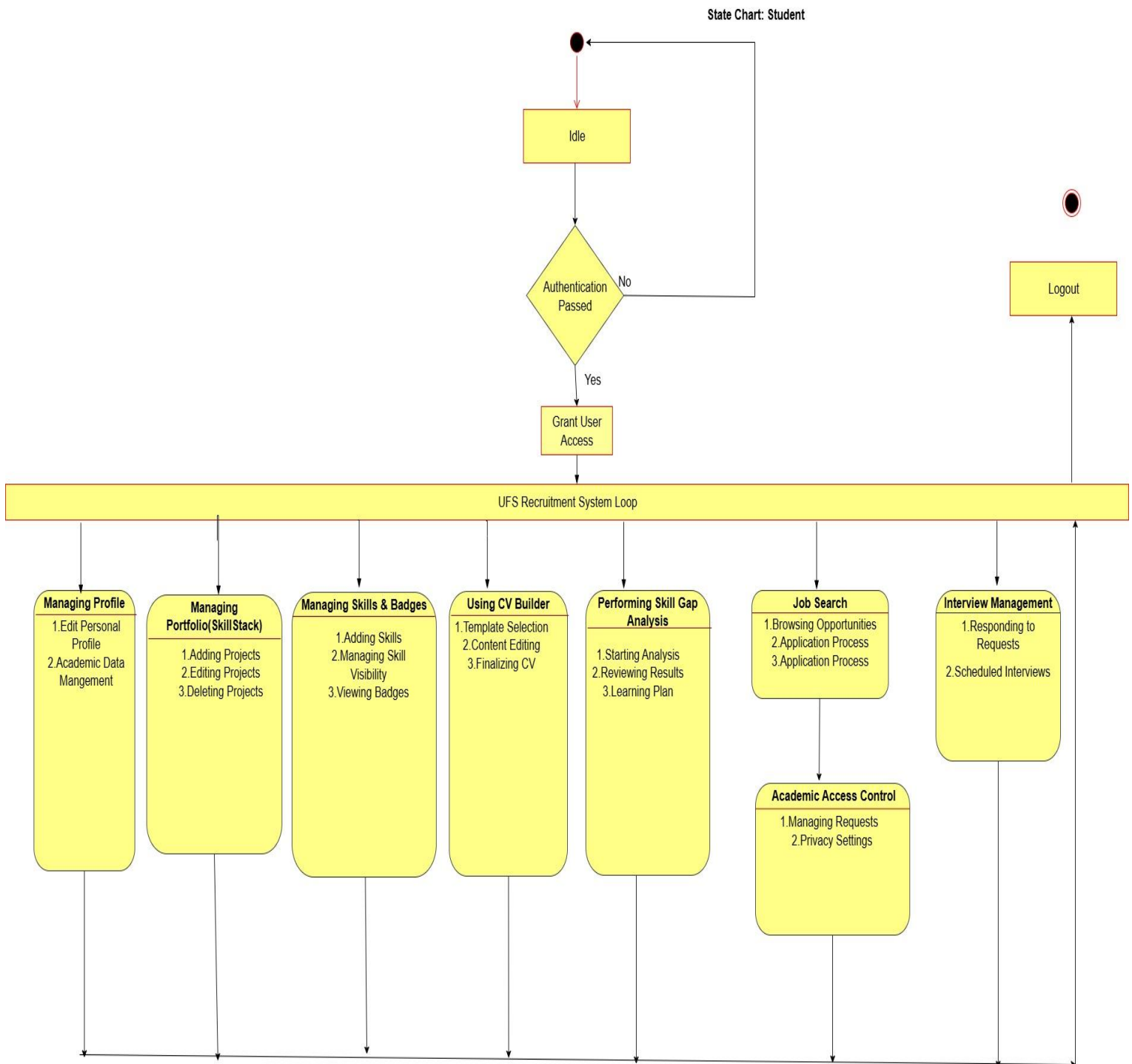


Figure 42: Student State Chart

State Chart: Employer

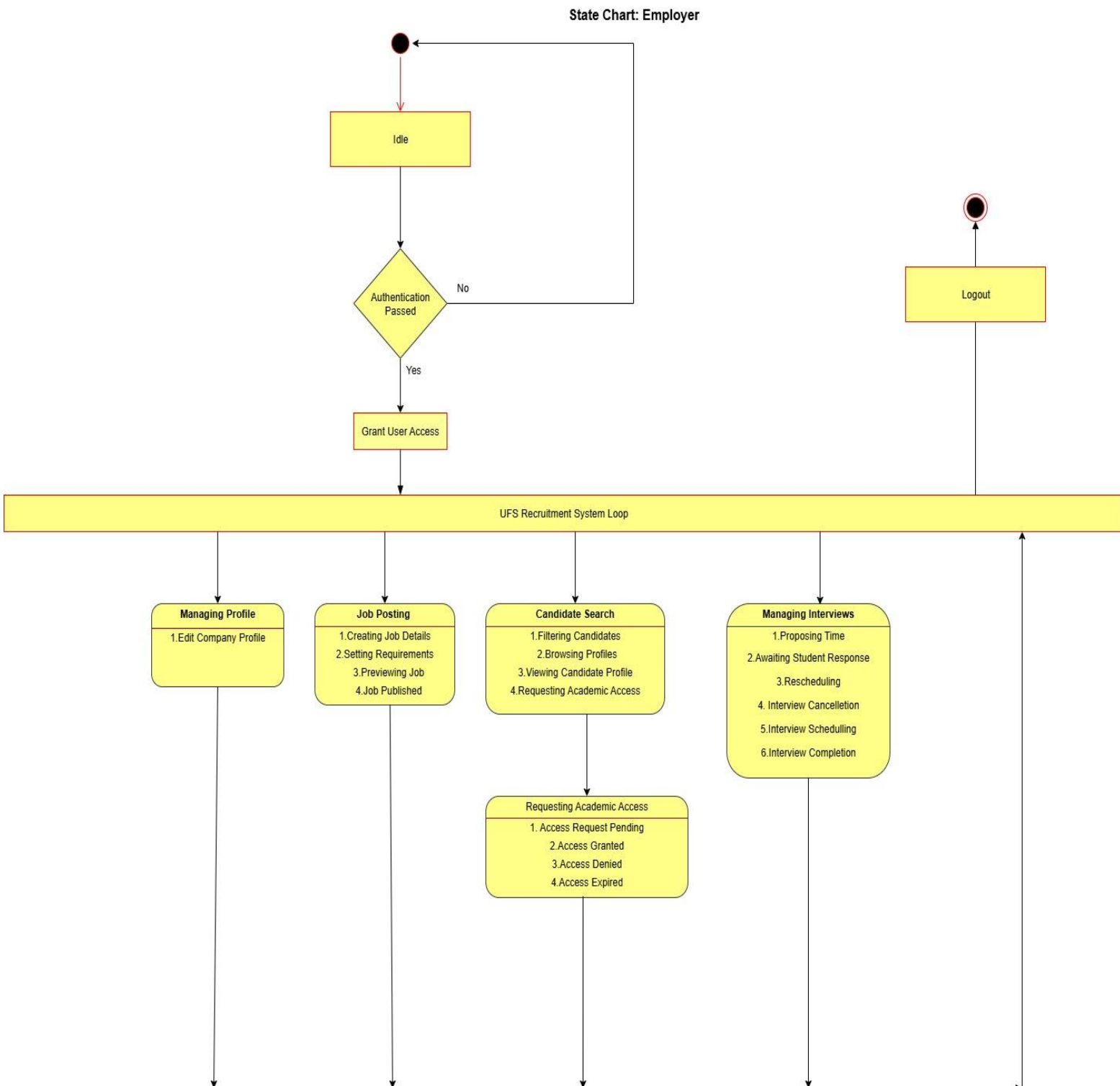


Figure 43: Employer State Chart

State Chart: Administrative

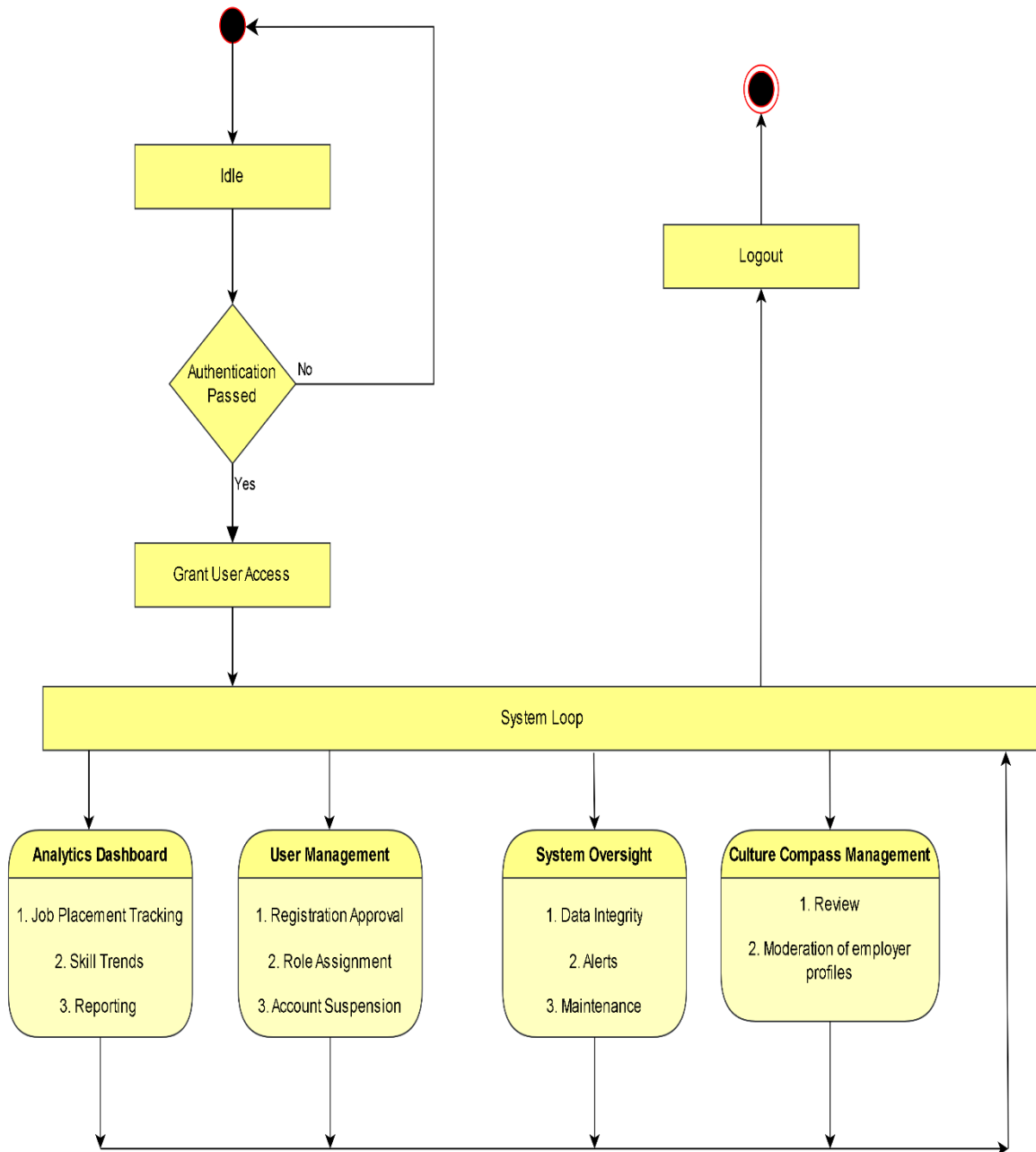


Figure 44: Admin State Chart

Pseudocode

Class: Student Registration

CLASS StudentRegistration:

ATTRIBUTES:

ufsDatabaseAPI: UFSDatabaseService

profileValidator: ProfileValidator

METHOD registerStudent (studentData: StudentRegistrationData) RETURNS
RegistrationResult:

BEGIN

// Step 1: Validate UFS credentials

verificationResult = ufsDatabaseAPI.verifyStudentCredentials(

studentData.studentNumber,

studentData.ufsEmail)

IF NOT verificationResult.isValid THEN

RETURN RegistrationResult (success: false, error: "Invalid UFS credentials")

END IF

// Step 2: Validate password strength

IF NOT isValidPassword (studentData.password) THEN

RETURN RegistrationResult (success: false, error: "Password does not meet
requirements")

END IF

// Step 3: Create user account

user = CREATE User WITH:

email: studentData.ufsEmail,

password: hashPassword(studentData.password),

role: "Student",

status: "Active"

```
// Step 4: Create student profile with academic data
profile = CREATE StudentProfile WITH:
    userId: user.id,
    studentNumber: studentData.studentNumber,
    degreeProgram: verificationResult.degreeProgram,
    major: verificationResult.major,
    academicData: verificationResult.courses
// Step 5: Import academic records if consented
IF studentData.consentAcademicImport THEN
    academicRecords =
ufsDatabaseAPI.getAcademicRecords(studentData.studentNumber)
    profile.transcript = academicRecords
    profile.gpa = calculateGPA(academicRecords)
END IF
// Step 6: Save to database
userDatabase.save(user)
profileDatabase.save(profile)
// Step 7: Send welcome notification
notificationService.sendWelcomeEmail(user.email, "Student")
RETURN RegistrationResult(success: true, userId: user.id, profileId: profile.id)
END METHOD
```

Class: Analytics Dashboard

CLASS AnalyticsDashboard:

ATTRIBUTES:

dataCollector: DataCollectorService

reportGenerator: ReportGenerator

visualizationEngine: VisualizationEngine

METHOD generatePlacementAnalytics(timeRange: TimeRange, filters: AnalyticsFilters) RETURNS AnalyticsReport:

BEGIN

// Collect placement data

placementData = dataCollector.getPlacementData(timeRange, filters)

// Calculate key metrics

totalStudents = placementData.totalStudents

employedStudents = placementData.employedStudents

placementRate = (employedStudents / totalStudents) * 100

// Get trend data

trendData = dataCollector.getPlacementTrends(timeRange)

// Generate visualizations

placementChart = visualizationEngine.createBarChart(

data: trendData,

title: "Placement Rate Over Time",

xAxis: "Month", yAxis: "Placement Rate %")

skillDemandChart = visualizationEngine.createPieChart(

data: placementData.topSkills,

title: "Top In-Demand Skills")

// Compile report

report = CREATE AnalyticsReport WITH:

```
placementRate: placementRate,  
totalApplications: placementData.totalApplications,  
averageApplicationTime: placementData.averageApplicationTime,  
topEmployers: placementData.topEmployers,  
charts: [placementChart, skillDemandChart],  
generatedDate: currentDateTime()
```

```
RETURN report
```

```
END METHOD
```

```
METHOD exportAnalyticsReport(report: AnalyticsReport, format: string)  
RETURNS ExportResult:
```

```
BEGIN
```

```
SWITCH format:
```

```
  CASE "PDF":
```

```
    pdfData = reportGenerator.generatePDF(report)
```

```
    RETURN ExportResult(success: true, data: pdfData, format: "PDF")
```

```
  CASE "CSV":
```

```
    csvData = reportGenerator.generateCSV(report)
```

```
    RETURN ExportResult(success: true, data: csvData, format: "CSV")
```

```
  CASE "EXCEL":
```

```
    excelData = reportGenerator.generateExcel(report)
```

```
    RETURN ExportResult(success: true, data: excelData, format: "EXCEL")
```

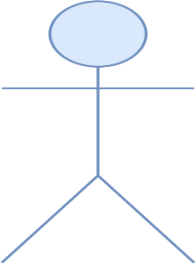
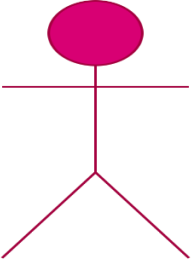
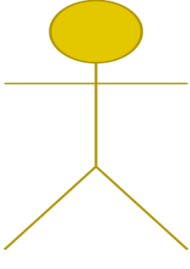
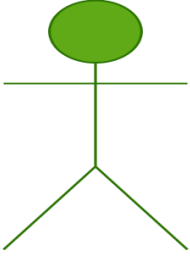
```
  DEFAULT:
```

```
    RETURN ExportResult(success: false, error: "Unsupported format")
```

```
END SWITCH
```

```
END METHOD
```

Sequence Diagrams

System Users			
Disclaimer: This serves as a reminder of users as u get into the sequences.			
Student	Employer	Administrator	UFS Database System
			

Sequence Diagram 1: Login

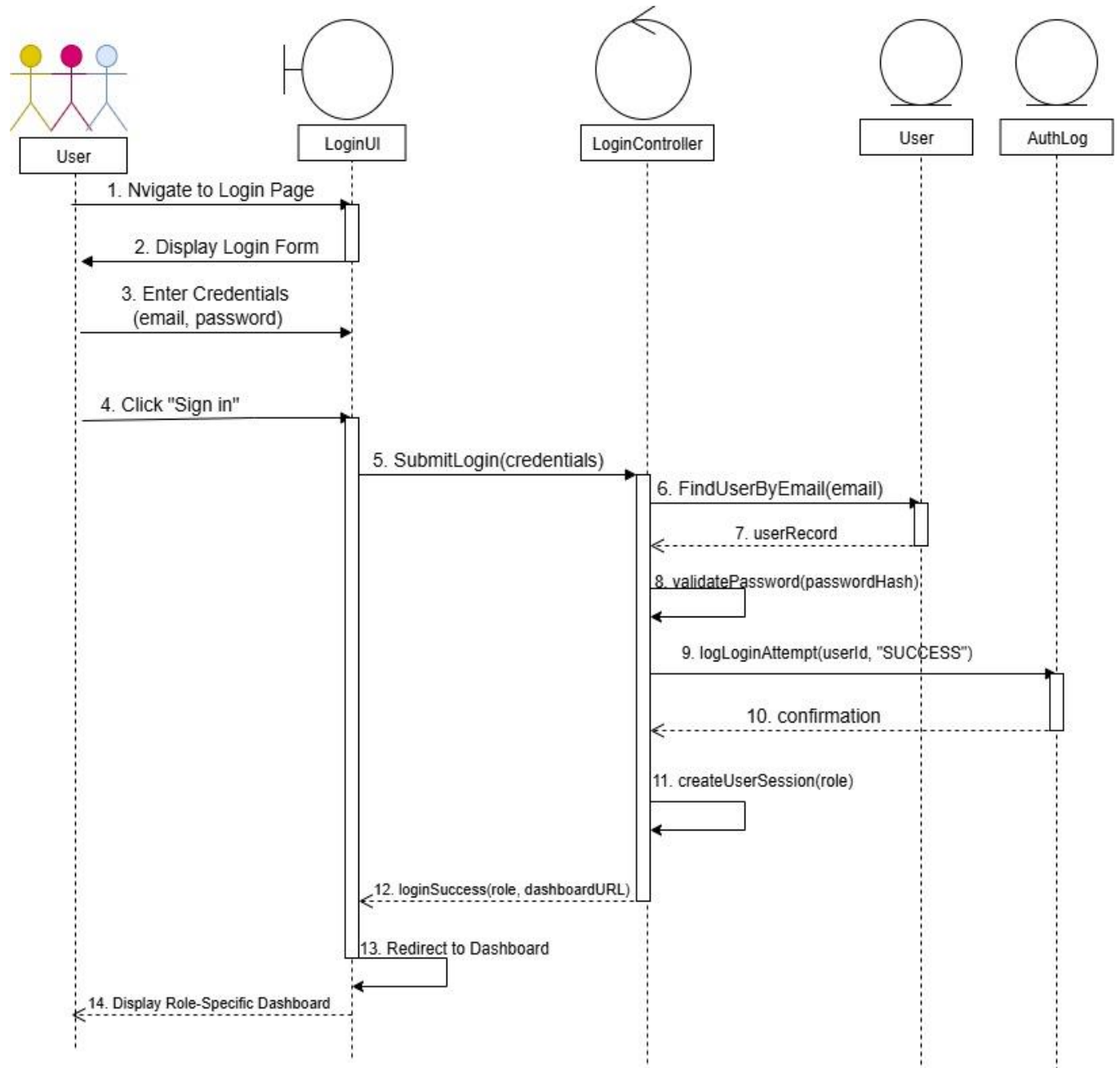


Figure 45: Sequence Diagram 1 (Login)

Sequence Diagram 2: Registration

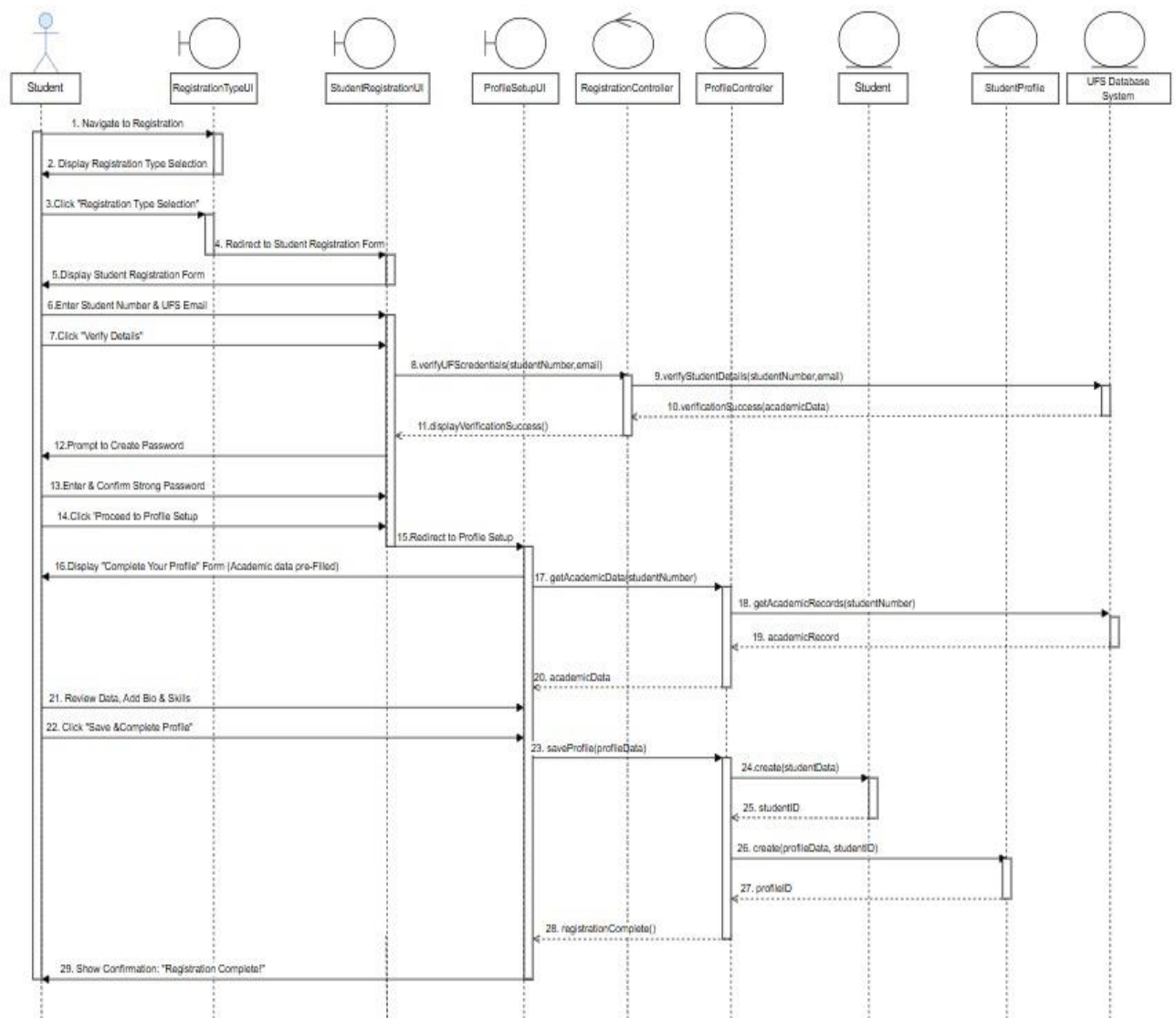


Figure 46: Sequence Diagram 2 (Registration)

Sequence Diagram 3: Employer Registration

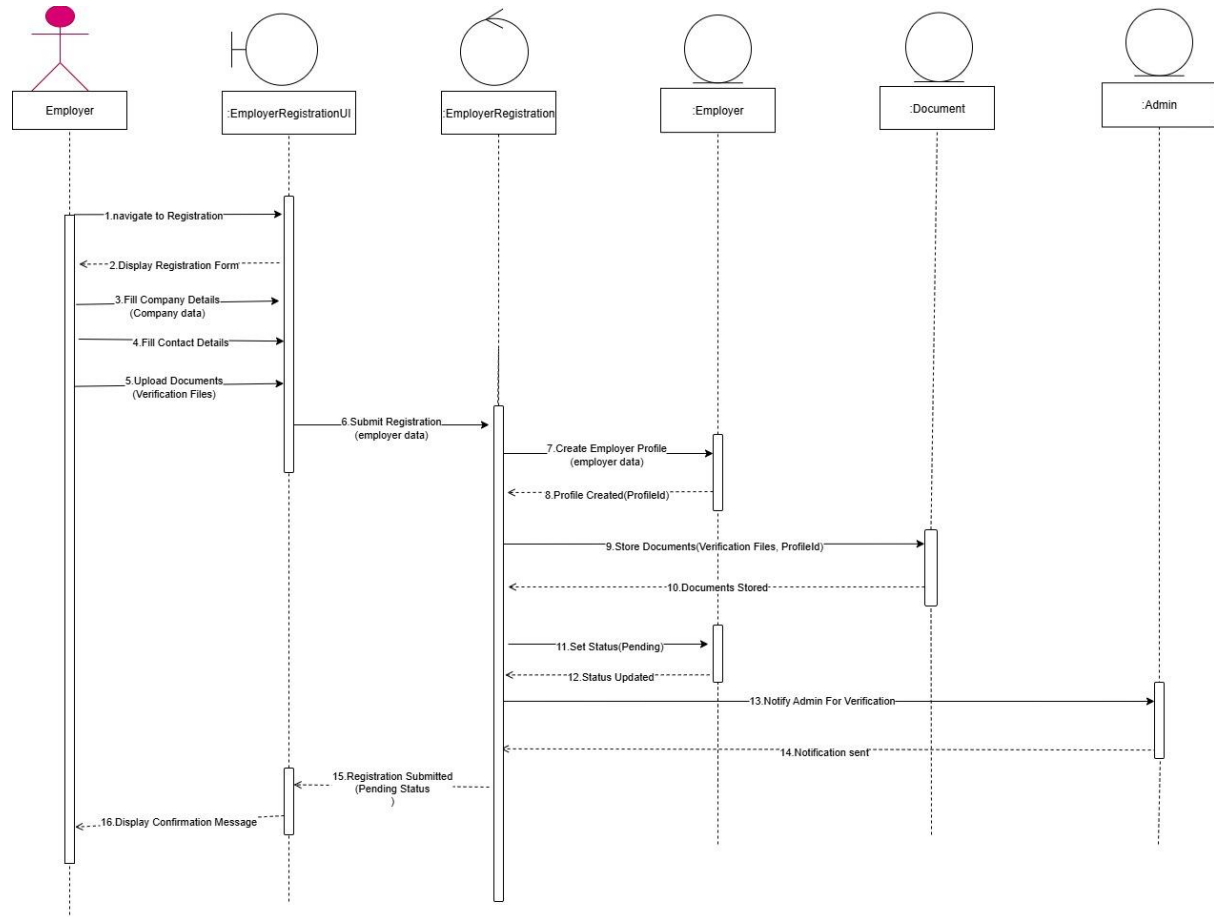


Figure 47: Sequence Diagram 3 (Employer Registration)

Sequence Diagram 4: User Management

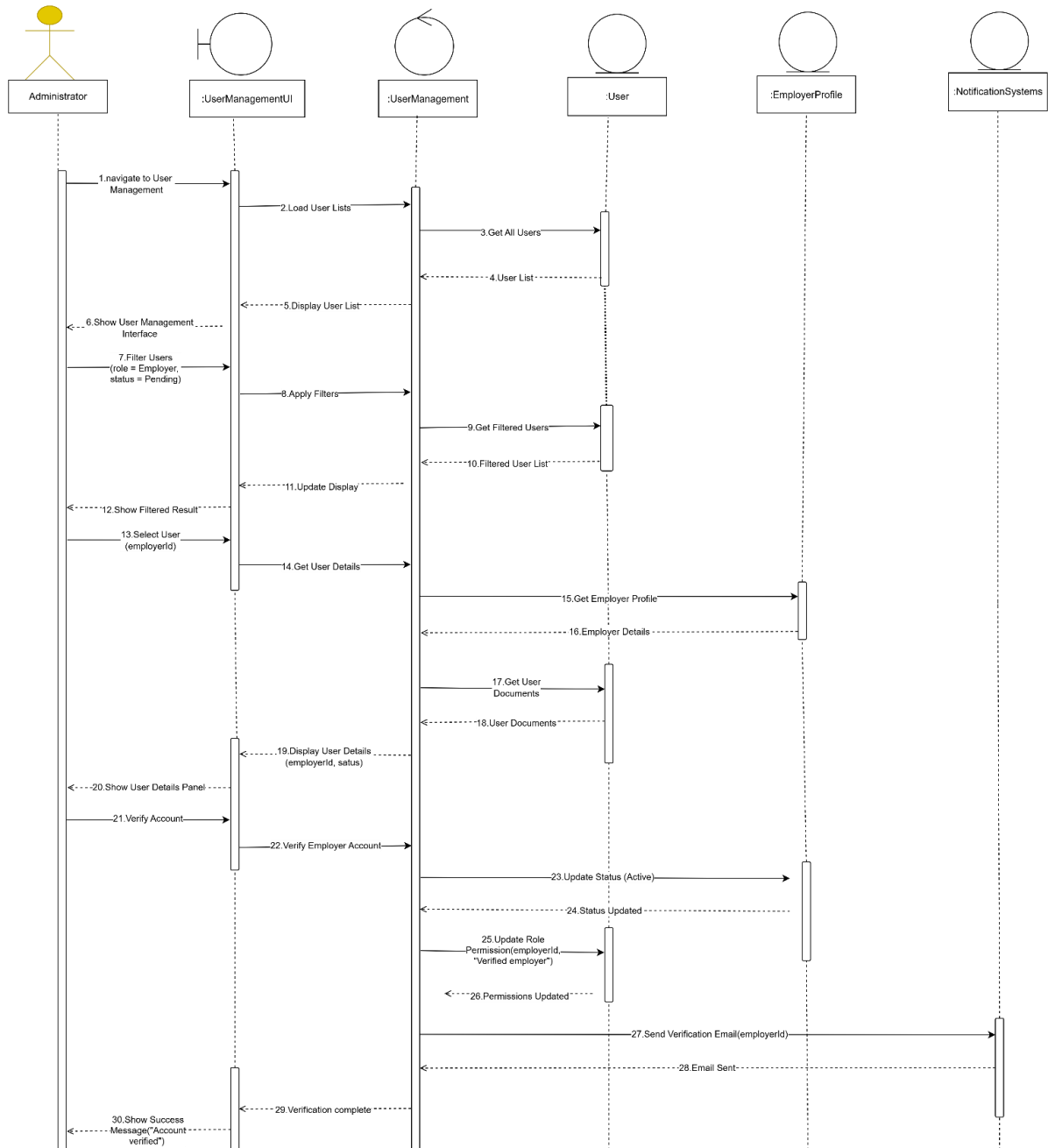


Figure 48: Sequence Diagram 4 (User Management)

Sequence Diagram 5: Manage Portfolio

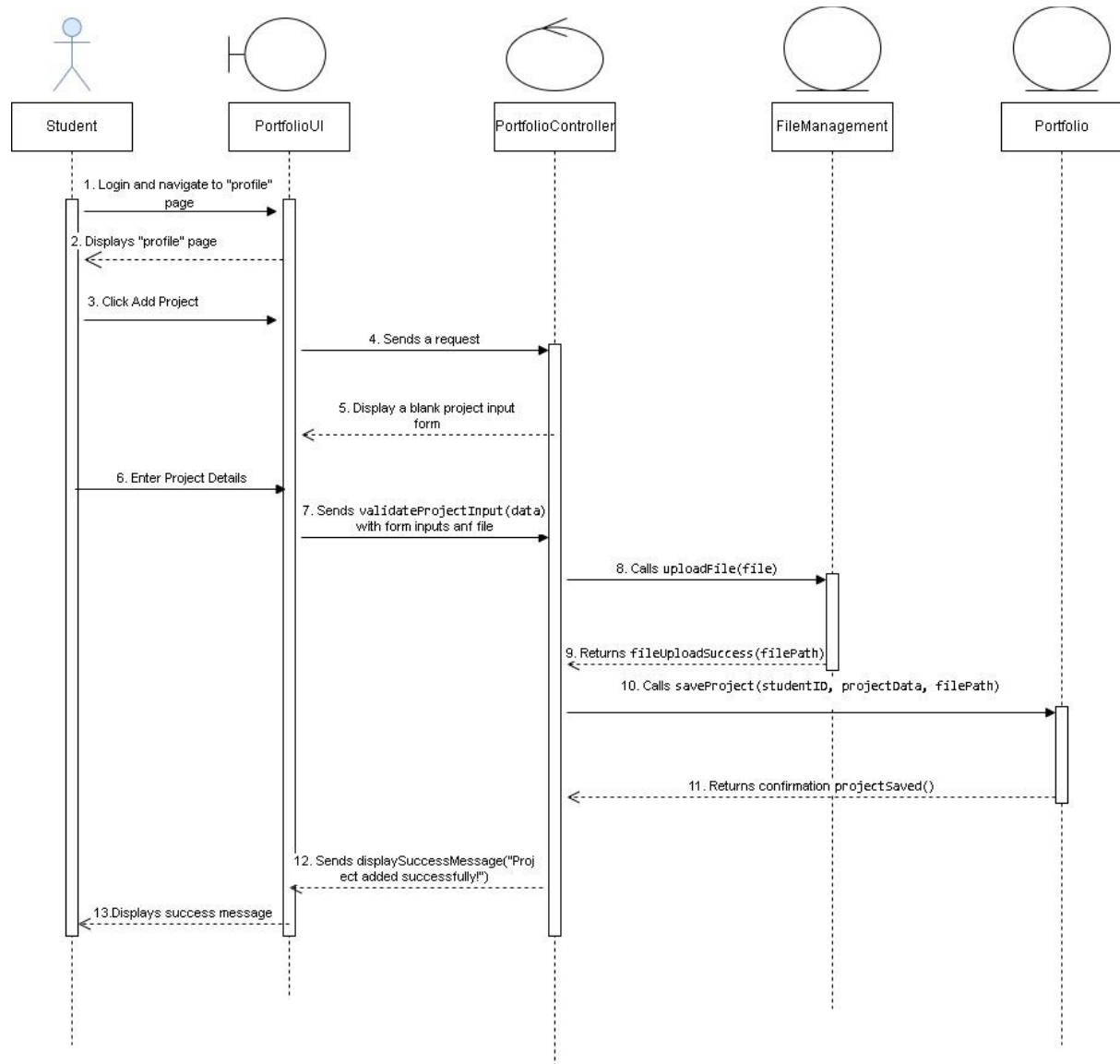


Figure 49: Sequence Diagram 5 (Manage Portfolio)

Sequence Diagram 6: Manage Skill Tags & Badges

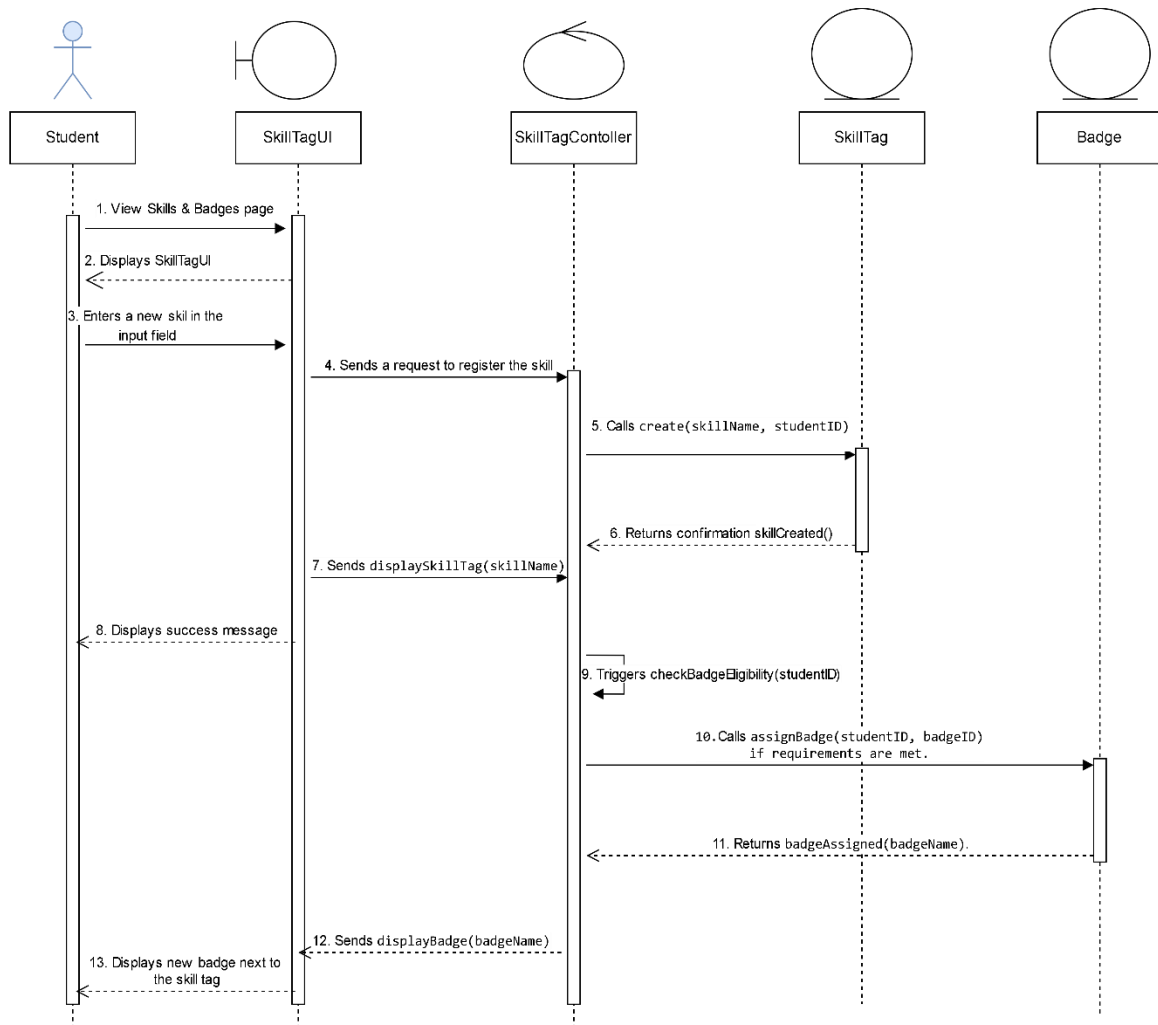


Figure 50: Sequence Diagram 6 (Manage Skill Tags & Badges)

Sequence Diagram 7: Use CV Builder

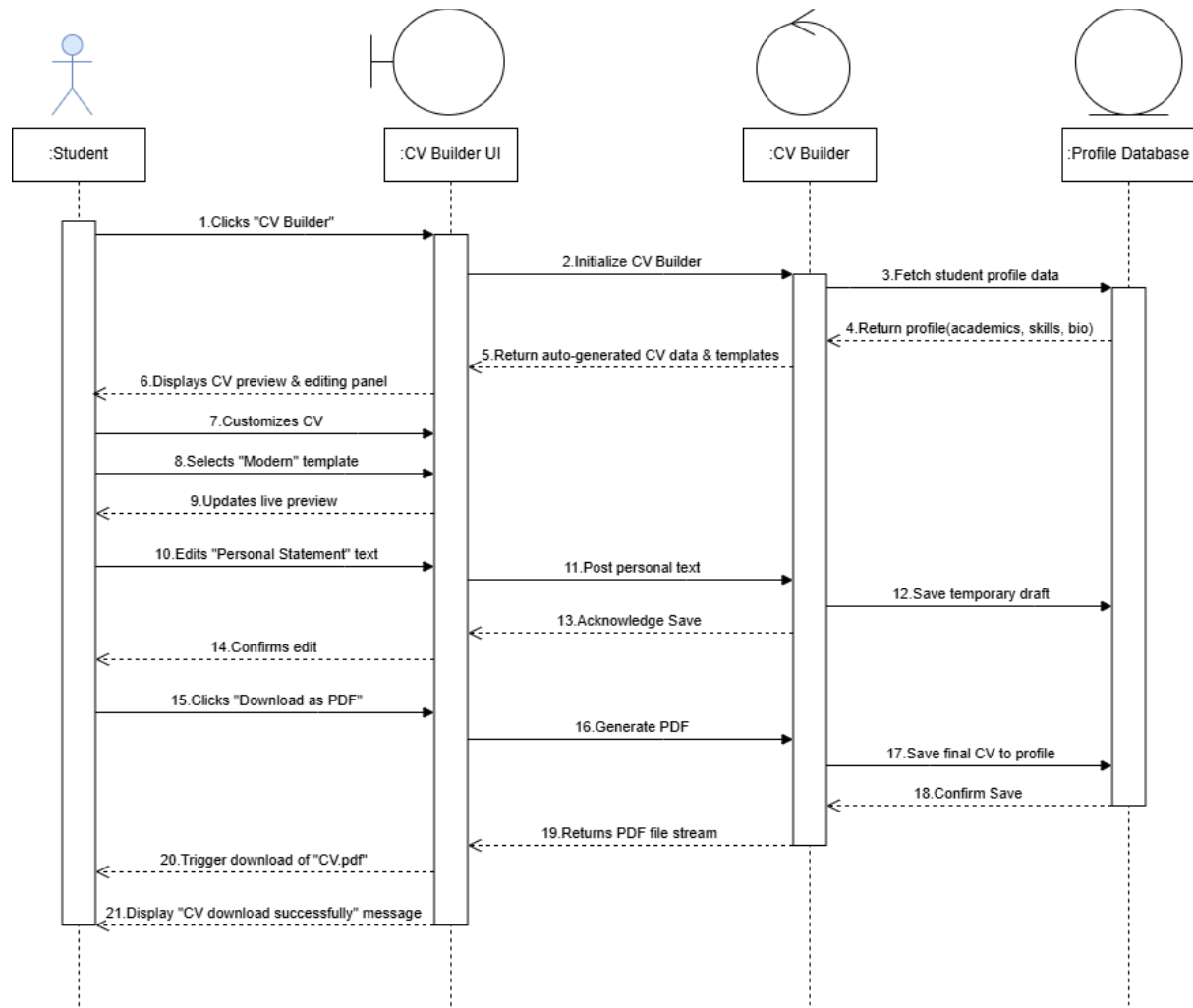


Figure 51: Sequence Diagram 7 (Use CV Builder)

Sequence Diagram 8: Skill Gap Analysis

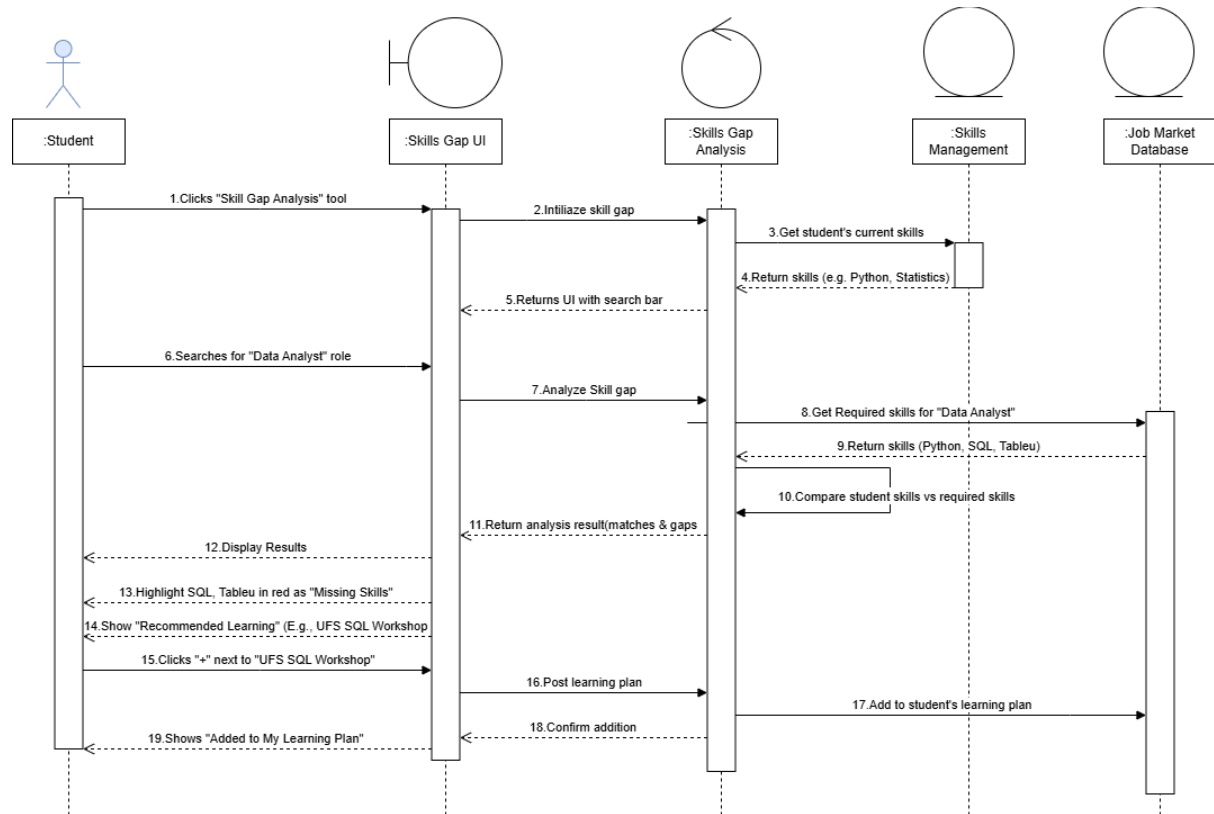


Figure 52: Sequence Diagram 8 (Skill Gap Analysis)

Sequence Diagram 9: Post Job Opportunity

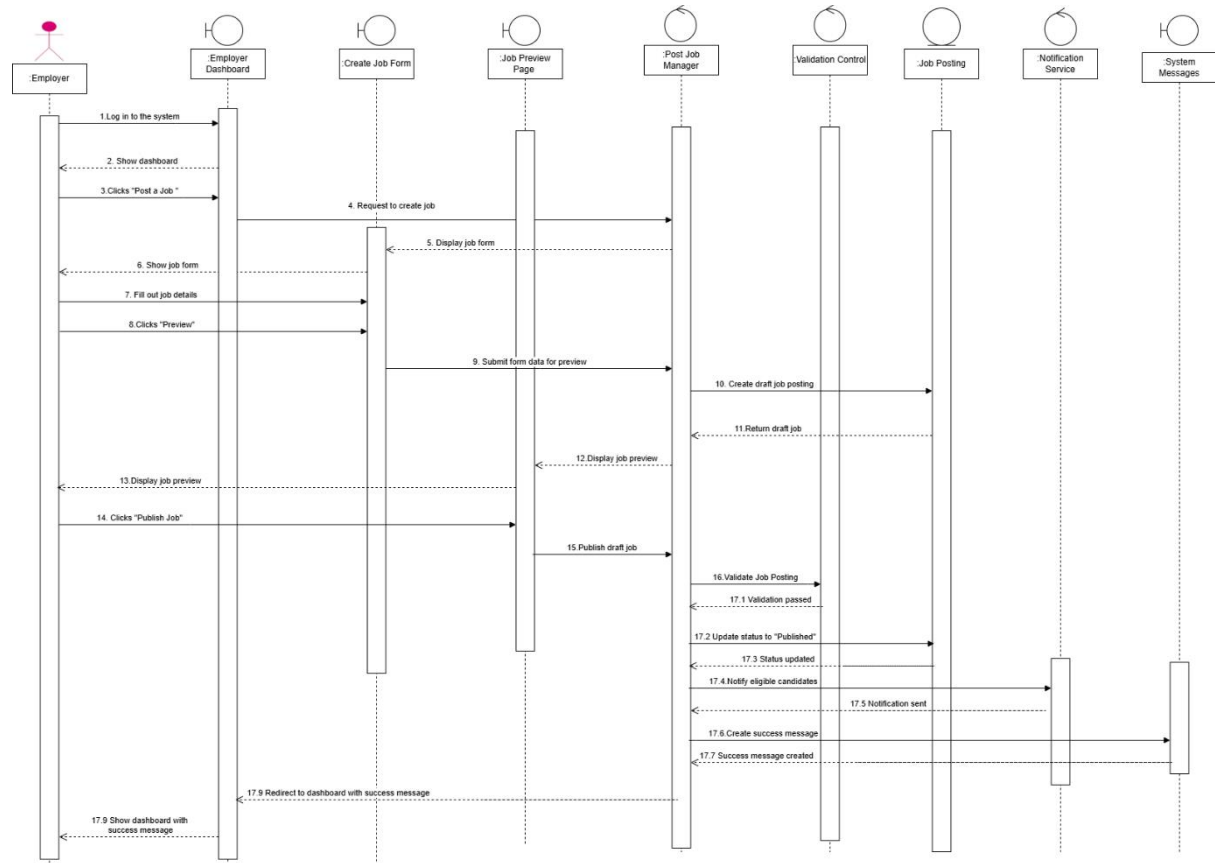


Figure 53: Sequence Diagram 9 (Post Job Opportunity)

Sequence Diagram 10: Job search & Application

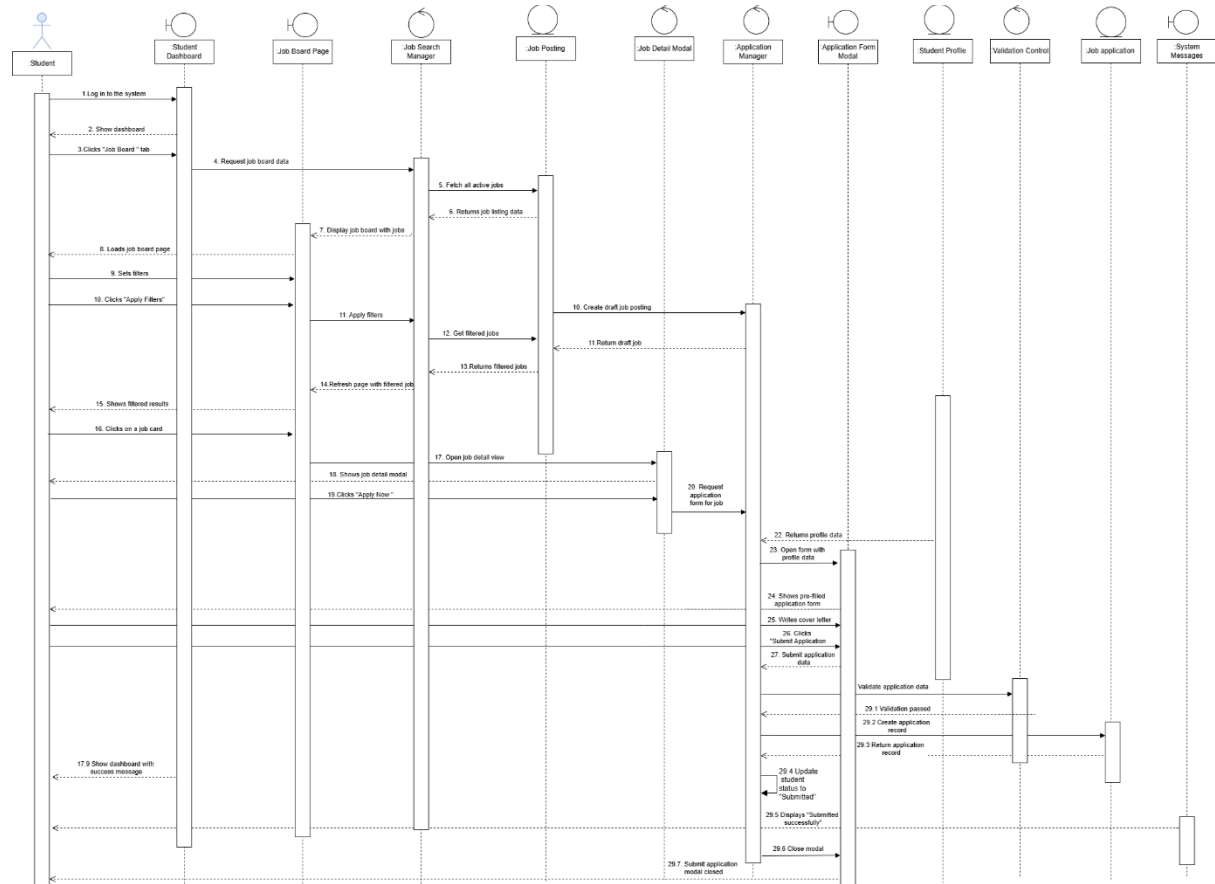


Figure 54: Sequence Diagram 10 (Job search & Application)

Sequence Diagram 11: Search For Candidate

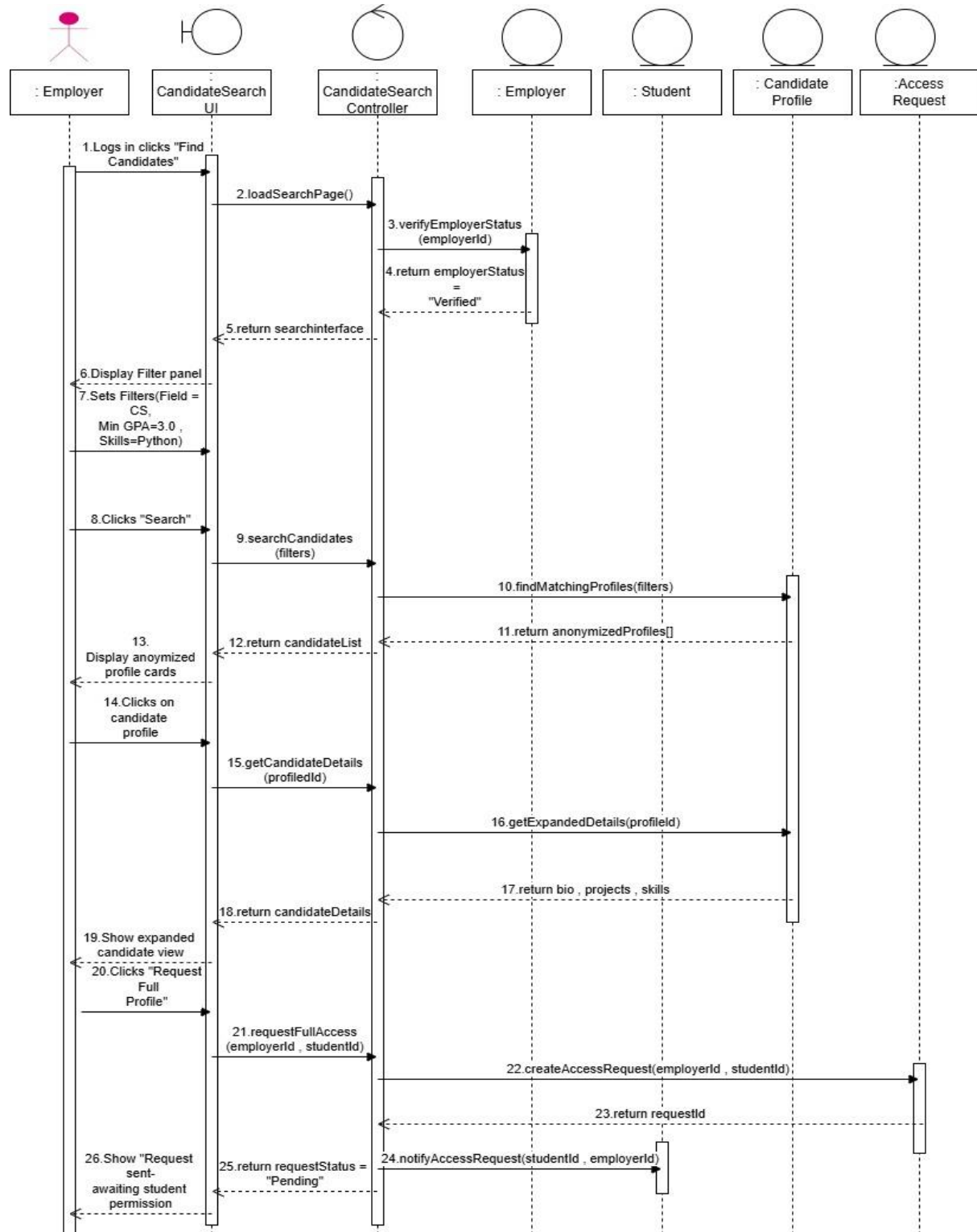


Figure 55: Sequence Diagram 11 (Search For Candidate)

Sequence Diagram 12: Schedule Interview

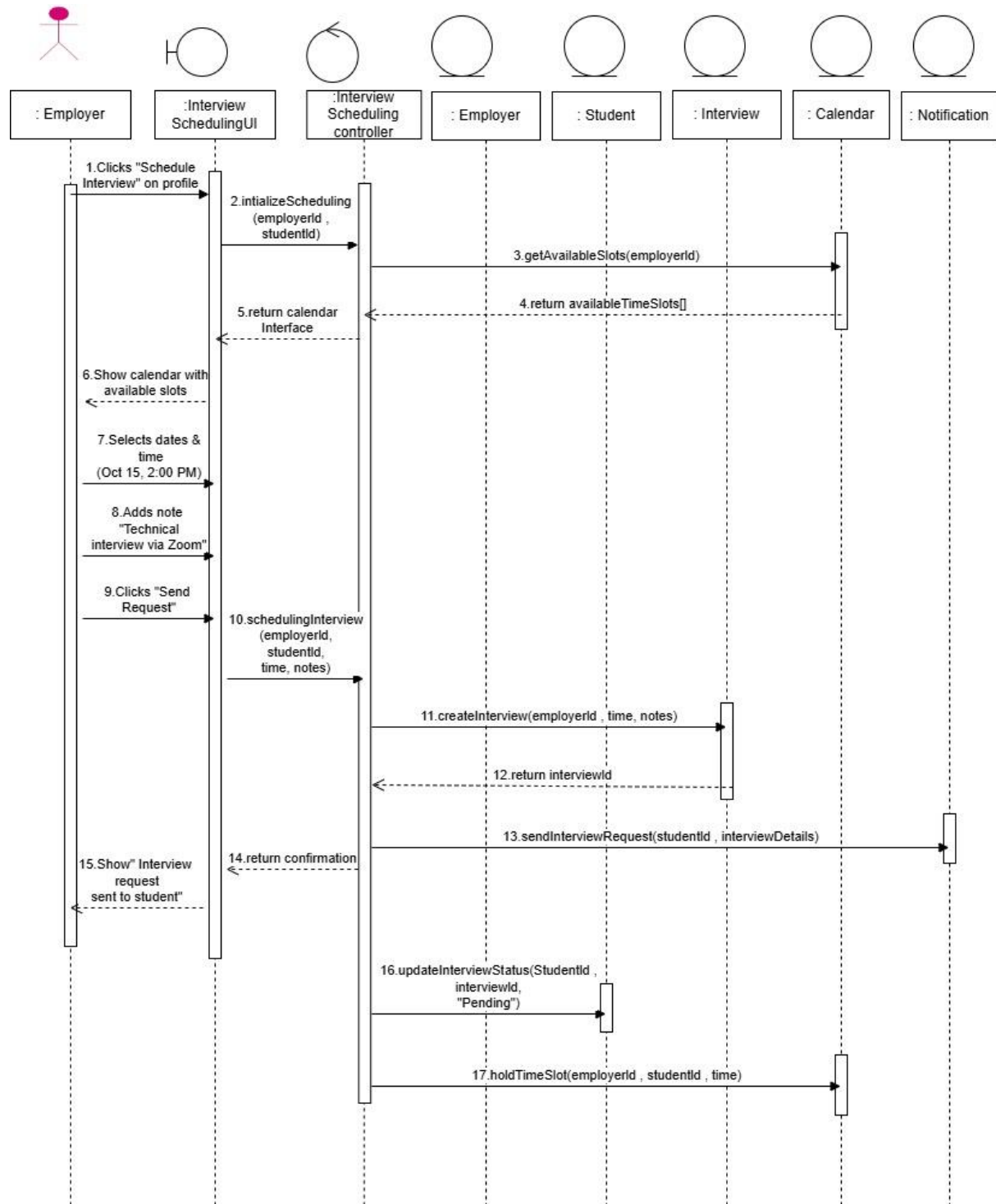


Figure 56: Sequence Diagram 12 (Schedule Interview)

Sequence Diagram 13: Interview Management

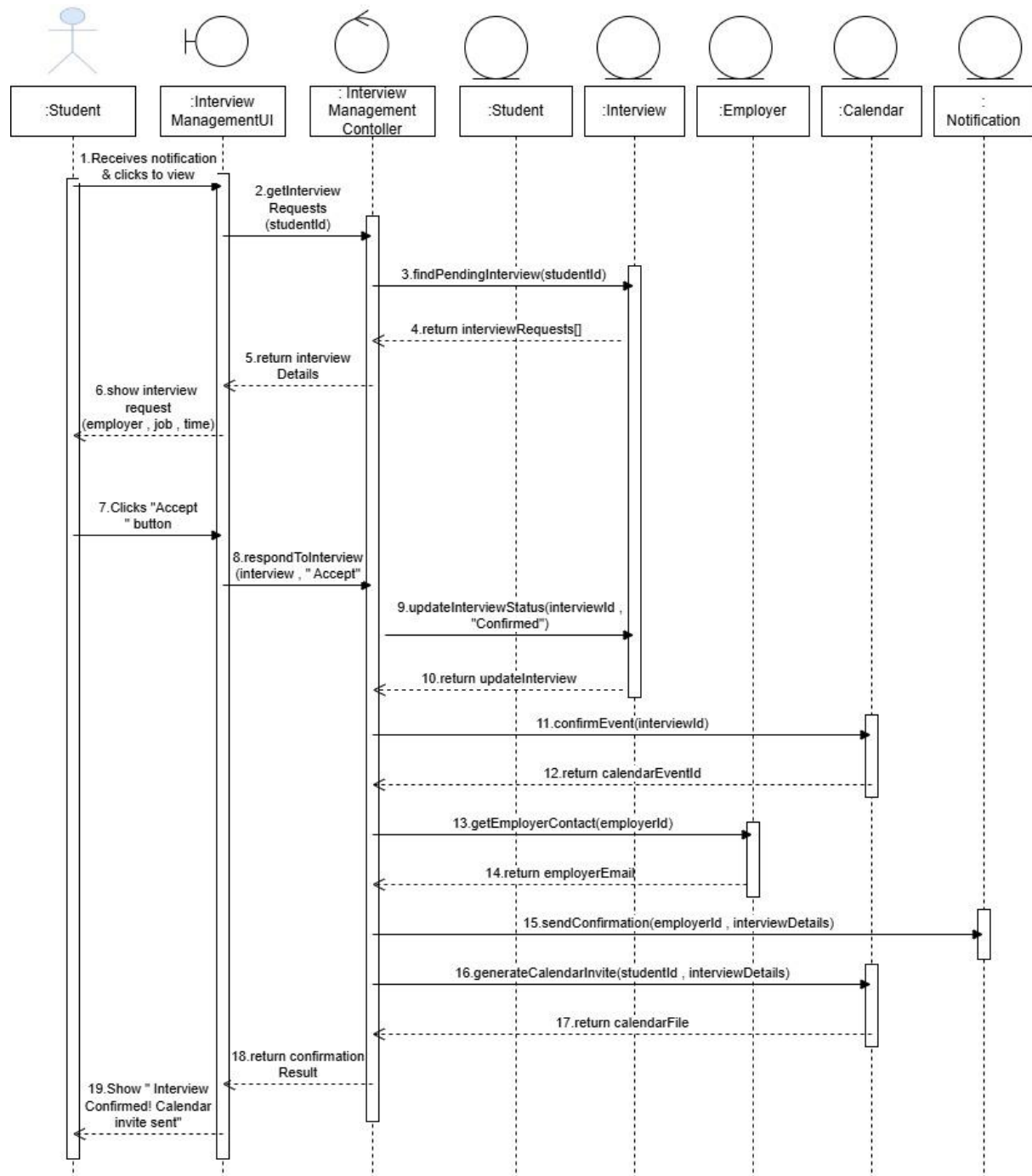


Figure 57: Sequence Diagram 13 (Interview Management)

Sequence Diagram 14: Request Academic Access

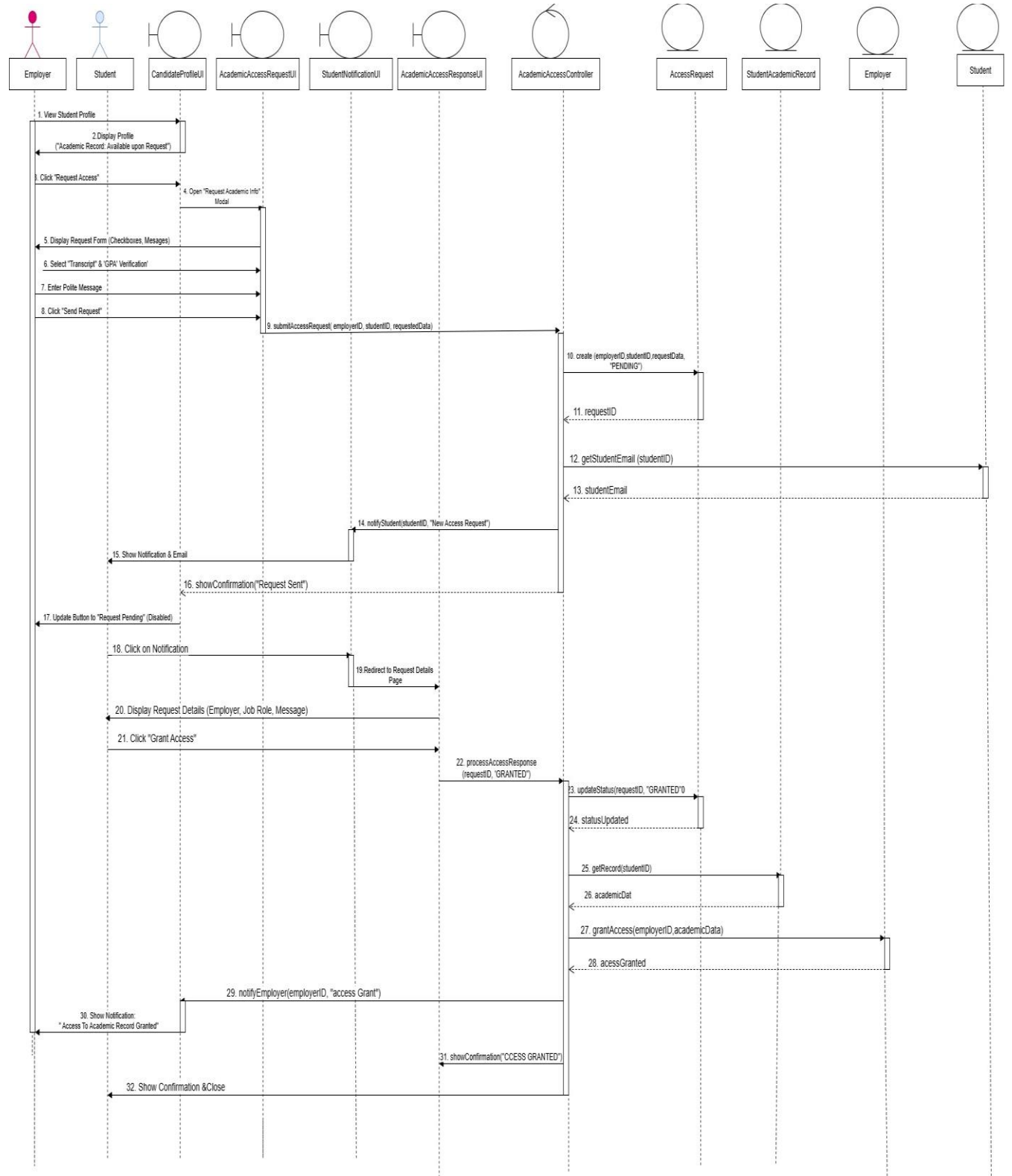


Figure 58: Sequence Diagram 14 (Request Academic Access)

Sequence Diagram 15: View Analytics Dashboard

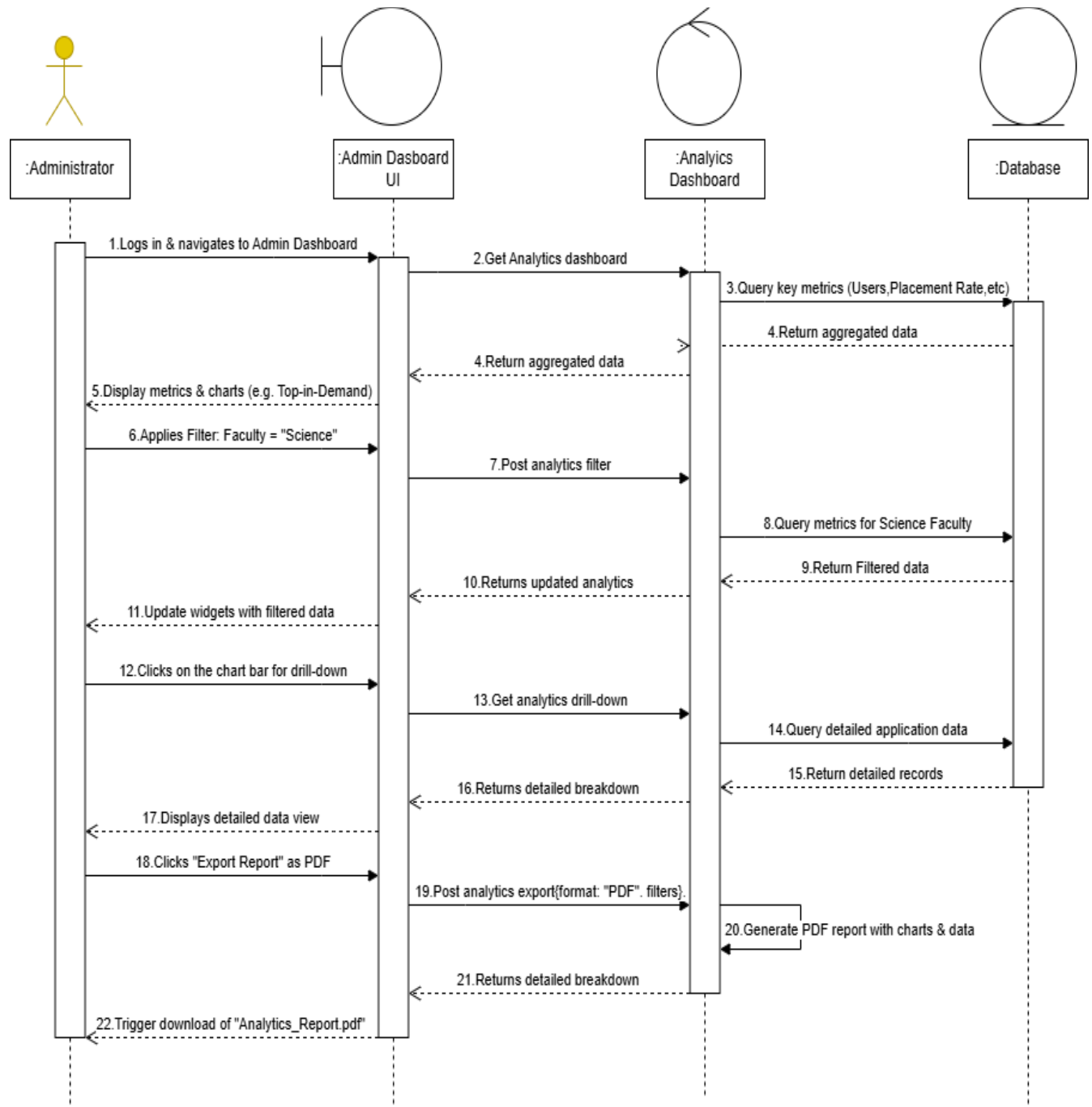


Figure 59: Sequence Diagram 15 (View Analytics Dashboard)

Sequence Diagram 16: Use Support Chatbot

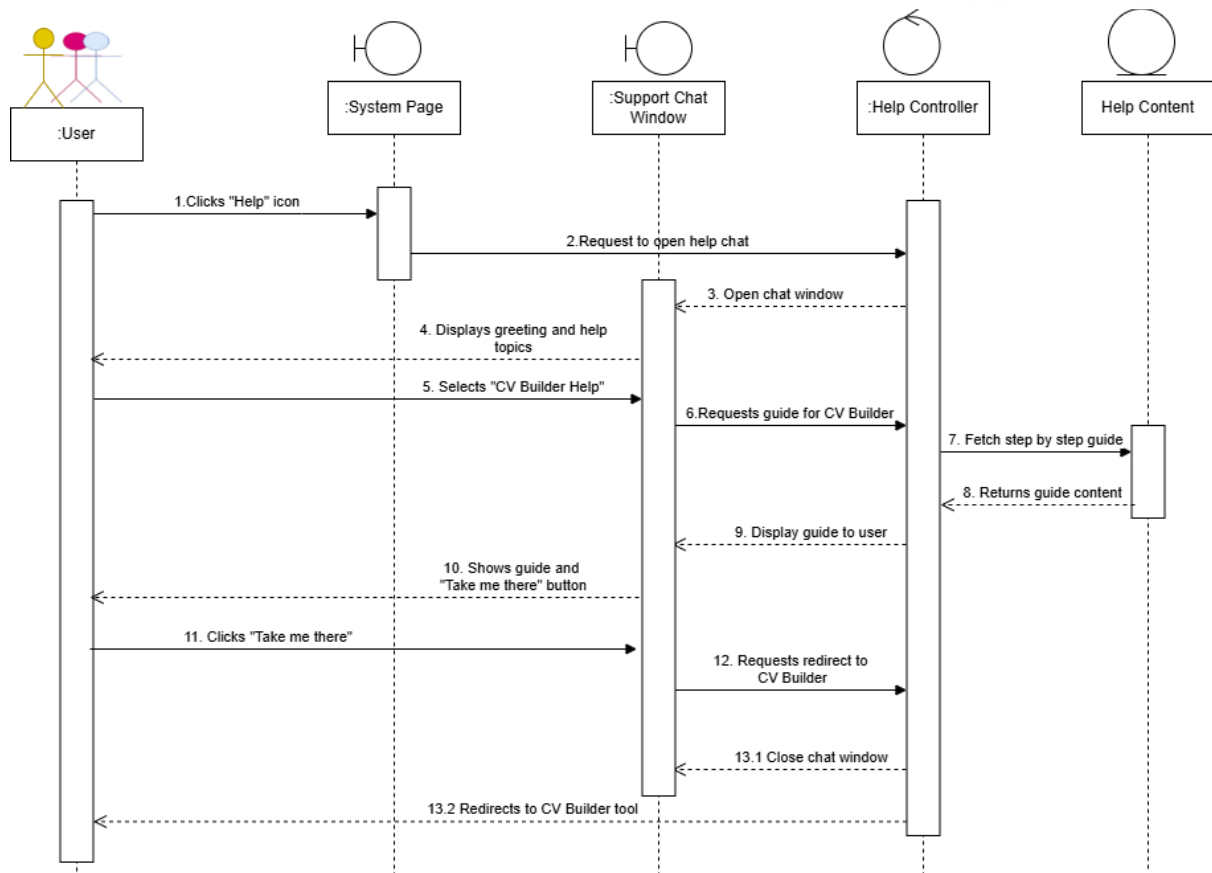


Figure 60: Sequence Diagram 16 (Use Support Chatbot)

Chapter 6: Record Keeping

Introduction

Record keeping is essential for tracking project progress, resource allocation, and team contributions throughout the UFS Recruitment System development. This chapter documents the time invested across all project phases, providing transparency and accountability for the 1,185-hour project timeline.

Completed Work (Documentation Phase):

Name	Requirements	Analysis	Design	SPMP/Docs	SUBTOTAL
V. Bera	16	14	15	26	71
K. Diseko	11hrs30min	12hrs20min	16	14	53hrs50min
S. Mngxozana	17	13	17	27	74
L. Neku	12	12hrs20min	16	19	59hrs20min
D. Nkosi	11hrs30min	12hrs20min	15	14	52hrs50min
O. Ramaru	14	13	16	20	63
TOTAL	82	77	95	120	374

Planned Work (Implementation Phase):

Name	Implementation	Testing	Deployment	SUBTOTAL
V. Bera	53hrs30min	75	6hrs40min	135hrs10min
K. Diseko	53hrs30min	75	6hrs40min	135hrs10min
S. Mngxozana	53hrs30min	75	6hrs40min	135hrs10min
L. Neku	53hrs30min	75	6hrs40min	135hrs10min
D. Nkosi	53hrs30min	75	6hrs40min	135hrs10min
O. Ramaru	53hrs30min	75	6hrs40min	135hrs10min
TOTAL	321	450	40	811

Grand Total Project Hours: 374 + 811 = 1,185 hours

Conclusion for Chapter 6

The documented 374 hours of completed work across Requirements, Analysis, Design, and SPMP phases demonstrate systematic progress and effective team collaboration. This record-keeping framework provides a solid foundation for monitoring the remaining 811 planned hours and ensures transparency throughout the project's implementation, testing, and deployment phases.

Appendices

4.1 Glossary of Terms

- **UFS Recruitment System:** The unified web and mobile platform designed to connect UFS students with employers.
- **API (Application Programming Interface):** Application Programming Interface enables secure data exchange between systems.
- **CV Builder Tool:** A feature that helps students auto-generate professional CVs.
- **POPIA:** The Protection of Personal Information Act, South Africa's data privacy law governing the handling of user data.
- **SkillStack:** A portfolio showcase of student projects, coursework, and achievements.
- **Skill Gap Analysis:** A tool that compares a student's skills against job market demands and suggests learning resources.
- **UFS Database:** The centralised university database holding academic records, integrated via secure APIs
- **Use Case:** A description of how a user (actor) interacts with the system to achieve a specific goal.

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**Additional research materials and team brainstorming notes. (2025).
Unpublished internal documentation.**



THANK YOU